

EFC Gases & Advanced Materials

Controls Standard

Date: October 2, 2026
Author: A. Zinman
Revision: H

Revision History

Table 1: Revision History

Rev	Date	Author	Description
A	September 21, 2026	A. Zinman	Initial rough draft.
B	September 21, 2026	A. Zinman	This was a test of the revision process.
C	September 22, 2026	A. Zinman	Added terminology and process definitions, selected panel-design guidance, scale and NAT-device comparisons, sequence guidance, common-system guidance for PDC compressors, and simplified SCADA/HMI standards.
D	September 24, 2026	A. Zinman	Reworked the guide for new controls hires, added PHA and interlock/permissive-table guidance, and expanded the analog-input terminal options.
E	September 30, 2026	A. Zinman	Aligned controls inputs and outputs with project execution phases 0–8; added the Stage 5A handoff, development and construction work, field wiring responsibility, and FAT Checkout timing.
F	October 1, 2026	A. Zinman	Simplified the overview and governance text; rewrote the design-process stages as summaries with numbered input, work, and output lists; trimmed contractor submittals; added PHA, setpoint, and future-practice notes; reorganized PLC hardware; added Phoenix Contact screw-terminal alternates, plain-language SCCR and enclosure-heat guidance with worked examples, protocol-converter and temperature-transmitter options, M580-as-gateway guidance with a network diagram, the OnLogic CL260 workstation, monitor tables, and Class I Div 2 equipment guidance; added the control transformer sizing appendix; added chapter/section page headers.

Table 1 (Revision History) continued

Rev	Date	Author	Description
G	October 1, 2026	A. Zinman	Converted the design-process stage lists to simple tables; moved the PHA explanation into Section 5.3; added E-stop requirements (dual-channel, illuminated, PLC feedback, reset button, PLC-driven force-guided relay) and the FGR definition; added grounding and bonding guidance; moved the SCCR, enclosure heat, and transformer calculations into a new Important Equations appendix; moved panel networking to Chapter 8; expanded the VFD section with a drive table, control methods, and Safe Torque Off; added thin-client hosts and an Ignition server section; added software versions, the Control Expert Classic license, and legacy ICONICS notes; added the approved vendors list; added a Modbus addressing appendix for the M580 and Productivity2000; made all tables full width.

Table 1 (Revision History) continued

Rev	Date	Author	Description
H	October 2, 2026	A. Zinman	Renamed the master file and added a build script that produces the full PDF and a separate PDF for each appendix; formalized the How to Use section and changed approvals to the Lead/Senior Controls Engineer; added the deliverables master list, permissive setpoints, an alarm list example, IPC and drawing-document terms, and footnotes citing the standards used with a Standards Referenced table; added OnLogic (pending); updated VFDs (Invertek), solenoid alternates, the air prep unit, scales (Mettler Toledo, Laumas TLB4), EKI gateways, industrial monitors and keyboards, and moved the NAT table into Chapter 6; reorganized servers and operator stations; rewrote source control, review/MOC, online changes, backup, and change testing; revised networking (protocol standards with OPC UA, corrected M580 topology, tiered security, hardening, time server) and simplified the accounts chapter; added Ignition login/SSO notes, data-type and function-block sections; added appendices for the Perspective screen design standard (replacing the scaling procedure), Edge Panel integration, electrical drawings, Class I Div 2 design, alarm documentation, data types, function blocks, Git, change testing, a sample FAT, and Cato/Claroty.

Lettered revisions are used before the initial approved release. Numbered revisions begin after formal approval.

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1. Overview

1.1 Purpose

This guide explains how EFC approaches the design, implementation, testing, commissioning, documentation, operation, and maintenance of industrial control systems. It is written primarily for engineers who are new to EFC projects and is intended to help them build systems that are safe, supportable, repeatable, secure, and consistent.

1.2 Scope

The guide covers PLCs, remote I/O, industrial control panels, operator workstations, HMI/SCADA systems, industrial networks, instrumentation interfaces, packaged-equipment integration, and the related project documents. It applies to internal engineering work and to contractors unless a project specification gives a more stringent requirement.

This document does not replace the adopted electrical code, the requirements of the Authority Having Jurisdiction (AHJ), a project risk assessment, a process hazard analysis, or a machine-safety/functional-safety assessment. The project engineer must identify the governing jurisdiction and applicable editions at project initiation.

1.3 How to Use This Guide

This guide does not need to be read cover to cover. Start with the parts that apply to the current stage of the project and return to the rest as needed.

Where to start

- **Selecting parts or designing a panel:** Chapter 6 (Hardware).
- **Writing PLC or SCADA code:** use the PLC and SCADA standards chapters and the related appendices.
- **Working with a contractor or panel shop:** the Working with Contractors chapter lists what they are expected to provide*[Future: A document will be made that will be a standard Panel Build Contract and well as one for the Electrical Contractors used. These will be attached as appendices when available.]*.

What the key words mean

- **Must** — required for safety, code compliance, security, or a basic EFC practice. If a requirement cannot be met, discuss it with the Lead/Senior Controls Engineer before proceeding.

- **Should** — the expected practice. A different approach may be used when the project requires it, provided the reason is documented.
- **May** — a permitted option.

How to read the notes in this guide

- *[Orange bracketed text]* marks an open question or an item that still needs a decision. It should not be treated as settled.
- *[Future: blue bracketed text]* marks something planned for later. It is not part of the current process.
- *TBD* in red marks a description that has not been supplied yet.
- A green **NEW** tag marks content that was recently added.
- Blue text is a link. Links to manufacturers and outside references are provided for convenience; keep a controlled copy of any datasheet the project depends on.
- Footnotes identify the standard (and edition) where text is taken directly from a code or standard.

When the guide does not fit the project

Every project is a little different. If something in this guide does not fit, or appears to be wrong, raise it with the Lead/Senior Controls Engineer. Changes for a single project are recorded as a deviation (see Section 1.4); changes that should apply to all projects go into the next revision of this guide.

1.4 Governance

Controls Engineering owns this guide, and the Lead/Senior Controls Engineer (or someone they choose) approves changes to it. We review it at least once a year, or sooner if it is required.

If a project needs to do something differently, write it down in the design basis as a deviation. Include what requirement you're not meeting, why, what the risk is, what you're doing instead, who approved it, and when it will be closed out. A deviation only applies to that one project; it doesn't change the guide for anyone else. *[Future: Create a standard deviation document.]*

1.5 Controlled Layers

Controls documentation must be maintained in four layers:

1. Engineering requirements.
2. Approved hardware and software catalog, including lifecycle status *[Future: to be added]*.
3. Version-controlled reusable PLC, SCADA, and drawing libraries.
4. Project-specific design, test, configuration, and validation records.

2. Terminology

Alarm
An operator notification requiring a defined response within a defined time to prevent or mitigate an undesirable consequence.

Warning
An advisory condition that increases awareness but does not meet the alarm definition.

Fault
A detected abnormal condition of a device, signal, communication path, or control function.

Control narrative
A written description of how a control system is intended to operate, including operating modes, sequences, permissives, interlocks, alarms, operator actions, failure responses, and interfaces with other equipment.

Table of Interlocks and Permissives
A project document that lists the conditions that allow equipment to operate and the conditions that stop or prevent operation. It normally identifies the initiating condition, affected equipment or action, required response, reset behavior, and related notes. The project team may begin the table earlier, but it cannot be finalized until the PHA is complete and its controls-related recommendations have been incorporated.

Process Hazard Analysis (PHA)
A structured review used by the project team to identify process hazards, evaluate existing safeguards, and recommend additional protection where needed. PHA results may create or change alarms, permissives, interlocks, and trips.

Permissive
A non-safety interlock: a condition that must be satisfied before an action (opening a valve, starting a pump) may start or continue. A permissive is not a rated safety function and is not a substitute for one.

Permissive setpoint
The value a permissive compares against. The operator may adjust it from SCADA, but only within a minimum and maximum that are hard-coded in the PLC and displayed on the SCADA screen. Example: the maximum pressure at which the vacuum pump lines may be opened. See Section 5.6.

Control interlock
Logic that prevents or stops an action to protect normal operation or equipment. It is implemented

in the basic process control system unless the risk assessment requires independence.[1]

Equipment protective interlock
A protective action intended principally to prevent equipment damage. Its required independence and integrity must be established by risk assessment.

Emergency-stop function
A complementary protective measure intended to avert or reduce an existing hazard. It is not a substitute for safeguarding or risk reduction by design.

Force-Guided Relay (FGR)
A relay whose contacts are mechanically linked so that if one contact welds shut, the others cannot change state. A safety relay can monitor one of the linked contacts to detect that fault. Used wherever a relay is part of an E-stop or other safety circuit (see Section 6.3.3).

Trip
An automatic transition to a defined safe or protective state.

Bypass/Override
An authorized temporary defeat or substitution of a protective condition or output. It must be controlled, visible, time limited, and logged.

Inhibit
Prevention of an alarm or protective action from operating while its initiating condition may still exist.

Suppression
Prevention of an alarm based on a designed operating state or condition.

Shelving
A time-limited operator action that temporarily removes an alarm from normal presentation while retaining its record.

Safe state
The process or equipment condition defined by the risk assessment as producing an acceptable level of risk.

First-out
Identification and retention of the initiating condition in a group of protective trips.

Management of Change (MOC)
The controlled review, authorization, implementation, validation, and documentation of a change.

Short-Circuit Current Rating (SCCR)
The maximum prospective short-circuit current at a stated voltage that an assembly or component can safely withstand under specified conditions. The panel SCCR must be at least equal to the available fault current at its installation point.

Industrial PC (IPC)
A computer built for the plant floor or panel mounting rather than an office: fanless or filtered

[1] EFC currently uses interlocks in the standard PLC to serve as its safety instrumented functions (SIFs); we do not yet design separate, integrity-rated SIFs. This is expected to change when safety PLCs are added to our systems, and this guide will be updated then.

cooling, a wide operating temperature range, DIN-rail or panel mounting, 24 VDC power, industrial-grade SSDs, and long product availability. EFC uses IPCs for Ignition Edge Panel stations, thin clients, and rugged gateways (Section 6.5).

Single-line diagram (SLD)
A simplified drawing of the power system that shows each circuit as one line: the incoming supply, disconnects, protective devices, transformers, and major loads with their ratings. See Section 6.3.8 and Appendix D.

Electrical schematic
The detailed circuit drawing of a panel or system, drawn on numbered lines (rungs), showing every device, contact, coil, terminal, and wire number and how they are connected.

Cable schedule
A table listing every field cable: its tag, type, size, number of conductors, from and to points, route, and length.

UDT / DDT
User-Defined Type; called a Derived Data Type (DDT) in Modicon Control Expert. A reusable data structure that bundles the data for one piece of equipment or signal into a single variable (Section 10.4).

DFB
Derived Function Block: a reusable block of PLC logic in Control Expert, used with a matching DDT (Section 10.8).

FAT/SAT
Factory Acceptance Test and Site Acceptance Test, respectively.

3. Design Process

[Future: this chapter needs a thorough review before it is treated as final.]

3.1 Getting Started

Collect the information that will shape the controls work as it becomes available: project goals and schedule, P&IDs, equipment and valve lists, process descriptions, layout drawings, utility and power information, area classification, and network requirements. Keep a list of open questions, assumptions, and who will answer them. The bulk of controls engineering starts near the end of Stage 3; Stage 2 is for early discussions and planning.

3.2 Project Execution Phases

Use the project execution phase numbers and names below when planning controls work. These are practical handoffs, so work may overlap. Each stage starts with a short summary, followed by tables of the **required inputs**, the **work performed**, and the **outputs**. Check the inputs before promising the outputs. Stage 5A is an added controls handoff between Construction and Commissioning; it does not renumber the company phases.

3.2.1 Stage 0 — Project Evaluation

Define high level commercial and operational requirements, and develop initial business case and associated CAPEX/OPEX assumptions.
There is no meaningful controls involvement at this point.

Required Controls Inputs
1. N/A

Controls Work Performed
1. N/A

Engineering Work Performed
1. N/A

Controls Outputs
1. N/A

Outgoing Master Scope Document: Project Objective Document (POD) **Project Maturity:** 3-5% **Engineering Maturity:** 1%

3.2.2 Stage 1 — Project Initiation

The Project Manager/Engineer should engage the end customer to develop understanding of specific customer needs. If needed the Controls Engineer can be brough in to assist with this effort

Required Controls Inputs

1. Approved project direction
2. Team contacts
3. Overall project goal
4. General schedule
5. What is known about the equipment and site

Controls Work Performed

1. Get read into the project: goals, schedule, site, and equipment.
2. Record open controls questions and key interfaces (other systems, vendors, utilities).
3. Confirm who owns the P&IDs, Control Narrative, Interlock Matrix, and layout drawing.
4. Identify the information needed for preliminary engineering.

Relevant Engineering Deliverables

1. Estimated Target Timeline

Controls Outputs

1. List of open controls questions and key interfaces
2. Confirmed owner of the P&IDs
3. Confirmed owner of the Control Narrative
4. Confirmed owner of the Interlock Matrix
5. Confirmed owner of the layout drawing

Outgoing Master Scope Document: Project Charter **Project Maturity:** 3-5% **Engineering Maturity:** 5-10%

3.2.3 Stage 2 — Preliminary Engineering

4. Working with Contractors

4.1 Scope and Responsibility

The purchase specification must identify the contractor's design responsibility, applicable codes and editions, required certifications, supplied hardware/software, interfaces, exclusions, schedule, review milestones, testing, commissioning support, training, warranty, and turnover deliverables. Acceptance by EFC does not relieve the contractor of code compliance or design responsibility.

4.2 Required Submittals

Before fabrication or programming, contractors must submit the basis of design, component data, schematics, panel layout, I/O list, and bill of material.

The following submittals are **optional** and are requested only when the project calls for them:

- SCCR calculation (see Section 6.3.2)
- Enclosure heat calculation (see Section 6.3.6)
- Network architecture

Deviations from this standard must be clearly listed; silence does not constitute approval.

4.3 Panel Fabrication

The purchase order must state whether a UL Listed industrial control panel is required. When listing is not required, panels must still follow the documented UL 508A design approach and this standard, but drawings and nameplates must not imply a UL Listing that does not exist. Fabricators must be qualified for the voltage, SCCR, environmental, and hazardous-location requirements of the project.

4.4 Approved Vendors

The vendors below are approved for controls hardware, services, and support. Use the contact listed where there is one; otherwise call the main number.

Table 4.1: Approved Vendors

Company	Location	Contact	Phone	What We Use Them For
MDCI Automation	Hatfield, PA	Jeff DiTrolio	*[to be added]*	*[to be added]*
ADI (Automation Distribution)	2398 North Penn Rd, Hatfield, PA 19440	Art McEvoy	888-600-3080 (main)	*[to be added]*
United Electric Supply	1111 N. Broad St, Lansdale, PA 19446 (closest branch to Hatfield)	—	215-368-1960	*[to be added]*
ifm efector, inc.	1100 Atwater Dr, Malvern, PA 19355 (US headquarters)	—	800-441-8246	*[to be added]*
WIAutomation	Online; operated by Vision Automation SA, Lugano, Switzerland	—	+41 91 21 01 366 (customer support)	*[to be added]*
Neal Systems, Inc.	122 Terry Dr, Newtown, PA 18940	—	215-968-7577	*[to be added]*
AutomationDirect	3505 Hutchinson Rd, Cumming, GA 30040	—	800-633-0405 (sales); 770-844-4200 (tech support)	*[to be added]*
Airline Hydraulics	3557 Progress Dr, Bensalem, PA 19020	—	*[confirm]*	*[to be added]*
Kendall Electric	15 Colwell Ln, Conshohocken, PA 19428	—	610-832-9000	*[to be added]*
OnLogic *[pending approval]*	435 Community Dr, South Burlington, VT 05403	—	802-861-2300	Industrial PCs (IPCs)

[Add direct phone numbers for Jeff DiTrolio (MDCI) and Art McEvoy (ADI), MDCI's main number, and Airline Hydraulics' Bensalem number, and fill in what each vendor is used for. "Niel Systems" was taken to be Neal Systems, Inc., and "Kendal Electric" to be Kendall Electric; confirm. OnLogic's approval as a vendor is pending.]

5. Process Definition and Protective Functions

5.1 Permissives, Interlocks, and SIFs

A permissive is a non-safety interlock: a condition that must be satisfied before an action may start or continue (for example, header pressure below a limit before the vacuum pump lines may be opened). A control interlock prevents or stops an action to protect normal operation or equipment. Neither term, by itself, indicates that the function is safety rated.

A **Safety Instrumented Function (SIF)** is a protective function that uses one or more sensors, a logic solver, and one or more final elements to move a process to a defined safe state or keep it there. EFC currently implements this kind of protection as interlocks in the standard PLC rather than as separate, integrity-rated SIFs (see the footnote on *Control interlock* in Chapter 2). This is expected to change when safety PLCs are added to our systems.

5.2 Control Narrative

The control narrative must describe how the system is intended to operate. It must cover operating modes, startup, normal operation, shutdown, abnormal operation, restart following power or communication loss, manual and maintenance operation, permissives, interlocks, SIF interfaces, trips, alarms, operator actions, and interfaces with packaged equipment. Each output must have a defined de-energized condition and loss-of-communications response.

The process engineer normally owns the Control Narrative. Controls Engineering uses it, together with the P&IDs and equipment information, as the main basis for detailed PLC and SCADA development.

5.3 Table of Interlocks and Permissives

Begin the Table of Interlocks and Permissives as the process design develops so it can support early software planning and project discussions. Do not treat it as final until the PHA is complete.

How we use the PHA to build the interlocks.

1. The project team (process, controls, operations, and EHS) walks through the P&IDs one section at a time and asks "what could go wrong here?" — for example no flow, high pressure, high temperature, or a valve in the wrong position.

2. For each credible problem, the team writes down the cause, what would happen, and what already protects against it.

3. Where the existing protection isn't enough, the team makes a recommendation — often a new alarm, permissive, or interlock.
4. Controls adds each approved recommendation as a row in the Table of Interlocks and Permissives: what triggers it, what it does, and how it resets.
5. That row becomes part of the PLC program and is tested during FAT and commissioning.

A formal risk classification of each protective function (for example under ISO 13849 or IEC 61511) is not currently part of our process.

After the PHA, update the table to include the approved controls-related recommendations and have the project team review the result before using it as the final programming and test basis.

5.4 Protective Function Requirements

*[Future: this section describes a practice we plan to adopt. It is not currently part of the EFC design process. When adopted, it fits into Stage 4 — Detailed Engineering (Section **??**), alongside the Interlock Matrix, and the records would be checked during Stage 6 commissioning.]*

For each trip or safety function, the design record must state the initiating conditions, setpoint basis, voting, time delay, safe state, latching, reset location, restart behavior, first-out indication, bypass rules, loss-of-power behavior, loss-of-communication behavior, proof-test interval, and validation method.

Automatic reset or restart is prohibited where it can create a hazard. Reset must not initiate motion or process flow unless separately justified and risk assessed. Reset must only clear a latched condition after the initiating cause is no longer present.

5.5 Bypass and Override Control

[Future: this section should be added to our process in the future. It is not part of the current plan.]

Bypasses, overrides, and inhibits must require authorized access, a stated reason, a defined expiration, visible HMI indication, event logging, and shift/operations communication. Safety-function bypasses require the process defined by the applicable safety lifecycle and MOC policy. Software must not conceal an active bypass after restart.

5.6 Setpoints

A **setpoint** is a target or limit value the control system works with — for example a tank level the fill valve aims for, or the pressure at which an alarm goes off. A setpoint is a *number*. Permissives and interlocks (Section 5.1 and Chapter 2) are *logic*: a permissive decides whether an action is allowed to start or continue, and an interlock stops or blocks an action. Many permissives and interlocks compare a reading against a setpoint ("stop the pump if pressure is above the high-pressure trip setpoint"), but changing the setpoint only changes *when* the logic acts — the permissive or interlock logic itself is unchanged.

Because of that, treat the three kinds of setpoints differently:

- **Operating setpoints** (control targets and normal alarm limits) may be operator-adjustable. Each must have approved engineering limits, units, a default value, an access level, change logging, and validation.
- **Permissive setpoints** (the values permissives compare against, such as the maximum pressure at which the vacuum pump lines may be opened) may be adjusted by the operator, but only within a minimum and maximum that are hard-coded in the PLC and cannot be changed from SCADA. The limits are displayed on the SCADA screen next to the setpoint so the operator can see the allowed range. Each permissive setpoint must have units, a default value, an access level, and change logging, and the PLC must reject or clamp any entry outside the limits (Section 10.3.1). Where a hard-coded limit comes from an equipment rating or the PHA, it must be traceable to that source in the same way as a protective setpoint, so the operator can only make the permissive *more* conservative, never less. Each permissive setpoint is listed in the Table of Interlocks and Permissives with its default, limits, and source. (Example in the current library: `DDT_Station.HMI_SP_Max_Vac_Press`, Appendix H.)
- **Protective setpoints** (the values that trigger interlocks and trips) must be traceable to the PHA or an equipment limit and must not be adjustable from the HMI unless specifically authorized by the risk assessment and MOC process.

5.7 Alarm Management

[Alarm management at EFC is currently informal. A formal alarm-management procedure should be put in place; until then, use the requirements below as guidance.]

The project must apply the alarm lifecycle in ANSI/ISA-18.2.[1] Each alarm must have a unique identifier, cause, consequence, required response, priority, maximum response time, setpoint basis, deadband, delay, operating-state applicability, shelving/suppression rules, and test method. Priority must be based on consequence severity and available response time, not equipment importance alone.

Alarm systems must detect and manage chattering, fleeting, stale, standing, and frequently occurring alarms. Suppression by design must be state based and documented. Shelving must be time limited, access controlled, visible, and auditable. Alarm performance must be reviewed after startup and periodically during operation.

Example alarm list entry

An alarm list (the master alarm database) records every alarm and the decisions behind it. One example entry is shown below; Appendix G gives the full field list, more examples, and the governing standards.

[1] ANSI/ISA-18.2-2016, *Management of Alarm Systems for the Process Industries*; the international equivalent is IEC 62682:2022 (Ed. 2).

Field	Value
Tag / alarm ID	PT-210.HI
Description	Vacuum header pressure high
Setpoint / deadband	50 torr / 5 torr
On delay	10 s
Priority	Medium
Cause	Vacuum pump stopped or degraded, or a leak on the header
Consequence	Station evacuation stalls; product fill delayed
Operator response	Check the vacuum pump status and running current; inspect the header for an open valve or leak
Maximum response time	15 minutes

6. Hardware

6.1 PLC

6.1.1 Primary Selection — Modicon

Table 6.1: Modicon PLC Hardware — Primary Selection

Type	Part Number	Required Cable / Interface	Description
PLC CPU	BMEP582040	N/A	Modicon M580 standalone processor; 8 MB memory; supports up to 61 Ethernet devices and eight X80 remote I/O racks.
Power Supply	BMXCPS3500	N/A	Modicon X80 power supply; 100–240 VAC input; 36 W.
Ethernet Communication	BMENOC0301	Ethernet	Modicon M580 Ethernet network option module. Can also be used to separate networks (see Section 8.1.2).
OPC UA Communication	BMENUA0100	Ethernet	Modicon M580 OPC UA communication module.
Remote I/O Drop	BMXCRA31210	Ethernet/IP	X80 Ethernet RIO drop adapter with service port and advanced features.
Serial Communication	BMXNOM0200	RS-232 / RS-485	Two-port Modbus serial communication module.
Digital Input, 16-point	BMXDDI1602	BMXFCC203 / ABE7H16CR21	24 VDC discrete input module.
Digital Input, 32-point	BMXDDI3202K	BMXFCC203 / ABE7H16CR21	32 positive-logic, 24 VDC discrete inputs.
Digital Output, 16-point	BMXDDO1602	BMXFCC203 / ABE7H16CR21	24 VDC transistor output module.
Digital Output, 32-point	BMXDDO3202K	BMXFCC203 / ABE7H16CR21	32 positive-logic, 24 VDC transistor outputs.
Digital Output, 64-point	BMXDDO6402K	BMXFCC203 / ABE7H16CR21	64 positive-logic, 24 VDC transistor outputs.
Analog Input, 8-channel	BMXAMI0810	BMXFTA300	High-level analog input module; 0–20 mA, 4–20 mA, and ±10 V.

Table 6.1 continued

Type	Part Number	Required Cable / Interface	Description
Analog Output, 8-channel	BMXAMO0802	BMXFTA302 / ABE7CPA02	Eight-channel X80 analog output module.
RTD / Thermocouple Input	BMXART0814	BMXFCW301S	Eight-channel isolated low-level analog input module (see Section 6.4.7).
Ethernet Backplane, 12-slot	BMEXBP1200	N/A	Modicon X80 12-slot Ethernet backplane.
Standard Backplane, 12-slot	BMXXBP1200	N/A	Modicon M340/X80 12-slot DIN-rail backplane.
Backplane Extension Kit	BMXXBE2005	N/A	Modicon M340/X80 backplane extension kit.
Digital I/O Cable	BMXFCC203	40-way to dual HE10	Two-meter prefabricated digital I/O cord set.
Digital I/O Sub-base	ABE7H16CR21	HE10	Passive 16-channel ABE7 connection sub-base.
Analog Input Cable	BMXFTA300	28-way to SUB-D 25	Three-meter analog input cord set.
Analog Output Cable	BMXFTA302	20-way to SUB-D 25	Three-meter analog output cord set.
Analog Distribution Base	ABE7CPA02	SUB-D 25	Passive distribution base for eight analog channels.
RTD / TC Cable	BMXFCW301S	40-way to flying leads	Three-meter shielded low-level analog cord set.
Fiber Media Converter	MC 1000-MM ST	100Base-TX / multimode ST fiber	Phoenix Contact 10/100 Mbps multimode fiber-optic media converter.

6.1.2 Low-Cost Option — AutomationDirect Productivity2000

For low-cost applications, the standard PLC is the AutomationDirect Productivity2000 P2-622 CPU.

Table 6.2: AutomationDirect Productivity2000 — Low-Cost Option

Type	Part Number	Notes
CPU	p2-622	50 MB memory; 2 Ethernet ports; 2 serial ports (RS-232/485); USB-C; microSD slot; small OLED display. Needs Productivity Suite v4.0.1 or later.
DC Power Supply	P2-02DC	24 VDC input. One per base.

Table 6.2 continued

Type	Part Number	Notes
7-Slot Base	P2-07B	Holds the power supply, CPU, and seven I/O modules. DIN-rail or panel mount.
Analog Input	P2-08AD-1	8 channels, 4–20 mA, 16-bit. Needs external 24 VDC.†
Analog Output	P2-08DA-1	8 channels, 4–20 mA, 16-bit. Needs external 24 VDC.†
Digital Input	P2-16ND3-1	16 points, 12–24 VDC, sinking or sourcing; 2 commons.†
Digital Output	P2-16TD1P	16 points, 12–24 VDC sinking; 0.25 A per point; short-circuit protected.†

† *Each I/O module also needs a P2-RTB / P2-RTB-1 terminal block or a ZIPLink cable.*

Productivity2000 may be selected for a stand-alone, non-safety application with limited I/O, no high-availability requirement, a simple network interface, and no requirement for M580/X80 compatibility. Modicon M580 must be used when remote I/O, system growth, network segmentation, advanced diagnostics, redundancy, large memory, or integration with an existing Modicon architecture is required. The basis of design must record the selection. Remote I/O and termination hardware not listed above require technical approval before design release.

6.1.3 Panel Layout Consistency

Where a project includes multiple panels performing a similar function (e.g., a column control panel repeated across several production lines, or system variants that share a common core), panel layout — DIN rail arrangement, sizing, and general internal organization — should be kept as consistent as possible across those panels, even when their I/O content differs.

Rule of thumb: Design to the largest/worst-case variant for that panel type, and let smaller variants carry unused space rather than designing a unique layout for each. This trades a small amount of panel real estate for a significant reduction in fabrication variability, which matters most when panels are outsourced to a panel builder — a consistent design lets the builder run a repeatable process instead of building bespoke panels project to project.

This applies at the panel-type level, not across unrelated equipment — the goal is consistency among panels that do the same job, not one universal panel layout for the whole system.

[A Panel Layout Design Guide is in development. Once it is added as an appendix, a cross-reference to it will be placed here.]

6.1.4 Local vs. Remote I/O Placement

Whether a field device is wired to local or remote I/O must be based on lifecycle cost, cable length and voltage drop, analog signal integrity, fault containment, availability, environmental and hazardous-area classification, maintenance access, network loading, spare capacity, local power, and safe behavior following a communication failure. Physical distance is one input, not the sole criterion:

- **Local I/O:** used when the device/area is close enough to the primary panel that direct wiring is practical.

- **Remote I/O:** used when distance makes direct wiring impractical (cable run length, voltage drop, install cost). A remote drop is placed near the equipment cluster it serves (e.g., near a feed or fill section) and connected back to the primary panel via the plant network.

This decision should be made per functional area (e.g., column, feed, fill) based on the facility layout, not assumed to be the same across every project. This decision cannot be made from the P&ID alone — it requires the system layout drawing (see Stage 4 in Section **??**) showing physical distances between functional areas.

6.2 Panel Components

6.2.1 Terminal Blocks

This subsection lists the standard WAGO terminal blocks approved for use in panel builds under this standard, organized by use case. Except where noted, all part numbers are DIN-rail mount, push-in CAGE CLAMP® termination with push-button release.

Table 6.3: Terminal Blocks (WAGO)

Use Case	PN	Description	Size	Color	Rating	Cost
24 VDC Distribution	2202-1404	4-conductor through terminal block, push-button	2.5 mm^2	Blue	24 A / 800 V	$5.87
Fused Terminals (1/4 x 1-1/4")	2206-1631/1000-541	2-conductor fuse terminal block, pivoting fuse holder, blown-fuse LED (12–30 V)	6 mm^2	Grey	10 A (fuse sets protection)	Quote
Fused Terminals (5 x 20 mm)	2202-1811/1000-541	4-conductor fuse terminal block, pivoting fuse holder, blown-fuse LED (12–30 V)	2.5 mm^2	Grey	6.3 A (fuse sets protection)	Quote
Analog Outputs	2202-2201	Double-deck through terminal block, push-button	2.5 mm^2	Grey	24 A / 800 V	Quote
Analog Inputs	280-560	3-conductor sensor terminal block (triple-deck), colored conductor entries	2.5 mm^2	Grey	20 A / 400 V	Quote
Analog Inputs (Alternate)	2020-5311	3-conductor sensor terminal block for PNP sensors, pluggable signal level	1 mm^2	Grey	13.5 A / 250 V	Quote
Digital Inputs	2202-1201	2-conductor through terminal block, push-button	2.5 mm^2	Grey	24 A / 800 V	$2.46
Digital Outputs	857-304	Relay module, 1 changeover (SPDT) contact, 24 VDC coil, status LED	6 mm wide	Grey	6 A (contact)	$17.02
Ground – Single	2202-1207	2-conductor ground terminal block, push-button	2.5 mm^2	Green-Yellow	24 A / 800 V	$7.28
Ground – Double	2202-1407	4-conductor ground terminal block, push-button	2.5 mm^2	Green-Yellow	24 A / 800 V	$15.18

Costs are budgetary single-unit snapshots retained for estimating. "Quote" indicates that procurement must obtain current pricing. The 857-304 is an active relay module; its size is module width. Separate terminal colors/markers must be selected for DC common, 120 VAC line and neutral, and externally energized circuits. Ratings must be checked against conductor size, circuit protection, temperature, and panel SCCR.

Accessories

Table 6.4: WAGO TOPJOB S Accessories

Part Number	Description
2002-1291	End and intermediate plate, 0.8 mm thick
209-191	Separator, for Ex e / Ex i applications
2002-2292	End plate / end cover, for 2002 double-deck terminal blocks (orange)
2009-115	WMB-Inline marking strips (for Smart Printer), white, 5 mm width, 1,500 pieces per roll
2002-410	Cross-connector, pluggable, 10-pole, for 2.5 mm^2 TOPJOB S terminal blocks, 25 A max, IP20

Screw-Terminal Alternates (Phoenix Contact)

[WAGO does not make screw-clamp terminal blocks; its range is spring-clamp (push-in CAGE CLAMP®) only. When a project, customer, or panel shop requires screw terminals, use the Phoenix Contact UT (screw connection) equivalents below.]

Table 6.5: Phoenix Contact Screw-Terminal Alternates to WAGO Items

WAGO PN	Phoenix Contact Type	Order No.	Notes
Terminal blocks (Table 6.3)			
2202-1404	UT 2,5-QUATTRO BU	3044555	4-conductor feed-through, blue, 2.5 mm^2 — 24 VDC distribution.
2206-1631/1000-541	UK 10-DREHSILED 24 (6,3X32)	3005646	Fuse terminal for 6.3 x 32 mm (1/4 x 1-1/4") fuses, 24 V blown-fuse LED.
2202-1811/1000-541	UT 4-HESILED 24 (5X20)	3046090	Fuse terminal for 5 x 20 mm fuses, 24 V blown-fuse LED.
2202-2201	UTTB 2,5	3044636	Double-level feed-through, grey — analog outputs.
280-560	UT 2,5-3L	3214259	Three-level terminal block — analog inputs (sensor wiring).
2020-5311	UT 2,5-3L	3214259	No direct pluggable screw equivalent; use the same three-level block.
2202-1201	UT 2,5	3044076	2-conductor feed-through, grey — digital inputs.
857-304	PLC-RSC-24DC/21	2966171	Relay module, 1 changeover contact, 24 VDC coil, screw connection — digital outputs.
2202-1207	UT 2,5-PE	3044092	2-conductor ground terminal, green-yellow.

Table 6.5 continued

WAGO PN	Phoenix Contact Type	Order No.	Notes
2202-1407	UT 2,5-QUATTRO-PE	3044568	4-conductor ground terminal, green-yellow.
Accessories (Table 6.4)			
2002-1291	D-UT 2,5/10	3047028	End cover for UT 2,5 / UT 2,5-QUATTRO blocks.
209-191	ATP-UT	3047167	Partition (separating) plate for UT blocks.
2002-2292	D-UTTB 2,5/4	3047293	End cover for UTTB 2,5 double-level blocks.
2009-115	UC-TM 5	0818108	Blank terminal markers, 5 mm — print with the Phoenix marking system.
2002-410	FBS 10-5	3030213	10-position plug-in bridge (jumper) for 5.2 mm pitch UT 2,5 blocks.

These are suggested equivalents for size and function. Before ordering, check wire range, current/voltage rating, color, and that accessories fit the exact block family. Fuse terminals and relay modules in particular should be checked against the circuit they protect or switch.

6.3 Industrial Control Panel Design

This section explains the main design checks for a control panel in plain terms. If something here doesn't make sense or doesn't fit your project, ask before you design around it.

6.3.1 Codes, Listing, and Nameplate

Three documents set most of the rules for our panels: NFPA 70 (the National Electrical Code), NFPA 79 (electrical equipment on industrial machinery), and UL 508A (how industrial control panels are built and rated). Design panels using the UL 508A approach.

[Future: formally identifying which edition of each code applies (and any local changes) is something we plan to do for each project. It is not current practice.]

A panel is only *UL Listed* if a UL-certified shop builds it and applies a UL label. The purchase specification should say whether the project needs a listed panel. If it doesn't, the panel and drawings must not say or imply that it is UL Listed.

Every panel needs a nameplate that shows: supply voltage, phase, and frequency; full-load current; SCCR (see Section 6.3.2); enclosure type rating (for example NEMA 4X); and the drawing number.

6.3.2 SCCR and Circuit Protection

What SCCR is. If a short circuit happens somewhere downstream, a huge burst of current flows for a split second before a fuse or breaker clears it. The panel's **Short-Circuit Current Rating (SCCR)** is the largest of these fault currents the panel can survive without catching fire or blowing

apart. It is given in kA (thousands of amps) at a stated voltage, for example "SCCR: 65 kA at 480 VAC."

The rule: the panel's SCCR must be *equal to or greater than* the available fault current where the panel is installed.[1] The panel is only as strong as its lowest-rated power component, so one weak part (often a power distribution block or an unrated component) sets the rating for the whole panel.

How to calculate it: see Appendix C.1 for the step-by-step method, the equation, a worked example, and links to more detail.

Circuit protection. Every circuit needs the right protection: feeders, branch circuits, motors (overloads), control circuits, and transformers (see Section 6.3.6 and Appendix C.3). Each protective device must be able to interrupt the fault current it could see. Show the protection on the schematics.

6.3.3 Disconnects and Emergency Stops

Disconnect. Each panel needs a way to shut off all incoming power, usually a door-mounted handle. It should be interlocked with the door so the door can't be opened with power on, except by a qualified person using the defeat feature.

Emergency stop (E-stop). An E-stop is for emergencies only, not for normal process stops. Where the risk assessment calls for E-stops, they must meet the requirements below.

E-stop devices and circuits

1. **The device.** Red mushroom-head (palm) button on a yellow background, with direct-opening (positive-opening) contacts and mechanical latching: once pressed, it stays pressed until someone deliberately twists or pulls it to release it. These are the core requirements of ISO 13850 and NFPA 79; buy devices rated to IEC 60947-5-5.[2]

2. **Dual-channel.** Every E-stop circuit must be **dual-channel**: each E-stop has two separate normally closed safety contacts, wired as two independent chains into a safety relay (or, in the future, a safety controller). The safety relay must monitor both channels and detect a fault between them, such as one channel stuck closed or the two channels shorted together ("cross-fault detection"). With suitable rated components this arrangement is typically used for ISO 13849-1 Category 3 or 4 (PL d or e); the risk assessment sets the level actually required.[3]

3. **Stop category.** The E-stop must perform a Stop Category 0 (remove power immediately) or Category 1 (controlled stop, then remove power) as defined in NFPA 79 / IEC 60204-1.[4] The risk assessment decides which. For VFD-driven motors, see Safe Torque Off in Section 6.4.1.

[1] UL 508A, *Industrial Control Panels*, 3rd edition (2018, revised through 2025), Supplement SB; NFPA 70-2023, 409.22.

[2] ISO 13850:2015, *Safety of machinery — Emergency stop function*; NFPA 79-2024, 10.7 (devices for stop and emergency stop); IEC 60204-1:2016+A1:2021, 10.7.

[3] ISO 13849-1:2023, *Safety-related parts of control systems — Part 1: General principles for design.*

[4] NFPA 79-2024, 9.2 (stop categories and emergency stop); IEC 60204-1:2016+A1:2021, 9.2.2 (categories of stop functions).

4. **Illuminated.** Every E-stop must light up when it is pressed, so anyone can see at a glance which one stopped the system. The light is an indicator only; it is not part of the safety function and the button must still look red-on-yellow when unlit.
5. **Signal back to the PLC.** Every E-stop must have its own auxiliary contact wired to a PLC input, so the PLC and HMI can show exactly which E-stop was pressed. This contact is for indication only and is *not* part of the safety chain.
6. **No automatic restart.** Releasing an E-stop or resetting the safety relay must never restart anything on its own; it only allows a separate, deliberate start.

[Future: once safety PLCs are adopted, the separate auxiliary contact back to the PLC may no longer be needed. A safety PLC can tell which E-stop was pressed when each E-stop is wired to its own pair of safety inputs, depending on how the circuit is designed.]

E-stop reset button

Every E-stop system has a **blue, illuminated reset pushbutton** with two normally open (NO) contacts and a light:

- **Contact 1** is wired directly to the safety relay's reset input. This is what actually resets the safety circuit; the PLC is not in that path.
- **Contact 2** is a PLC input, so the PLC knows a reset was requested (for logging, HMI messages, and sequence logic).
- **The light** is driven by a PLC output:
  - **Off** while any E-stop is pressed.
  - **Blinking** once every E-stop has been released: a reset is required.
  - **Steady on** once the reset has been accepted and the safety relay's outputs are on (system healthy and ready to run).
- To drive the light, the PLC needs each E-stop's auxiliary signal and a status signal from the safety relay showing that its outputs are on.
- Set the safety relay for **monitored manual reset**: it accepts a reset only when the button is pressed *and released*, so a stuck or jumpered button can't reset the circuit by itself. A normally open contact works with this.

IEC 60204-1 allows blue (or white, grey, or black) for reset buttons and does not allow green, so a blue reset button is correct[5] (Schneider Electric — pushbutton colors per IEC 60204-1).

[IEC 60204-1 indicator colors use blue for "action required," which matches the blinking state. A steady "ready" indication is more often green or white; the steady-blue state is an EFC choice and should be confirmed during review.]

[5] IEC 60204-1:2016+A1:2021, 10.2.1 (colours of push-button actuators) and 10.3.2 (colours of indicator lights).

PLC-initiated emergency stop

Each E-stop system includes one **force-guided relay (FGR)** controlled by a PLC output, so the PLC can trigger an emergency stop (for example on a critical interlock).

- Wire it fail-safe: the PLC keeps the FGR energized during normal operation, and its contacts sit in the E-stop chain, one contact in each channel so the circuit stays dual-channel. The PLC trips the system by dropping the relay out; losing PLC power or the output also trips it.
- Use one of the FGR's mechanically linked contacts for monitoring (in the safety relay feedback loop or to a PLC input) so a welded contact is detected.
- Because a standard PLC drives it, this adds another way to stop the system but is not counted as a rated safety function (see the footnote on interlocks in Chapter 2).

6.3.4 Power Distribution and Segregation

Keep high voltage separate. When you can, put 240/480 VAC equipment (drives, motor starters, the main disconnect) in a different enclosure from the 24 VDC and 120 VAC controls. That way someone can work on the PLC without being exposed to 480 V. If that's not practical, use barriers, spacing, separate wire routing, and clear labels.

Keep wiring types apart. Route power, control, analog, safety, and network wiring separately to reduce electrical noise and the chance that one fault takes out several systems. Show shield termination, grounding, and cable entry on the drawings. Never put a switch or fuse in the protective ground (PE).

6.3.5 Grounding and Bonding

Grounding connects the electrical system to earth; **bonding** ties all metal parts together so that a fault trips the breaker or fuse instead of leaving metal live. The requirements come from NEC Article 250.[6]

- Bond every metal part that could become energized (enclosure, back panel, door, DIN rails, subpanels) to the ground bar; use a bonding jumper across door hinges.
- Bring an equipment grounding conductor in with the incoming power, sized from NEC Table 250.122, and make bonding connections to bare metal (remove paint or use bonding hardware).
- Bond neutral to ground in one place only: X2 to the ground bar at the control transformer (NEC 250.30).
- Never switch or fuse a ground conductor; use green or green/yellow insulation.
- Ground cable shields at the panel end only, unless the device manual says otherwise.

Further reading: NFPA 70 (NEC), Article 250 — free read-only access from NFPA; NFPA grounding and bonding fact sheet; Mike Holt — Grounding and Bonding (sample chapter).

[6] NFPA 70, *National Electrical Code*, 2023 edition: Art. 250, including 250.30 (separately derived systems) and Table 250.122 (equipment grounding conductor size).

6.3.6 Control Power, Heat, and Capacity

Sizing power. Size power supplies and transformers for the steady load, the start-up (inrush) surge, and future additions. Leave at least **20% spare** power-supply capacity and panel space unless the project says otherwise.

Control transformers. Most panels fed from 480 VAC use a control transformer to make 120 VAC for power supplies, coils, fans, lights, and a service receptacle. The basic rules:

- On a 480 V delta supply (no neutral), connect a single-phase control transformer across two of the phases.
- Bond the secondary's X2 terminal to the panel ground bar; X1 is the fused hot (see Section 6.3.5).
- Fuse both primary lines and the X1 secondary, using time-delay fuses on the primary.
- Size for the steady load plus 25%, then check that it can handle the inrush when coils pull in.

Appendix C.3 gives the sizing method, the NEC fuse limits, and a worked 480 VAC to 120 VAC example.

Heat. Everything in the panel gives off heat. If the inside gets too hot, electronics fail early — most PLCs, power supplies, and PCs are rated to about 40–55 °C inside the panel. Check the heat at the hottest expected ambient temperature, and also think about condensation, dust, corrosive air, washdown, and how filters will be maintained. In short:

- Add up the heat given off by everything in the panel.
- Compare it with how much heat the enclosure can shed through its walls on its own.
- If the enclosure can't shed enough, add cooling: a filtered fan when the room air is clean and cooler than the panel needs to be, an air-to-air heat exchanger when the enclosure must stay sealed, or an air conditioner when the room is as hot as (or hotter than) the panel is allowed to get.

Appendix C.2 has the equations, a table for choosing the cooling method, a worked example, and links to more detail.

6.3.7 Conductors, Identification, and Termination

Wire. Pick wire size, insulation, and temperature rating to match the current, the terminal it lands on, and the bend room available.

Numbering. A wire's number is the drawing line where that signal starts. A relay's or contactor's tag is the drawing line where its coil is. Label every wire and terminal at both ends. Wires that stay live when the panel disconnect is off (for example, powered from another panel) must be clearly marked, usually with orange (or yellow) wire and a warning label.

Colors. Show the wire color scheme on the project drawings and use it consistently:

- 24 VDC control (+): **blue**.
- Protective ground (PE): **green/yellow**.

- Neutrals, AC control wiring, and externally powered wiring each get their own distinct, code-compliant color.

6.3.8 Electrical Drawing Package

The electrical package produced in Stage 4 (Section **??**) is a set of related documents, each with its own job:

- **Single-line diagram (SLD):** the power system on one page, with each circuit drawn as one line — incoming supply, disconnect, protective devices, transformers, and major loads with their ratings. It is used to check SCCR and protection (Section 6.3.2).
- **Electrical schematics:** the detailed circuit drawings on numbered lines (rungs), showing every device, contact, coil, terminal, and wire number. Wire and device numbering follows Section 6.3.7.
- **Panel layout:** a scaled drawing of the enclosure door and back panel showing where every component, DIN rail, wire duct, and terminal strip goes, plus the nameplate.
- **Wiring and interconnection diagrams:** point-to-point connection drawings for the panel shop and the electrician, including the terminal plan and connections between panels and field devices.
- **Cable schedule:** a table of every field cable — tag, type, size, conductor count, from/to, route, and length.
- **Supporting lists:** the I/O list, terminal plan, and electrical bill of materials.

Appendix D describes each document in more detail, with short examples, the differences between them, and when each is used.

6.4 Other Hardware

6.4.1 VFDs

VFD selection must be approved by Controls and Electrical Engineering. Required features include an industrial Ethernet protocol compatible with the project, hardwired Safe Torque Off where required by the risk assessment, local disconnect coordination, motor overload protection, line/load reactor or filtering evaluation, fault history, parameter backup, and documented behavior on communication loss and power restoration. The project must document harmonic, cable-length, motor-insulation, grounding, and EMC considerations.

Table 6.6: VFD Options

Drive	Power / Voltage	Communications	STO Rating	Best Fit	Price
AutomationDirect DURApulse GS20	1/4–30 hp; 120, 230, 460, 575 VAC	Modbus RTU (RS-485) built in; EtherNet/IP + Modbus TCP card (GS20A-CM-ENETIP)	SIL 2 (TÜV), two STO inputs	Lowest cost; fans, pumps, conveyors; 100 kA SCCR	$190–$1,377 (list, Oct 2026)

Table 6.6 continued

Drive	Power / Voltage	Communications	STO Rating	Best Fit	Price
Allen-Bradley PowerFlex 525	0.5–30 hp (0.4–22 kW); 100–600 VAC	EtherNet/IP built in; Modbus RTU (RS-485)	PL d / SIL 2, Cat. 3	Rockwell systems (Rare Gas column, Helium system); wide local support	$$ (est.)
ABB ACS580	0.75–500 kW (1–700 hp); 208–600 VAC	Modbus RTU built in; EtherNet/IP and Modbus TCP via FENA-21 adapter	SIL 3 / PL e	General purpose, including large motors; strong pump and fan features	$$$ (est.)
Schneider Electric Altivar Process ATV630	0.75 kW up to large frame sizes; 208–690 VAC	Modbus TCP Ethernet port and Modbus RTU built in; EtherNet/IP via VW3A3720/3721 card	SIL 3 / PL e (verify per frame)	Best match for Modicon M580 systems (same vendor, DTM support)	$$$ (est.)
Invertek Optidrive P2 ODP-2-34100-3HF42-MN	Frame 3: 10 hp (7.5 kW), 18 A; 380–480 VAC 3-phase; internal EMC filter and brake transistor; IP20	Modbus RTU built in; EtherNet/IP via OPT-2-ETHNT-IN plug-in (preferred); Modbus TCP via OPT-3-MTPEG-IN (backup)	SIL 2 / PL d, Cat. 3	Specified drive for 10 hp, 480 V applications	$1,390 (distributor, Oct 2026)

Price tiers are estimates for drives of 10 hp and under: $ = budget, $$ = mid-range, $$$ = premium; actual prices are shown where known. Get a current quote for the actual horsepower and options. Confirm the STO rating and power range for the exact catalog number before ordering. For the Invertek P2, the OPT-2-ETHNT-IN EtherNet/IP module is the standard; the OPT-3-MTPEG-IN is an external Modbus TCP to Modbus RTU converter that connects to the drive's RJ45 port and is the approved backup only when the OPT-2-ETHNT-IN cannot be sourced. Using the backup means the drive is mapped with Modbus registers instead of EtherNet/IP (Appendix F).

How the PLC controls the drive

- **Analog plus hardwired I/O** (4–20 mA or 0–10 V speed reference, with hardwired run/stop and fault signals). Simple, works with any drive and any PLC, and easy to troubleshoot with a meter. But it uses several I/O points per drive and gives no diagnostics (no motor current, fault codes, or parameter access). Use for very simple applications or when matching existing equipment.
- **EtherNet/IP.** Full control and status over one network cable: speed, start/stop, current, faults, and parameters. The M580 scans EtherNet/IP drives through the DTM Browser. This is the standard method for new installations (Section 8.2).
- **Modbus TCP** (or Modbus RTU on RS-485). The backup method: used when a drive or its EtherNet/IP option is not available, or when converting an existing Modbus RTU system. Widely supported and simple to set up. Data comes as register numbers rather than named tags, so it needs a documented register map (see Appendix F).

Whichever method is used, define what the drive does when communication is lost (for example stop, or hold last speed) and document it. Network control is for normal operation only; safety stopping is done with Safe Torque Off.

Safe Torque Off (STO)

STO is a built-in drive safety function that stops the drive from producing torque by blocking the power stage. The incoming power stays connected, which allows a faster restart than opening a contactor.

- **Wiring:** the drive's two STO inputs are wired to two outputs of the safety relay (or, in the future, a safety controller), one per channel, matching the dual-channel E-stop circuit in Section 6.3.3. When the E-stop chain opens, STO removes torque.
- **Stop category:** wired directly, STO gives a Stop Category 0 (the motor coasts). For a Stop Category 1, use a safety relay with time-delayed outputs: the PLC ramps the drive down first, then STO is applied.
- **PLC status:** the PLC reads the drive's STO status (over the network or from a drive output) so the HMI can show why the drive won't run.
- **Limits:** STO is not electrical isolation. The motor and drive terminals can still be live, so lockout/tagout at the disconnect is still required for maintenance.

[Future: with safety PLCs, STO can be commanded from safety outputs or over a safety network (for example CIP Safety on drives that support it), instead of from a hardwired safety relay.]

6.4.2 Solenoids

1. Need to standardize airline size, preferably metric (4 or 6mm), since most solenoid providers use metric.
2. SMC EX260 Ethernet/IP Operated Solenoid Bank — standard solenoid bank for most projects using EtherNet/IP for I/O communication. Must incorporate a dump valve upstream of the solenoid to be controlled by the system.
   (a) SY300 and SY5000 series have imperial options.

Approved Manifold Base Model — SS5Y3-10S

SS5Y3-10S (SY3000 Series Manifold, Side Ported, EX260 Integrated-type for Output Serial Transmission System) is the standard solenoid bank manifold for most projects using EtherNet/IP for I/O communication. The final part number is built per-project from the configuration fields below; the ones marked "standard" should be held constant across projects, and the ones marked "project-specific" are selected based on the panel's actual valve count and layout.

Table 6.7: SS5Y3-10S Manifold Configuration Fields

Field	Selection	Status
SI unit	EA — EtherNet/IP, 32 outputs, M12	Standard

Table 6.7 continued

Field	Selection	Status
A, B port size	C6 — 6 mm push-in fitting	Standard (change only if flow requires)
Mounting	DIN rail (D / A / B)	Standard
Valve stations	2–24	Per project
P, E port location	U, D, or both sides	Per project
SUP/EXH block	Internal pilot, internal pilot with silencer, or external pilot	Per project
DIN rail length	Standard, or specified length 3–24	Per project

The manifold must include documented safe-state behavior and a separately controlled upstream dump valve where required by the risk assessment. Spare stations and output capacity must be included where practical.

Alternate Manifolds

Two alternates are approved. The Festo terminal is the Ethernet backup when the SMC EX260 manifold is not available. The SMC D-sub manifold is the option for panels that drive valves with discrete PLC outputs instead of an Ethernet I/O network.

Table 6.8: Solenoid Manifold Options

	SMC SS5Y3-10S + EX260 (primary)	Festo VTUG + CPX-AP-I (Ethernet backup)	SMC SS5Y3-10F (discrete)
Control signal	EtherNet/IP (SI unit EA)	EtherNet/IP; the same interface runs Modbus TCP when set by its rotary switch	One PLC discrete output per coil through a 25-pin D-sub cable
Interface part	Integrated EX260 SI unit	Bus interface CPX-AP-I-EP-M12 (8086610) with valve terminal interface VAEM-L1-S-24-AP (8081923)	25-pin D-sub connector: F1 = top entry, F2 = side entry (IP40)
Valves / size	SY3000, 10 mm	VTUG, 10 mm width (14 and 18 mm also available)	SY3000, 10 mm (same valves as the primary)
Stations	2–24	Up to 24 valve positions (48 coils)	2–24
Fittings	C6 — 6 mm push-in	QS 6 mm push-in (QS 4 also available on 10 mm)	C6 — 6 mm push-in
Example part	SS5Y3-10S… (built per Table 6.7)	VTUG-10-… (configured per project)	SS5Y3-10F1-06B-C6

	SMC SS5Y3-10S + EX260 (primary)	Festo VTUG + CPX-AP-I (Ethernet backup)	SMC SS5Y3-10F (discrete)
Why choose it	Standard: one cable for all valves, diagnostics over the network	Different vendor when SMC lead times or availability are a problem; works with M580 and Productivity2000	Small valve counts, no Ethernet I/O network, or where hardwired outputs are preferred
Cautions	—	Separate configuration table and EDS file; confirm the full order code with Festo	A double-solenoid valve uses two outputs; include this in the I/O count

The dump-valve and documented safe-state requirements above apply to all three options.

6.4.3 Standard Air Prep Unit

Each pneumatic system uses a standard SMC air preparation assembly (filter, regulator, and lubricator or filter-regulator, "FRL") with an electrically operated dump valve and a pressure sensor. The dump valve is the upstream dump valve required for the solenoid manifold above: when it is de-energized it exhausts downstream air so the valves cannot move. The pressure sensor confirms supply pressure to the PLC and is used as a permissive for pneumatic operations.

Table 6.9: Standard Air Prep Unit

Item	Description	Last Cost
SMC FRL assembly *[check]*	Modular SMC air preparation unit (AC20/AC30 series) with a solenoid residual-pressure-release (dump) valve (VP-series modular type) and a digital pressure sensor/switch (ISE20 series)	$295.53 (12/3/2025)

[The exact SMC part number is still needed (the reference photo did not come through). Add it to this table and confirm the port size, filter rating, regulator range, dump-valve coil voltage, and pressure-sensor output.]

6.4.4 Scales

Table 6.10: Approved Scale and Weight-Indicator Options

Device	Capabilities	EtherNet/IP	Modbus TCP
Rice Lake SCT-2200	DIN-rail weight transmitter for connecting load-cell weighing systems to a PLC or supervisory network. Intended for control-panel integration rather than use as a full operator scale terminal.	Supported through the dedicated SCT-2200 EtherNet/IP fieldbus module.	Supported through the dedicated SCT-2200 Modbus TCP/IP fieldbus module. Specify the required protocol module when ordering.
Minebea Intec Combics 3	Graphic-display industrial indicator for more complex weighing applications. Supports checkweighing, totaling, product memory, optional filling functions, statistics, and optional second weighing points.	Available using the B10 EtherNet/IP fieldbus option. Confirm the complete indicator and interface configuration when ordering.	Native Modbus TCP support was not confirmed in current manufacturer literature. The optional Ethernet TCP/IP interface must not be assumed to provide Modbus TCP.
Mettler Toledo weighing system (IND780/IND700 terminal or ACT350 transmitter)	One platform from gram-level precision bases up to 2,000 kg floor scales. The IND780 terminal runs up to four scales (analog, POWERCELL, IDNet, or SICS bases) plus a sum scale from one unit. The ACT350 is a DIN-rail single-scale transmitter; the ACT350xx version is rated Class I Div 2.	IND780: optional EtherNet/IP card. ACT350: EtherNet/IP version (30076690).	IND780: optional Modbus TCP card. ACT350: no Modbus TCP version (PROFIBUS DP, PROFINET, or EtherNet/IP only).

Resolution depends on the scale base, not the indicator. No single scale reads 1 g up to 2,000 kg (that would be 2,000,000 divisions; legal-for-trade industrial scales are limited to about 10,000 divisions). Use a precision base for gram-level weighing (typically up to 30–60 kg) and a floor scale with 0.2–0.5 kg divisions for heavy loads, both on the same terminal family so they work the same way in the PLC.

[A distributor lists the IND780 as discontinued (September 2026) and replaced by the IND700. Confirm with Mettler Toledo before specifying.]

Custom Scale Setups

A custom setup may be used instead of a packaged indicator: a standalone scale base (load cells or a weighing platform with no indicator) wired to a DIN-rail load-cell transmitter in the control panel. EFC has used the Laumas TLB4 family from AutomationDirect. Each model accepts four load-cell inputs and has relay outputs and a Modbus RTU port; the network or analog interface is

chosen by model number, not by configuration.

Table 6.11: Laumas TLB4 Load-Cell Transmitters (AutomationDirect)

Part Number	PLC Interface	Price
TLB4	16-bit analog output (current or voltage) and Modbus RTU	$308
TLB4ETHETCP	Ethernet TCP/IP with web server, and Modbus RTU	$454
TLB4MODBUSTCP	Modbus TCP and Modbus RTU	$518
TLB4ETHEIP	EtherNet/IP and Modbus RTU	$588

Prices are AutomationDirect list prices (October 2026). All models: 4 load-cell inputs (expandable to 16 cells), 24-bit input, 2 logic inputs, 3 relay outputs, 12–24 VDC, DIN-rail or panel mount (TLB4 specifications). Choose TLB4ETHEIP to follow the EtherNet/IP default (Section 8.2).

6.4.5 Protocol Converters

Use a protocol converter (gateway) when a device only speaks a serial or other protocol that the PLC or SCADA can't read directly — most often an instrument with a Modbus RTU (RS-485) port that needs to appear on Modbus TCP. Use the simple Modbus gateway when that's all you need; use a Red Lion when you need other protocols, more ports, or local data logging. Modbus addressing on the M580 and Productivity2000 is covered in Appendix F.

Table 6.12: Protocol Converter Options

	Advantech EKI-1221/1222/1224-CE	Red Lion DA10D	Red Lion DA30D
Best for	Simple Modbus RTU/ASCII ↔ Modbus TCP; pick the model by the number of serial ports needed	Basic multi-protocol conversion	Multi-protocol conversion with logging, web server, or expansion
Serial ports	1, 2, or 4 × RS-232/422/485 (EKI-1221, -1222, -1224)	1 × RS-232, 1 × RS-232/485	2 × RS-232, 1 × RS-485
Ethernet	2 × 10/100 RJ45	1 × 10/100	1 × 10/100
Protocols	Modbus serial (master/slave) and Modbus TCP (client/server)	300+ industrial drivers	300+ industrial drivers
Capacity / features	Up to 16 client / 32 server connections per serial port	SD card; no data logging or web server	SD card, data logging, web server, virtual HMI, expansion slot (e.g., fieldbus cards)
Configuration	Web browser / utility	Crimson software	Crimson software
Power	12–48 VDC, redundant dual inputs	24 VDC ±20%	10–30 VDC

	Advantech EKI-1221/1222/1224-CE	Red Lion DA10D	Red Lion DA30D
Hazardous location	UL/cUL Class I Div 2, Groups A–D; ATEX Zone 2	None listed	UL Class I Div 2, Groups A–D
Mounting	DIN rail or panel	DIN rail	DIN rail
Approx. price	$283 / $392 / $534 (AutomationDirect, Oct 2026)	Quote	Quote
Links	AutomationDirect: 1221, 1222, 1224; product insert, manual	Datasheet	Datasheet

6.4.6 Panel Networking

How panel networks are laid out — including when to use a NAT/security appliance and when to use the Modicon M580 itself as the gateway — is covered in Section 8.1.1. The appliance options are listed below. The ranking is a starting point; final selection must consider company cybersecurity requirements, availability, support, and the required number of NAT mappings and ports. A device with a stateful firewall is preferred, because it lets one box both hide the local network and restrict what can reach the exposed devices (Section 8.6).

Table 6.13: NAT and Security Appliance Options for Local Controls Networks

#	Device	Firewall	Fit for the Application	Reasoning and Cautions
1	Siemens SCALANCE S615	Stateful	Best overall fit when the device is both the security boundary and the 1:1 NAT appliance.	Strong industrial-security features and traffic control, with more configuration effort than a NAT-only device.
2	Moxa NAT-102	Basic filtering	Strong fit for a compact, straightforward 1:1 NAT deployment.	NAT-focused; well suited when ease of deployment matters more than advanced firewall functions.
3	Weidmüller IE-SR-4GT (2873910000)	Stateful (SPI)	Strong fit where Gigabit ports, NAT, and VPN are useful.	Four Gigabit ports, source/destination NAT, IPsec and OpenVPN. Suitable for a formal network boundary.
4	Hirschmann EAGLE20/EAGLE30	Stateful	Excellent fit for higher-security or more complex installations.	Advanced firewall and routing, but usually more costly and complex than a small system needs.

Table 6.13 continued

#	Device	Firewall	Fit for the Application	Reasoning and Cautions
5	Phoenix Contact FL NAT 2008	Basic filtering	Good fit when several local switch ports and NAT are wanted in one device.	Convenient eight-port format. Confirm that its filtering meets the plant security standard.
6	Allen-Bradley 1783-NATR	Limited	Functional for straightforward 1:1 NAT.	Easy to apply, but firewall capability is limited. DLR and Rockwell-integration advantages do not benefit the M580 application.
7	Schneider Electric TCSE-FEC23F3F21	Stateful	Modicon Networking industrial firewall/router with NAT and VPN.	**Not for new designs:** end of commercialization December 1, 2027.
8	Schneider Electric ConneXium Tofino TCSE-FEA23F3F22	Deep packet inspection (transparent bridge)	Industrial firewalling rather than NAT.	**Not for new designs:** end of commercialization December 1, 2027; not a natural fit for 1:1 NAT.

6.4.7 Temperature Sensors & Transmitters

Wiring temperature sensors to the PLC. There are two ways to get an RTD or thermocouple reading into the PLC:

1. **Directly into a temperature card.** Wire the sensor straight to an RTD/thermocouple input card such as the Modicon BMXART0814 (8 channels). There is no extra device per sensor, but the card itself is fairly expensive and uses a rack slot.

2. **Through a transmitter.** A small DIN-rail transmitter next to the terminals converts the sensor signal to 4–20 mA, which goes to an ordinary analog input card. Each sensor needs its own transmitter, but the 4–20 mA signal is less sensitive to noise and long wire runs, and it can share an analog card with pressure and flow signals.

How to choose: it mostly comes down to how many temperature signals the panel has. With only a few, transmitters are cheaper than buying a dedicated temperature card. Once there are enough temperature signals to fill most of a card (roughly six or more in one panel), a temperature card is usually the better buy. Also consider spare slots, distance from the sensor, and whether a 4–20 mA signal is needed by anything else.

*[Use **3-wire RTDs** as our standard. The third wire lets the PLC card or transmitter cancel out the resistance of the lead wires, which would otherwise read as extra temperature. 2-wire RTDs should not be used except for very short runs; 4-wire RTDs are only needed for high-accuracy measurements.]*

Table 6.14: Temperature Transmitter Options

	AutomationDirect ProSense SC6-7102 (current)	Phoenix Contact MINI MCR-2-RTD-UI / -TC-UI	Acromag TT230 series (TT235 / TT233)
Input	RTD or thermocouple (one universal model)	RTD (2/3/4-wire) model or thermocouple model	TT235: isolated RTD; TT233: thermocouple/mV
Output	4–20 mA, loop powered (5.5–35 VDC)	Configurable current/voltage (4–20 mA, etc.)	4–20 mA, loop powered
Power	From the loop	24 VDC supply (or DIN-rail power bus)	From the loop (12–32 VDC)
Isolation	Yes	Yes (3-way)	TT235 isolated; check TT233
Setup	PC software	DIP switches or software	USB software or mobile app
Hazardous location	Hazardous-location rated (FM, UL)	UL Class I Div 2, ATEX/IECEx	UL Class I Div 2, ATEX/IECEx Zone 2
Why choose it	Current standard; one part for both sensor types; low cost ($200, Oct 2026)	Very compact; same family as other Phoenix signal conditioners; DIP-switch setup with no PC	Slim 12.5 mm width; loop powered like the SC6-7102; easy USB setup
Links	Product page, specs	2902049 (RTD), 2902055 (TC)	TT235, TT230 guide

6.5 Servers, Computers, and Operator Stations

6.5.1 Ignition Perspective Server

The site SCADA server runs the full Ignition gateway (Perspective, alarming, historian, and device connections). It should be a server-class machine; operator PCs and thin clients (below) only display screens. Inductive Automation's Server Sizing & Architecture Guide gives these starting points:

Table 6.15: Ignition Gateway Sizing (Inductive Automation Guidance)

Size	CPU	RAM	Tags	Clients	Typical Use
Small	2 cores	2–4 GB	500–5,000	2–10	Edge Panel, HMI, small SCADA
Medium	4 cores	4–16 GB	10,000–50,000	20–50	Medium full SCADA or MES

Size	CPU	RAM	Tags	Clients	Typical Use
Large	8–16 cores	8–32 GB	75,000–150,000	50–100	Large full SCADA or MES

EFC minimum for a site Perspective server:

- Server-class CPU (Intel Xeon, AMD EPYC, or equivalent, as Inductive Automation recommends) with at least 4 cores.
- 32 GB ECC RAM, which leaves headroom above the "medium" guidance for growth and the 8.1 to 8.3 upgrade.
- Two SSDs in RAID 1 (mirrored), at least 960 GB each, plus redundant power supplies.
- Windows Server or a supported Linux LTS release, confirmed against the Ignition version's supported-OS list.
- The historian/SQL database on its own server, as Inductive Automation recommends for all sizes, plus a UPS, scheduled gateway backups, and a redundant (backup) gateway for critical systems.

Table 6.16: Ignition Server Options

Server	Configuration	Form Factor	Best For	Price
HPE ProLiant ML30 Gen11 (P72995-005)	Xeon E-2434 (4 cores, 3.4 GHz), 32 GB, 2 × 960 GB SSD on MR216i-p RAID controller, 2 × 500 W power supplies	Tower	Small/medium site server in an office or server room; meets the EFC minimum as configured	est. $4,500
Dell PowerEdge T360	Xeon E-2400 (up to 8 cores) or Xeon 6300 (up to 12 cores), up to 128 GB ECC; configure 32 GB and 2 × SSD in RAID 1	Tower	Medium site server with room to grow	From $5,299 (Dell.com base price, Oct 2026)
OnLogic Karbon 800 series	12th-gen Intel Core, configurable RAM and dual SSDs; fanless and wide-temperature	Rugged box PC	Gateway that must live on the plant floor or in a control panel; small/medium systems	est. $3,500–$5,000

The Karbon 800 uses a desktop-class Intel Core CPU rather than the server-class CPU Inductive Automation recommends; use it only where a rugged, fanless unit is required. "est." prices are estimates; get current quotes.

6.5.2 Default Operator PC

The standard operator PC is the OnLogic CL260 industrial PC (IPC). It is configured to run Ignition Edge Panel (a local Ignition gateway plus the operator display on the same PC) on both Ignition 8.1 and 8.3, and meets or beats the previous CL250 configuration in every area.

Table 6.17: Approved Ignition Operator Workstation Hardware

System	Ignition Application	Hardware Overview	Cost
OnLogic CL260	Ignition Edge Panel 8.1 and 8.3 (local gateway plus Perspective/Vision client on the same PC). Also suitable as a thin-client workstation for a central gateway. Not intended as a central gateway or database server.	**CPU:** Intel N250, 4 cores (higher-performing option of the two offered) **Memory:** 8 GB LPDDR5-4800 (soldered; the only size offered) **Storage:** industrial M.2 NVMe SSD, 256 GB preferred, 128 GB minimum **OS:** Windows 11 IoT Enterprise LTSC 2024 **Video:** 2 × USB-C with DisplayPort 1.4 (supports 4K at 60 Hz); needs a USB-C to DisplayPort cable **Network:** 2 × 1 GbE **Power:** 12–24 VDC terminal block (run from panel 24 VDC; no adapter needed) **Mounting:** DIN rail (MTD109 bracket)	Quote

Ignition's published minimum is a dual-core processor, 4 GB RAM, and 10 GB of free disk (Ignition 8.3 install guide; 8.1 install guide). Edge Panel runs the gateway and the client on one PC, so 8 GB and an SSD with room for logs and backups give comfortable headroom. Do not use both network ports to bridge the plant and local networks. Confirm the selected Windows release against the Ignition version's supported-OS list before ordering, and get current pricing from OnLogic (CL260 technical documentation).

6.5.3 Thin-Client Hosts

A thin client is a small PC or IPC that runs Perspective Workstation and displays screens served by a central Ignition gateway; it does not run its own gateway. Requirements:

- A 64-bit Windows, Linux, or macOS install. ARM-based systems and "zero clients" are not supported by Perspective Workstation.
- Graphics and a DisplayPort output able to drive the standard 32" 3840 × 2160 monitor at 60 Hz (Section 6.5.4).
- Wired Ethernet, an SSD, and at least 8 GB of RAM.

The OnLogic CL260 in Table 6.17 can also be used as a thin client; choose it when the PC must mount on a DIN rail inside a panel.

Table 6.18: Thin-Client Host Options

Model	Specs	Video	Best For	Price
Lenovo ThinkCentre M70q Gen 5 Tiny	Intel Core i5-14400T, 16 GB DDR5, 256 GB NVMe SSD, Windows 11 Pro	DisplayPort + HDMI; up to 3 monitors	Control rooms and offices; VESA mount behind the monitor	$1,104 (CDW, Oct 2026)
HP Elite t655 Thin Client	AMD Ryzen Embedded R2314 (4 cores), 16 GB DDR4, 256 GB NVMe SSD, Windows IoT Enterprise	3 × DisplayPort 1.4	Control rooms; locked-down, managed thin-client fleet	est. $750
OnLogic ML100G-56	Intel Core Ultra 5 125U, 8 GB, 256 GB SSD, fanless; Windows 11 IoT or Linux	Confirm 4K at 60 Hz on the ordered configuration	Plant floor; DIN-rail or wall mount, dusty areas	est. $1,100

"est." prices are estimates; get a current quote. The HP t655 ships with Windows 10 IoT Enterprise on some configurations; order a Windows 11 IoT image where offered, or confirm the support end date of the version supplied.

6.5.4 Default Monitor

*[Any screen selected must have a **16:9 aspect ratio** (e.g., 1920 × 1080 or 3840 × 2160) to work properly with the current SCADA screen design standard (see Appendix A).]*

Standard Thin-Client Setup

For a normal operator station (thin client or Edge Panel PC in a control room or clean area), use a generic 32" non-touch UHD monitor at 3840 × 2160.

Table 6.19: Standard Thin-Client Monitors

Part / Model	Item	Description	Cost
Generic	32" UHD monitor	Any 32", 16:9, 3840 × 2160, non-touch monitor with DisplayPort input and VESA mount.	—
Samsung LU32J590UQNXZA (example)	Samsung 32" UJ59 4K UHD	31.5" VA panel, 3840 × 2160, 16:9, 60 Hz; HDMI and DisplayPort; VESA mountable. Example of a generic monitor that meets the requirement.	—

Industrial Monitors

For panel-mounted or plant-floor stations, use an industrial touch monitor. The Hope Industrial 23.8" panel-mount monitor is the first choice; the others are alternates for budget, availability, or

hazardous-location needs. All options are 16:9.

Table 6.20: Industrial Monitors

Model	Display	Video / Power	Enclosure	Class I Div 2	Cost
Hope Industrial HIS-ML23.8-FPVB (first choice)	23.8", edge-to-edge, PCAP multi-touch	DisplayPort; 100–240 VAC or 10.8–26.4 VDC	NEMA 4X, IP65/66/69/69K	Yes	$1,420
AutomationDirect C-more CTM-24W-M-PM	23.8", 1920 × 1080, PCAP multi-touch, 200 nits	VGA and HDMI (no DisplayPort); USB-B for touch; 12 VDC	IP65 front; panel or VESA mount (brackets sold separately)	**No**	$852
Pepperl+Fuchs VisuNet FLX	19" (5:4) or 21.5" (1920 × 1080, 16:9); use the 21.5"	Remote Monitor (built-in thin client) or Edge Workstation (built-in PC or thin client)	Industrial, stainless options	Yes (also Zone 2/22)	Quote
Exor eX721M web panel	21.5", 1920 × 1080, PCAP multi-touch	No video input: runs Perspective in its built-in Chromium browser; 24 VDC	IP66 front; −20 to +60 °C	Yes (also ATEX/IECEx Zone 2/22)	Quote

Notes: The CTM-24W-M-PM is the budget option for unclassified areas only; it needs a 24-to-12 VDC converter (or a separate supply) when run from panel power, and a DisplayPort/USB-C-to-HDMI cable or adapter for the CL260 and thin clients. The VisuNet FLX is the hazardous-location step up, and its built-in thin client removes the need for a PC in a purged enclosure. The Exor eX721M is a web panel rather than a monitor: it replaces both the monitor and the thin client, but Perspective in its embedded browser (performance, touch behavior, and the screen scaling in Appendix A) must be bench tested before it is standardized. Hope prices are from a Hope Industrial quote (October 2026); AutomationDirect price is list (October 2026).

Industrial Keyboards

Mountable industrial keyboards with a built-in touchpad. Order the screen protector, DisplayPort cable, and USB cable with the monitor as needed.

Table 6.21: Industrial Keyboards

Model	Mounting	Touchpad	Sealing	Class I Div 2	Interface	Cost
Hope Industrial KB-R3-WMT-BLK-4-USB (first choice)	Wall mount, fold-down	Yes	NEMA 4, IP65/66	Yes	USB, 3.8 m cable	$1,150
iKey PM-102-SS	Panel mount (cut-out), stainless	Yes	NEMA 4X, IP66	**No**	USB or PS/2	$674

Table 6.21 continued

Model	Mounting	Touchpad	Sealing	Class I Div 2	Interface	Cost
Pepperl+Fuchs EXTA3-*-K9	Panel mount or stainless desk housing	Yes (resistive)	IP66 front	Yes (also Zone 2/22)	USB	Quote

The Pepperl+Fuchs EXTA3 requires its SK-PC-D2-UU1-10-HS nonincendive USB barrier in a Division 2 area, and pairs with the VisuNet FLX for a single-vendor hazardous-location station. The iKey keyboard is for unclassified areas only. Hope price is from a Hope Industrial quote (October 2026); iKey price is a distributor price (October 2026).

6.5.5 Class 1 – Div 2 Zone Hardware

A Class I, Division 2 (C1D2) area is one where flammable gas or vapor is *not* normally present but could be if something goes wrong, such as a leaking fitting or a release during maintenance. Equipment there must not be able to ignite that gas. The area-classification drawing shows which areas are C1D2 and which gas groups and temperature code (T-code) apply.

Basic rules:

- Everything installed in the area must be marked for Class I, Division 2 (or Zone 2), the right gas group, and a suitable T-code.
- Putting ordinary equipment in a box does not make it suitable, unless the box is a purged or pressurized enclosure built for that purpose.
- The wiring method and conduit seals are part of the rating.
- The area-classification drawing and the AHJ have the final say.

Ways to make a system C1D2 safe: (1) buy C1D2-rated devices; (2) use nonincendive (low-energy) circuits; (3) use intrinsic safety barriers; (4) use a purged/pressurized (Type Z) enclosure; or (5) keep the panel out of the area and run only rated field devices into it. Appendix E explains how the design rules differ from a normal panel.

Table 6.22: Class I, Division 2 Equipment Examples and Use Cases

Component Type	Example Product(s)	Typical Use Case	What to Check
Operator display & keyboard	Hope Industrial HIS-ML23.8 with KB-R3 keyboard; Pepperl+Fuchs VisuNet FLX with EXTA3 keyboard; Exor eX721M web panel (Tables 6.20 and 6.21)	Operator station on the plant floor in a C1D2 area	HazLoc marking on the label; a separate PC driving a monitor must be outside the area or in a purged enclosure (the VisuNet thin client and the Exor web panel avoid this)
Purge / pressurization system	Pepperl+Fuchs Bebco EPS Type Z	Using an ordinary PC, VFD, or other non-rated equipment inside a panel in the area	Type Z for Division 2; clean air or N_2 supply; loss-of-pressure alarm
Intrinsic safety barriers / isolators	Pepperl+Fuchs K-System or Phoenix Contact MACX MCR-EX isolators	Field sensors in a Division 1 area wired to a panel in Division 2 or a safe area	Entity parameters of barrier, cable, and field device must match
Signal conditioners / transmitters	Phoenix MINI MCR-2, Acromag TT230, ProSense SC6-7102 (Table 6.14)	Temperature and analog signal conditioning in a panel located in the area	C1D2 marking and installation instructions (often require a suitable enclosure)
Protocol gateways	Red Lion DA30D; Advantech EKI-122x-CE (Table 6.12)	Serial instrument integration from inside a C1D2 panel	Both are rated Class I Div 2
Conduit seals	Crouse-Hinds EYS sealing fittings	Where conduit leaves the classified area or enters an enclosure, to stop gas traveling through the conduit	Seal location per NEC 501.15; fill with the approved compound after wiring
Cable and cord fittings	Hazardous-location cable glands (e.g., Crouse-Hinds or Hubbell HazLoc glands)	Bringing cable (instead of conduit) into enclosures	Fitting and cable type both listed for the area
Lighting, fans, and AC units	HazLoc-rated LED fixtures; C1D2-rated enclosure fans and air conditioners	Panel lighting and cooling for panels located in the area	Rated for the class/division and T-code; cooling sized per Section 6.3.6
Pushbuttons and pilot lights	Hazardous-location operators or nonincendive-rated devices	Local start/stop stations and indicators in the field	Rating on each device; nonincendive circuit documentation

Further reading: Pepperl+Fuchs — Purge and pressurization systems; Crouse-Hinds — EYS/EZS

sealing fittings.

7. Software Lifecycle

7.1 Approved Platforms

Schneider Electric Control Expert is the primary Modicon programming platform; Productivity Suite is used for approved Productivity2000 applications; and Ignition is the standard SCADA/HMI platform for new projects. Projects must use a company-approved release that remains supported by the vendor and is compatible with installed firmware and modules. The approved version register must be maintained separately from this standard so security and compatibility updates do not require a full standards revision.

7.1.1 Current Software Versions

Table 7.1: Programming and SCADA Software Versions

Platform	Software	Version	Notes
Schneider Modicon M580	EcoStruxure Control Expert Classic	v16.2	EFC standard PLC platform. License ordering below.
AutomationDirect Productivity2000	Productivity Suite	v4.7	Latest full release listed by AutomationDirect (October 2026). The P2-622 CPU needs v4.0.1 or later. Free download.
Allen-Bradley ControlLogix	Studio 5000 Logix Designer (Logix5000)	v38	**Not an EFC standard platform.** Used only to support the two existing systems that run on ControlLogix PLCs: the Rare Gas column and the Helium system. New projects use the M580 (or Productivity2000 for small systems) per Chapter 6.
Ignition (SCADA/HMI)	Inductive Automation Ignition	8.1	EFC standard SCADA/HMI platform; upgrade to 8.3 planned.

7.1.2 Control Expert Classic License

New engineering seats use a **Large (L)** license, which covers M580 programming. The older paper license (CEXSPUCZLSPMZZ) was discontinued in 2022; order the digital (node-locked) license instead.

Table 7.2: Control Expert Classic License Ordering

Part Number	Description	Last Quote
CEXSPUCZLSPAZZ	Control Expert Classic — Large (L) — single license (1 user), digital node-locked license	$2,380.67 (WIAutomation, October 2, 2026)

[If several engineers need seats, ask the distributor about team (multi-user) license options.]

7.1.3 ICONICS SCADA (Legacy)

Most of our legacy systems use **ICONICS** SCADA software (the GENESIS family). ICONICS is **not** used on new projects; Ignition is the standard SCADA/HMI going forward.

- Existing ICONICS systems are maintained as they are. Changes to them follow the same change, backup, and testing rules as the rest of this chapter.
- Moving a system from ICONICS to Ignition is done as a planned project, not piece by piece during other work.

[Current ICONICS version(s) in use: to be confirmed.]

7.2 Source Control and File Naming

There is currently no official source control system for PLC and SCADA software. Files are saved manually using the naming convention below.

File name format: `ProjectNumber_ShortDescription_Revision_YYMMDD`

- Example of a working file: `014_R23-Column-PLC_C_261002` (project 014, R23 column PLC program, revision C, saved October 2, 2026).
- The short description uses hyphens, not underscores, so the four fields stay easy to separate.

Revisions:

- During development, revisions are **letters** (A, B, C, . . .).
- The as-built software at the time of turnover to Operations is always **Rev 0**. Changes after turnover continue as Rev 1, 2, 3, . . .
- Development involves many changes, so the engineer uses their discretion on when a change deserves a new revision. The exception is a change that requires an MOC (Section 7.3): it always gets a new revision.
- When a revision is officially released, the engineer drops the date suffix from the file name. A file name with no date marks a released version (for example `014_R23-Column-PLC_0`); working files keep the date.

[Future: integrate Git for real revision control of PLC and SCADA software (see Appendix J).]

[Future: keep a companion document with the software that lists each revision and what changed between revisions.]

[Future: major and minor revisions are under consideration — for example, an MOC change creates a major revision while minor revisions are left to the engineer's discretion. To be determined.]

7.3 Review and Release

Interlocks and safety-related items must not be changed without an approved Management of Change (MOC). This includes interlock logic, protective setpoints, the hard-coded limits on permissive setpoints, bypass and override logic, and anything tied to the E-stop or safety relay circuits. Any such change always requires a new software revision (Section 7.2). This applies to software that has been turned over to Operations (Rev 0 and later); before turnover, interlocks are developed and reviewed through the process in Chapter 5.

Changes should be peer reviewed for logic, safe-state behavior, cybersecurity, alarms, performance, maintainability, and test coverage. Development files should not be placed in production without review, except under an approved emergency-change process. For each released revision, record what changed, how it was tested (Section 7.6), and who loaded it.

7.4 Online Changes and Forces

[This is not currently enforced. A policy will be reviewed and defined before it is put in place; until then, treat the text below as guidance.]

Online edits, forces, simulations, and overrides should be authorized, assessed for risk, logged, and clearly indicated on SCADA. In particular:

- Keep a running list of active forces and overrides, with who applied each one, why, and when it will be removed. Every force is removed, or made part of the controlled source, before the work is closed out; forces are never used as permanent logic.
- Set a time limit for every force or bypass, and review any that are still active at shift change.
- A force or bypass on an interlock or safety-related signal needs a second person to check it and an approved MOC (Section 7.3).
- After any change to production software, upload the running program, save a verified backup, and retest what the change affects (Section 7.6).[1]

7.5 Backup and Recovery

[This is not a formal policy. It is a proposed approach to discuss with the engineering group, to decide whether it should be adopted and, if so, how it should be implemented.]

[1]Sources for this section: OSHA 29 CFR 1910.119(l) (management of change); ISA/IEC 62443-2-1:2024, SPE 2 (configuration management and change control) and SPE 7 (event logging); ISA/IEC 62443-3-3:2013, SR 2.8 (auditable events) and SR 7.6 (network and security configuration settings); NIST SP 800-82 Rev. 3 (2023), App. F (configuration management overlay); IEC 61511-1:2016+A1:2017, 11.7 and 16.2 (bypasses, overrides, and forcing of safety functions); NFPA 79-2024, 18.7 and IEC 60204-1:2016+A1:2021, 18.7 (retesting after modification).

What should be backed up:

- PLC programs: the Control Expert `.STU`/`.STA` files and the Productivity `.adpro` file.
- The configuration actually running in the PLC, uploaded from the controller, so the backup matches the field.
- Ignition gateway backups (`.gwbk`) and project exports.
- Network switch and router (NAT/firewall) configurations.
- VFD parameter files.
- Database structure (schema) for historian and application databases.
- Certificates and license information.
- A short written recovery procedure.

Test the restore. A backup is only proven when it has been restored onto spare hardware or a virtual machine and shown to work; saving the file is not enough. This should be done once before turnover and then on a regular schedule.

Protect the backups. Keep backups where only authorized people can change them, so older versions cannot be overwritten.

Keep a copy away from the equipment. At least one copy should be stored away from the system it backs up. A copy kept only on the operator PC or server in the same panel or room would be lost with it in a fire, flood, or hardware failure. Released backups use the released file name without a date (Section 7.2).

7.6 Change Testing

This section covers testing of changes to a system that has already been built — during commissioning or after turnover. Initial testing of a new system is covered by the FAT Checkout (Appendix L) and commissioning (Chapter 3).

Any change to I/O, the electrical system, sequences, interlocks, permissives or their setpoint limits, alarms, or similar must be documented and tested before the equipment is returned to service. Record the change on the Change Test Record in Appendix K. Testing is scaled to the change: a text edit on a screen needs a quick check, while an interlock change needs a full functional test. Changes that require an MOC follow Section 7.3.

Where relevant, the test should cover:

- Normal operation of the changed function and of anything connected to it.
- Limits and boundary values, including permissive setpoint limits.
- Bad-quality, out-of-range, open-circuit, or shorted analog signals.
- Loss and restoration of communications and power, and restart behavior.
- Alarms, permissives, trips, and first-out indication.
- Manual modes and operator commands, including security roles.

- That a current backup was taken before the change and a new one after it.

Safety functions require independent validation appropriate to their lifecycle.

8. Networking and OT Cybersecurity

8.1 Architecture

Industrial networks are separated into zones (groups of devices that need the same level of protection) connected through controlled conduits, following the ISA/IEC 62443 approach and company IT/OT policy.[1] The aim is a level of protection that matches the system's size and risk, without making the system hard to operate and support.

Every panel:

- Keep each panel's devices on a local controls network, separated from the plant network by a NAT/security appliance or by the M580 itself (Section 8.1.1).
- Expose only the devices that must be reached from the plant network (usually the PLC, sometimes the HMI), and allow only the traffic they need.
- Keep unencrypted industrial protocols (Modbus TCP, EtherNet/IP) inside the controls zones; they are not routed to the business network or the internet.
- Change default passwords and disable unused services and switch ports (Section 8.6).

Larger or higher-risk systems (plant-wide SCADA, multiple panels sharing a network, remote access, or connections to the business network) also need:

- A firewall between zones with allow-only rules, and VLANs where several zones share switches.
- A data-flow list identifying each connection's source, destination, protocol, port, purpose, and owner. For a single panel, the NAT mapping list and the Stage 4 network map serve this purpose.
- Central logging of security events and a documented recovery plan.

8.1.1 Panel Networking

The normal application isolates a local controls network while exposing selected devices, such as the PLC, to the larger plant network through 1:1 NAT. Device Level Ring (DLR) support is not required for this use case. There are two ways to do this:

1. **A NAT/security appliance** between the plant network and the local controls network (Table 6.13 in Chapter 6). This is the most flexible option and the right choice when several devices need to be reached from the plant network.

[1] ISA/IEC 62443-3-2:2020 (security risk assessment, zones and conduits) and ISA/IEC 62443-3-3:2013 (system security requirements and security levels); NIST SP 800-82 Rev. 3 (2023), *Guide to Operational Technology (OT) Security*.

2. **The Modicon M580 itself as the gateway** (Section 8.1.2). This avoids an extra box when only the PLC's data needs to reach SCADA.

Figure 8.1 shows both layouts.

What makes layout (a) work:

- **Default gateway.** Every exposed device on the local network must have its default gateway set to the NAT appliance's inside (local) address. Without this, replies to the plant network have no route back.[2]
- **What crosses the NAT.** SCADA polling crosses 1:1 NAT without trouble: OPC UA, Modbus TCP, and EtherNet/IP explicit messaging. Multicast traffic does not translate through NAT, so EtherNet/IP I/O (implicit) connections between the PLC and its drives, valve banks, and remote I/O stay on the local network.
- **Standard panels.** Because each panel is hidden behind its own NAT, every panel can reuse the same local subnet template. This keeps panel designs, spares, and vendor setups the same from project to project; only the plant-side (outside) addresses are unique.

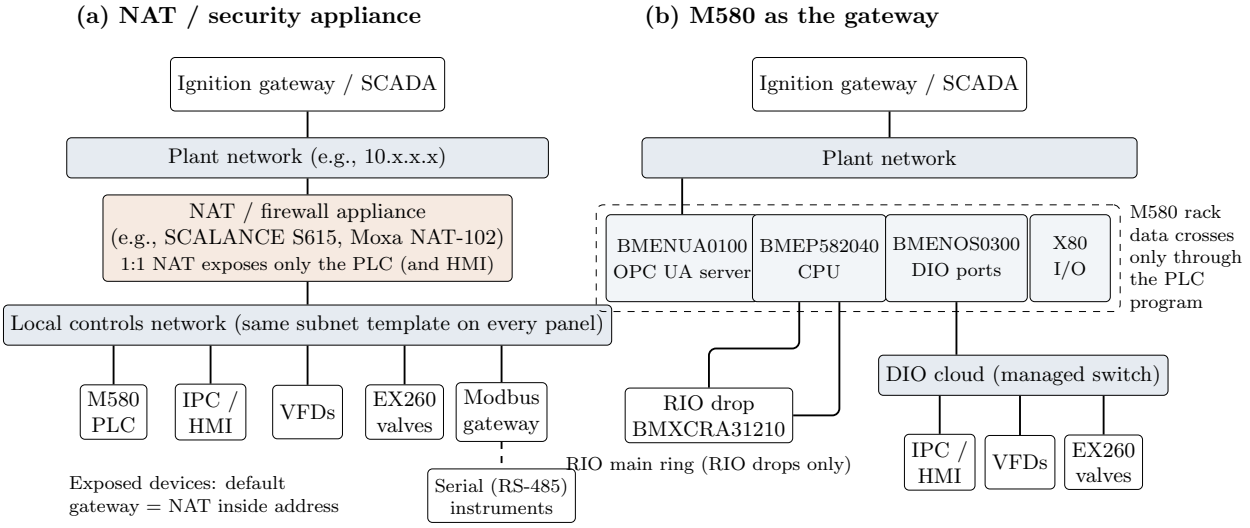


Figure 8.1: Typical panel network layouts. In (a) an appliance translates addresses so only selected devices are visible on the plant network. In (b) the M580 keeps the networks separate: SCADA reads PLC variables through the BMENUA0100 OPC UA server, remote I/O drops are on the RIO main ring, and distributed devices are on a separate DIO cloud connected through a BMENOS0300 (or a dual-ring switch).

8.1.2 Using the Modicon M580 as a Gateway

An M580 can act as the boundary between the plant network and the local controls network without a separate NAT device. The idea is simple: give the PLC separate network connections and let *only the PLC program* pass data between them.

[2]Rockwell Automation 1783-NATR user manual: all machine nodes must have their gateway IP set to the NATR private address. Rockwell/Cisco, *Deploying Network Address Translation within a Converged Plantwide Ethernet Architecture* (ENET-TD007).

1. **Local side.** Remote I/O drops go on the RIO main ring from the CPU's ports. Distributed devices (HMI, VFDs, valve banks, gateways) go on a separate DIO cloud, connected through a BMENOS0300 module, a dual-ring switch, or a BMENOC0301/11 — never directly on the RIO ring. Schneider does not permit RIO modules in a DIO cloud or DIO devices directly on the main ring.[3]

2. **Plant side (standard).** Add a BMENUA0100 OPC UA module and connect its control port to the plant network. Ignition connects to it as an OPC UA client and browses the PLC variables by name (Section 8.2). Configure the module's cybersecurity (it ships with a Secured/Standard mode rotary switch) and confirm in the module configuration that it does not forward traffic between its control port and the device network.

3. **Plant side (alternative).** Where Modbus TCP or EtherNet/IP explicit access is needed instead, add a BMENOC0301 and leave its *Ethernet backplane port disabled* (the default). Schneider's documentation states that the module is then isolated from the rest of the M580 Ethernet network, so the plant network can't see the local devices; the module still exchanges data with the CPU over the X Bus backplane. Bench test the access to CPU variables in this configuration before relying on it.

4. **Give SCADA the data it needs.** SCADA reads and writes only the PLC's own variables. Data from local devices reaches SCADA because the PLC program copies it into those variables.

Limitations. SCADA can't talk directly to a VFD or other local device this way — everything goes through the PLC. If you need to reach local devices directly (for example to configure a drive from the plant network), use a NAT appliance instead. The BMENOC0321 control-network module can route traffic between networks (IP forwarding), but that is a routed design, not an isolated one, so it needs the same security review as a NAT appliance. Confirm the exact module configuration in Control Expert for each project.

Further reading: Schneider Electric — The BMENOC0301/11 modules in Modicon M580 networks; Schneider Electric — BMENUA0100 OPC UA module.

8.2 Communication Protocol Standards

PLC to SCADA: OPC UA. Modicon M580 PLCs communicate with the Ignition SCADA server over OPC UA through the BMENUA0100 module.

- Ignition connects as an OPC UA client and browses the PLC variables by name, so no register maps are needed and each PLC data type maps directly to an Ignition UDT (Section 10.4).
- Expose variables in Control Expert under *Tools > Project Settings > General > PLC embedded data > Data dictionary*. Selecting *Only HMI variables* limits OPC UA access to variables with the HMI attribute set in the data editor; use it so SCADA only sees what it needs.
- Use certificate-based security (policy Basic256Sha256, sign and encrypt), named user accounts with the minimum permissions, and the module's Secured mode. Anonymous access is not used.

[3] Schneider Electric, *Modicon M580 Standalone System Planning Guide* (HRB62666) and *System Planning Guide for Complex Topologies* (NHA58892). CPUs ending in "40" (e.g., BMEP582040) support RIO; those ending in "20" support local and DIO only.

- The M580 CPU itself has only an OPC UA *client* (firmware 4.10 and later); the OPC UA *server* for SCADA is the BMENUA0100.

Device communications: EtherNet/IP by default, Modbus TCP as the backup. Communication between the PLC and drives, valve banks, scales, IPCs, and similar devices defaults to **EtherNet/IP**. **Modbus TCP** may be used:

- as a backup when a device does not support EtherNet/IP or its EtherNet/IP option is not available (for example the Invertek OPT-3-MTPEG-IN case in Section 6.4.1); or
- to convert an existing Modbus RTU (serial) system to Ethernet through a gateway such as the EKI-122x (Section 6.4.5).

Modbus addressing and setup are covered in Appendix F.

Exceptions.

- **Productivity2000** has no OPC UA server and Ignition has no native Productivity driver, so its SCADA link uses Modbus TCP with a documented register map.
- The two existing **ControlLogix** systems (Rare Gas column and Helium system) use Ignition's Allen-Bradley Logix driver.

8.3 Addressing

IP addresses and VLANs must be assigned through the company IP address management process. A project number must not be inserted directly into an octet unless the resulting address and subnet have been allocated. The subnet mask must create the intended boundary; a /16 network must not be treated as though different third octets are isolated. Static addresses, reserved addresses, gateways, VLAN IDs, DNS names, and ownership must be recorded in the network schedule.

Do not use the common vendor default subnets (192.168.0.x and 192.168.1.x) for local controls networks. Many devices and laptops ship on those subnets, and plugging one in causes address conflicts.

8.4 Network Equipment

Managed industrial switches must be used unless the basis of design documents why an unmanaged switch is acceptable. Required features must be selected for the application and may include VLANs, access control, port security, SNMPv3, redundant ring protocol, quality of service, multicast/IGMP management (needed where EtherNet/IP I/O traffic is present), time synchronization, diagnostics, configuration backup, and event logging. Unused switch ports and services must be disabled where supported. Unmanaged switches are never used on an M580 RIO ring.

8.5 Remote Access

Direct inbound access to PLCs, HMIs, or panel devices from the internet is prohibited. Remote access must use the company-managed secure access service, MFA, named accounts, authorization, session logging where available, and time-limited vendor access. Permanent vendor modems or

cloud tunnels require documented cybersecurity approval.[4]

For the parent company's network and remote access platforms, Cato Networks and Claroty xDome Secure Access, see Appendix M.

8.6 System Hardening and Maintenance

[Nothing is currently done to harden controls systems. A hardening plan should be discussed and adopted.]

The plan should cover:

- An asset list: what is installed, its firmware and software versions, and who owns it.
- Changing default passwords on every device.
- Turning off unused services (for example FTP, HTTP, and TFTP on PLCs) and unused switch ports.
- A practical approach to firmware and Windows updates that weighs security against uptime.

Vendor guidance: Schneider Electric, *Modicon Controllers Platform — Cyber Security Reference Manual* (EIO0000001999); Inductive Automation, Ignition Security Hardening Guide.

A NAT device is a good first step.

- **Hides the local network.** Only the devices you choose (usually the PLC, sometimes the HMI) can be reached from the plant network. VFDs, valve banks, gateways, and everything else stay invisible.
- **Protects devices that can't be hardened.** Many drives, gateways, and older devices can't be patched or locked down. Putting them behind NAT protects them without touching each one — a "compensating control" in ISA/IEC 62443 terms.
- **One place to control access.** With a firewall-capable appliance (Table 6.13), allow-only rules (for example, SCADA server to PLC on the OPC UA port only) and logging live in one box instead of being set on every device.
- **Keeps panels standard.** Every panel can reuse the same local subnet, so panel designs, spares, and vendor setups stay the same from project to project.

8.7 Time, Logging, and Recovery

Controllers, SCADA, workstations, network devices, and security systems should use an approved time source and a consistent time zone. Security and operational events must be retained according to company policy. The design must document recovery priorities, replacement configurations, backups, and the test interval for restoring critical OT assets.

[Future: set up a plant time server so all controls equipment is synchronized to one source. This will be investigated with IT.]

[4] CISA, NCSC-UK, FBI, et al., *Secure Connectivity Principles for Operational Technology (OT)*, January 2026; CISA et al., *Guide to Securing Remote Access Software*, June 2023.

- **Why it matters:** alarm and event timestamps, historian data, and the first-out records of trips (Chapter 10) can only be compared across systems if every device uses the same clock.
- **Likely approach:** IT provides an NTP server on the plant network, or the controls network gets its own (for example a GPS-referenced appliance). PLCs, Ignition servers, IPCs, and managed switches sync to it as NTP clients. The M580 CPU can be an NTP client (a BMENOC03x1 can also update the CPU time), and Productivity2000 CPUs sync over SNTP using the Network Time Service instruction.
- **Questions for IT:** which server to use, whether controls networks behind NAT can reach it, and which time zone or UTC convention to standardize on.

9. Accounts and Access Control

9.1 Identity Requirements

Named individual accounts should be used for normal engineering, administration, and operator access. Shared administrator accounts are limited to controlled emergency ("break-glass") use or platform constraints. Roles should follow least privilege and separate operator, maintenance, engineering, security administration, and system administration duties where supported.

[Individual accounts are not currently implemented. See Section 11.7 for the Ignition login discussion.]

[It would be good to adopt policies covering:

- *Logins and passwords: multi-factor authentication for remote access, storing passwords in a password manager (never in code, drawings, or documents), and changing default passwords.*
- *Shared and emergency accounts: service accounts and "break-glass" emergency logins, who owns them, and how they are stored.*
- *Account lifecycle: who approves access, periodic reviews, removing access when people leave, and expiration dates for vendor accounts.*
- *Handing over credentials at turnover, so EFC is not left depending on a contractor's login.*

]

10. PLC Standards

10.1 Program Architecture

Programs must be organized by equipment or functional area with a documented execution order. Each project must provide a main routine, input processing, equipment modules, sequences, alarm/protective logic, communications, output processing, diagnostics, and HMI interface. Safety-related logic must reside in the approved safety architecture and must be clearly separated from ordinary control logic.

IEC 61131-3 languages may be used according to purpose: ladder diagram for maintainable discrete/equipment logic, function block diagram for continuous control and reusable blocks, and structured text for calculations, data handling, and bounded iteration. Dense or obscure code must be avoided regardless of language.

10.2 Naming and Data Quality

Names must be descriptive, consistent, and traceable to P&IDs and electrical drawings. Tags exposed outside the controller must include engineering units, range, description, quality/validity, and source where supported. PLC booleans must use `BOOL`; real process values must use `REAL` unless precision requirements justify another type. Retentive data must be explicitly identified and minimized.

10.3 Operating Modes and Commands

Equipment modules must distinguish Off, Manual, Auto, Maintenance/Lockout, and Simulation where applicable. HMI requests must not drive physical outputs directly; logic must validate mode, permissives, interlocks, ownership, and command timing. Manual mode must not bypass protective functions. On startup, mode and command restoration must follow the risk assessment and must not create an unexpected start.

10.3.1 Setpoint Limits

Operator-entered values are checked in the PLC, not only in SCADA. For permissive setpoints (Section 5.6) the minimum and maximum are hard-coded in the PLC program; the PLC clamps or rejects any value outside them and reports the active limits to SCADA for display. The limits are changed only through a software revision, under MOC where Section 7.3 applies.

10.4 Data Types (UDTs)

A user-defined type (UDT) is a reusable data structure that bundles everything about one piece of equipment or one signal — commands, status, setpoints, alarms, and diagnostics — into a single variable. Modicon Control Expert calls these **DDTs** (Derived Data Types); UDT is the general term used by Ignition and most other platforms.

Why we use them:

- Every pump, valve, or analog input of the same kind has the same members with the same names, on every project.
- They pair with reusable function blocks (Section 10.8): the block holds the logic, the DDT holds the data.
- Programming, SCADA screens, and troubleshooting are faster because the structure is familiar.

Rules of thumb:

- Name types with the `DDT_` prefix (for example `DDT_Valve`) and use one consistent capitalization style for member names.
- Use the standard type library shared across projects; do not create project-specific copies of a standard type.
- Changing a type changes every instance of it. Treat changes to standard types like any other controlled change (Section 7.2), and test them (Section 7.6).
- Analog scaling must handle divide-by-zero, invalid configuration, underrange, overrange, and bad quality. Alarm comparison must use the scaled value only when quality is acceptable, unless a separate instrument-fault alarm is intended.
- Equipment types keep request, command, status, permissive, interlock, feedback, fault, warning, and runtime data as separate members.

The current data types, with their members, are listed in Appendix H.

10.4.1 Using UDTs Between the PLC and SCADA

- **One type on each side.** Each PLC data type has a matching Ignition UDT with the same member names and structure. A PLC instance (for example `P101` of type `DDT_Motor_VFD`) is brought into Ignition as one UDT instance, so faceplates and templates are built once per type and reused for every instance.
- **OPC UA.** With the M580 standard (Section 8.2), Ignition browses the PLC variables and their DDT structure directly through the BMENUA0100. Only variables marked with the HMI attribute in Control Expert are exposed, so mark the instances SCADA needs and leave internal working variables unmarked.
- **Read versus write.** SCADA writes only to request/command members (for example `HMI_Auto`, `HMI_CMD`) and operator setpoints. Status, interlock, and feedback members are read-only from SCADA. This matches the rule that HMI requests never drive outputs directly (Section 10.3).

- **Keep both sides in sync.** When a PLC type changes, the matching Ignition UDT is updated in the same revision and both are tested together.
- **Productivity2000.** Communication uses Modbus TCP, so each structure is mapped to consecutive registers and a register map is required (Appendix F).

10.5 Interlocks and First-Out

Interlock logic must use a structured condition type rather than parallel unstructured arrays. Each condition must include enable, raw condition, active, latched, bypassed, first-out, message/identifier, and timestamp where supported. The summary must include any-active, trip-active, reset-permitted, bypass-present, and first-out index. The number of conditions must be sized by the reusable library and not silently truncated.

Interlocks must fail to the defined state on bad or stale inputs when required by the risk assessment. Reset must not clear an active initiating condition. Bypasses must survive or clear on restart only as defined by the approved bypass policy.

10.6 Sequences

Sequences must use an explicit state model with named steps, entry actions, continuous actions, exit conditions, timeout, hold, abort, restart, and recovery behavior. The sequence structure must include active state, previous state, transition reason, elapsed time, timeout, permissive summary, faulted/held/complete status, and operator requests. Restart after download or power loss must be deterministic and risk assessed.

10.7 Communications and Diagnostics

Communications must include heartbeat or sequence checking where suitable, timeout, data-valid status, stale-data handling, and documented fail behavior. Raw received data must be separated from validated values used by control. Diagnostic status must be available to SCADA without flooding operators with non-actionable alarms.

10.8 Reusable Function Blocks

10.8.1 What a Function Block Is

A reusable function block is a block of PLC logic written once and used for every pump, valve, analog input, or sequence of the same kind, so each one behaves the same way. In Control Expert these are **DFBs** (Derived Function Blocks). The current DFBs are documented in Appendix I.

10.8.2 The Standard Object: DFB, DDT, and Ignition UDT

Each standard equipment type is a matched set of three parts:

- a **DFB** that holds the logic (for example `DFB_Valve`);
- a **DDT** that holds the data (for example `DDT_Valve`, Section 10.4); and
- an **Ignition UDT** with its faceplate and templates that present it on SCADA.

The three are designed, versioned, and changed together. A change to one is checked against the other two.

10.8.3 When to Make One

Make a DFB for logic that is used more than once or that represents a standard equipment type. One-off logic stays in the project program, where it is easier to read in place.

10.8.4 Design Rules

- Pass the equipment's DDT to the block as an input/output parameter, so the block reads and writes one structure.
- No direct I/O addresses or global variables inside the block; everything comes through its pins.
- Choose fail-safe default values for every input and parameter.
- Start each block with a header comment giving its purpose, revision, author, and date.
- Use the standard Off / Manual / Auto / Lockout mode pattern (Section 10.3).
- Keep protective logic visible: interlocks and E-stop inputs come in through pins (for example `Interlock` and `Estop` on `DFB_Valve`) so they can be seen and traced at the call, not buried inside the block.

10.8.5 Library Management

Standard DFBs are kept in a master library (the Control Expert type library). Projects use copies; project-specific changes never overwrite the master. A DFB's revision follows the revision approach in Section 7.2. Changing a DFB that contains interlock or protective logic requires an MOC (Section 7.3) and a Change Test Record (Appendix K).

[Formal library versioning is not current practice. Git (Appendix J) is the planned way to track library changes in the future.]

10.8.6 Testing

New or changed blocks are tested in the simulator before they are used, covering every mode, the interlock and E-stop inputs, and bad-quality inputs. The test is kept on file with the library.

10.8.7 Productivity2000

Productivity Suite does not use DFBs. Reusable logic is built with User Defined Instructions (UDIs, Productivity Suite v4.5 and later), which can be stored in a UDI library (v4.7 and later). The same rules apply: one UDI per equipment type, used with a matching data structure.

10.9 Common Systems

Common systems should use a consistent control structure, naming convention, HMI faceplate, alarm approach, and test procedure. Project-specific limits and options may vary, but the basic operator experience and PLC interface should remain recognizable from project to project.

10.9.1 PDC Compressor

The PDC compressor module should organize the package interface into operating mode, start/stop requests, permissives, active interlocks, running status, load state, process values, warnings, trips, first-out indication, and maintenance counters.

The control narrative and code should address, as applicable, suction and discharge pressure, stage temperatures, cooling-water flow and temperature, hydraulic or oil-system status, diaphragm or process-gas leak detection, purge/vent status, valve position, motor feedback, vendor-panel communications, minimum stop time, and restart following a trip or power loss. Setpoints and shutdown actions must come from approved PDC documentation, the process design, and the project hazard review rather than from generic defaults.

11. SCADA and HMI Standards

11.1 Ignition Architecture

Ignition is the preferred SCADA/HMI platform for new projects. The differences between a standard Ignition gateway and Ignition Edge Panel, and how the two are integrated, are covered in Appendix B. The project team should choose the Gateway, database, client, and redundancy arrangement based on system size, availability needs, performance, and recovery time. Operator workstations should normally run Perspective or Vision clients rather than host the production Gateway or database.

Record the tag providers, projects, databases, historian, alarm pipelines, identity provider, certificates, modules, backups, and important external dependencies. Use a development or test environment when practical instead of making untested changes directly in production.

11.2 Project and Tag Design

Use shared projects, inheritance, UDTs, templates, and common scripts to avoid duplicating the same work across projects. Tag names and folders should follow the PLC equipment model and use consistent units, ranges, descriptions, quality handling, alarms, and history settings.

Keep scripts readable and reusable. Log errors that will help with troubleshooting, but avoid repetitive logging that hides useful information. Where a tag, expression, binding, or shared function can do the job cleanly, prefer it over copying scripts into many components.

11.3 HMI Philosophy

Screens should make the current process condition easy to understand. Use color mainly to show abnormal conditions or items requiring attention. Do not rely on color alone; use text, symbols, and state indication as well. Navigation, faceplates, trends, units, decimal precision, manual controls, and alarm presentation should work the same way throughout the project.

Use a clear display hierarchy, normally including overview, area, unit, equipment-detail, and diagnostic screens. Operators should be able to see the active mode, command ownership, equipment state, alarms, interlocks, and bypasses before issuing a command. Use confirmation prompts for consequential actions without adding unnecessary prompts to routine work.

11.4 Perspective Design

Perspective screens follow the Perspective Screen Design Standard in Appendix A. Perspective views should be tested at the screen sizes and aspect ratios expected in service. Use responsive

containers where resizing is required. Docked views, popups, embedded views, parameters, and message handlers should have clear interfaces and should not depend on fragile component paths.

Startup and navigation scripts should be safe to run more than once, finish quickly, and handle missing tags or bad-quality data without breaking the view. Reusable views should not assume that every optional parameter is present.

11.5 Alarms

Configure Ignition alarms from the approved alarm list. Include the identifier, priority, message, cause, consequence, expected response, deadband, delay, shelving permissions, and any state-based suppression. Acknowledging an alarm does not clear the process condition that caused it.

Test alarm journals and notification pipelines, including loss and recovery of the database or notification service. Review nuisance, chattering, standing, and frequently occurring alarms after startup.

11.6 Historian and Data

Historize data that has a defined operating, quality, maintenance, regulatory, or analysis purpose. Choose the sampling mode, deadband, timestamp source, and retention period to match that purpose. Document store-and-forward behavior and how bad or uncertain quality is handled.

Database connections should use managed credentials and only the permissions they need. Use encrypted connections when supported by the approved architecture.

11.7 Security and Audit

Individual accounts

Every operator, engineer, and other user should have their own Ignition account and be logged in before using the system. Individual logins mean that actions such as setpoint changes, alarm acknowledgments, and bypasses are recorded against a person, and that each user gets only the access their role needs.

[Not currently implemented. This should be discussed in the near future, including how logins work on shared operator stations — for example automatic logout after inactivity and a fast way to switch users at the station.]

[Future: Ignition Perspective can use single sign-on (SSO) through an identity provider using OpenID Connect 1.0 or SAML 2.0, such as Microsoft Entra ID (Azure AD) or Okta. Users would log in with their existing company account rather than a separate Ignition password, and accounts would be added or removed centrally when people join or leave. This works for Perspective, Vision, the Designer, and the Gateway in both 8.1 and 8.3, and is worth looking into, possibly with the parent company's IT group.]

Roles and audit

Use the approved identity provider when practical and assign access by role. Typical roles include operator, maintenance, engineering, supervisor, and administrator. Audit important actions such

as setpoint changes, alarm shelving, recipe changes, and security administration.

OPC UA connections should use trusted certificates, a supported security policy (Basic256Sha256, sign and encrypt), named accounts with the minimum permissions needed, and no anonymous access (Section 8.2). Remove default accounts and unused connections before turnover.

11.8 Backup, Deployment, and Recovery

Create a Gateway backup before and after significant production changes and after commissioning. Record the Ignition version, installed modules, licenses, database dependencies, certificates, identity-provider settings, DNS/NTP needs, and restore steps.

For critical systems, demonstrate that the backup can be restored and that the project can be rolled back if a deployment fails. Keep the accepted backup with the project turnover package.

12. Reference Material

12.1 Standards Referenced

The project basis of design must identify the editions adopted by the jurisdiction and the editions contractually required. The table below lists each standard cited in this guide once, with the edition cited and where it is used. When a new edition is published, this is the one place to check what needs updating. Applicability must be determined per project.

Table 12.1: Standards Referenced

Standard (edition cited)	Subject and where used in this guide
Electrical and machinery	
NFPA 70 (NEC), 2023	National Electrical Code. Art. 250 grounding and bonding (Section 6.3.5); 409.22 SCCR (Section 6.3.2); 450.3(B) transformer protection (Appendix C.3); Art. 501 Class I locations (Appendix E). The 2026 edition is published; confirm the edition adopted by the jurisdiction.
NFPA 70E, company-adopted edition	Electrical safe work practices and risk controls.
NFPA 79, 2024	Electrical equipment of industrial machinery: stop categories and E-stop (Section 6.3.3), technical documentation (Appendix D), testing and verification (Appendix L, Section 7.6).
UL 508A, 3rd ed. (2018), revised through 2025	Industrial control panels: construction, markings, and SCCR (Supplement SB).
IEC 60204-1:2016+A1:2021	Electrical equipment of machines: stop categories, push-button and indicator colors (Section 6.3.3), verification and retesting.
ISO 13850:2015	Emergency stop function (Section 6.3.3).
ISO 13849-1:2023	Safety-related parts of machinery control systems.
IEC 62061:2021	Functional safety of machinery safety-related control systems.
Hazardous locations	
NFPA 496, 2024	Purged and pressurized enclosures (Appendix E).
UL 121201, 9th ed.	Nonincendive equipment for Class I and II, Division 2 locations (Appendix E).
UL 698A, 4th ed. (2018, reaffirmed 2022)	Industrial control panels relating to hazardous locations (intrinsically safe circuits).

Table 12.1 continued

Standard (edition cited)	Subject and where used in this guide
Documentation	
IEC 61082-1:2014	Preparation of documents used in electrotechnology (Appendix D).
IEC 81346-1:2022 and 81346-2:2019	Reference designations (Appendix D).
IEEE 315-1975 (inactive-reserved) / IEC 60617	Graphic symbols for electrical diagrams.
ISA-5.1, adopted edition	Instrumentation symbols and identification.
Process safety, alarms, and testing	
OSHA 29 CFR 1910.119	Process safety management: (i) pre-startup safety review, (l) management of change (Sections 7.4 and 7.6).
IEC 61511-1:2016+A1:2017 / ANSI/ISA-84	Functional safety for safety instrumented systems; bypass and modification rules.
ANSI/ISA-18.2-2016 and IEC 62682:2022	Alarm management lifecycle (Section 5.7 and Appendix G).
ANSI/ISA-101.01-2015	HMI philosophy, design, implementation, operation, and maintenance.
IEC 62381:2024 (Ed. 3)	Factory, site, and site integration acceptance tests (Appendix L).
ISA-88 series, where applicable	Batch-control models, terminology, and modular structure.
Cybersecurity	
ISA/IEC 62443-2-1:2024, 62443-3-2:2020, 62443-3-3:2013	Security program (configuration management, logging), zones and conduits, and system security requirements (Chapter 8, Section 7.4).
NIST SP 800-82 Rev. 3 (2023)	OT cybersecurity guidance. A later revision must not replace Rev. 3 in project requirements until issued as final and adopted by EFC.
CISA et al., *Secure Connectivity Principles for OT* (2026)	Remote and external connectivity to OT (Section 8.5).

Manufacturer manuals, certificates, and application notes must be retained for the actual product revision used. A web link is informational and does not replace a controlled project copy of required technical data.

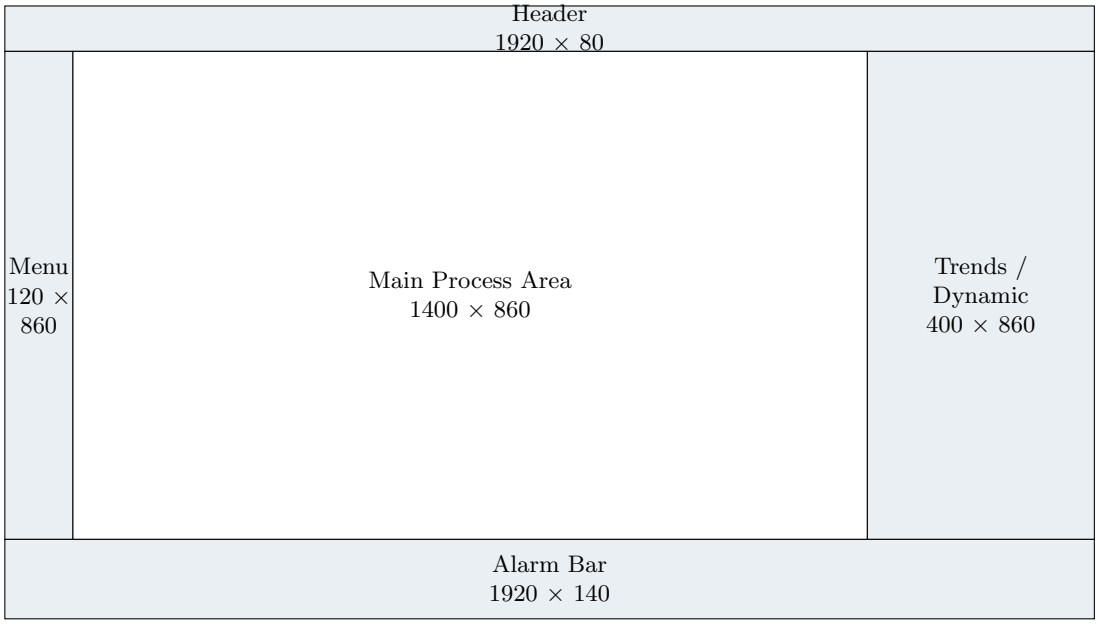
A. Perspective Screen Design Standard

Applies to: Ignition Perspective 8.1 (16:9 scalable layout and text scaling, final decisions). Target displays: 1920 × 1080 through 3840 × 2160. These techniques carry forward unchanged to the planned Ignition 8.3 upgrade.

A.1 Approach

All operator screens are built inside a single Shell view designed at 1920 × 1080. The shell's root is a Coordinate container in Percent mode with the aspect ratio locked to 16:9, so the entire interface scales proportionally and letterboxes on non-16:9 displays. Each screen region (header, menu, main area, trends, alarm bar) is an Embedded View placed in the shell. Perspective docks are not used for these regions, because docks are sized in fixed pixels and would not stay aligned with the scaled content.

A.2 Shell Layout



Region	Design size (px)	x	y	width	height
Header	1920 × 80	0	0	100%	7.41%
Menu	120 × 860	0	7.41%	6.25%	79.63%
Main process area	1400 × 860	6.25%	7.41%	72.92%	79.63%
Trends / dynamic window	400 × 860	79.17%	7.41%	20.83%	79.63%
Alarm bar	1920 × 140	0	87.04%	100%	12.96%

Percent values are the position properties to enter on each Embedded View in the shell's percent-mode root. The trends/dynamic window is 400 design px wide, which renders at 800 px on a 3840 × 2160 display (its largest size) and scales down proportionally on smaller displays. If a collapsible-trends variant of the shell is added later, its main area becomes 1800 × 860 and should use layouts designed for that size.

A.3 Container Settings

View	defaultSize	Root container	aspectRatio
Shell	1920 × 1080	Coordinate, mode = percent	16:9
Region views (header, menu, trends, alarm bar)	Equal to the region's slot size	Coordinate, mode = percent	Blank
Process screens (main area)	1400 × 860	Coordinate, mode = percent	Blank
Templates	The size they occupy on the screen	Coordinate, mode = percent	Blank

Only the shell locks the aspect ratio. Because the shell fixes the overall proportions, every slot keeps a constant shape; setting an aspect ratio on inner views would cause double letterboxing and gaps.

A.4 Navigation and Dynamic Content

Main area navigation. One page route, `/screen/:name`, points to the Shell as its primary view. The Shell has a `name` view parameter, and the main-area Embedded View's path uses the expression binding below. Menu buttons navigate to routes such as `/screen/Boiler`, so browser back, bookmarks, and direct links all work while the header, menu, and alarm bar remain in place.

```
"Screens/" + {view.params.name}
```

Trends / dynamic window. The right-hand Embedded View's path and params are bound to session custom properties (for example `session.custom.trend.view` and `session.custom.trend.params`). Clicking equipment on a process screen writes to these properties to change the panel's content. A message handler on the shell (`system.perspective.sendMessage`) is an acceptable alternative.

Menu. At 120 design px wide (about 80 px on a 1280 × 720 display), the menu uses icons with a short one-word label beneath each.

Alarm bar. At 140 design px tall, the alarm bar shows roughly four alarm rows plus a header row at about 28 design px per row.

A.5 Text Scaling Standard

All text, padding, borders, border radius, and icon sizes are sized in design pixels (dp): one pixel on a 1920 × 1080 display, scaled to the current screen. The unit is defined once as a CSS variable in a custom theme's variables file, or in the Advanced Stylesheet if the installed 8.1 build includes it:

```
:root { --dp:  min(0.052083vw, 0.092593vh); }
```

Any size is then written as `calc(N * var(--dp))`, where N is the size in pixels on a 1080p display. Because the unit is derived from the viewport rather than from the enclosing container, it gives the same result at any nesting depth: text in a template inside an embedded view inside the shell scales at exactly the same rate as the shell itself. Where the variable is unavailable, the equivalent direct form is `min(N/19.2 vw, N/10.8 vh)`.

Text sizes are applied through Style Classes rather than set on individual components:

Style class (suggested)	1080p size	fontSize using –dp	Direct equivalent
Text/Small	12	`calc(12 * var(--dp))`	`min(0.625vw, 1.111vh)`
Text/Label	14	`calc(14 * var(--dp))`	`min(0.729vw, 1.296vh)`
Text/Body	16	`calc(16 * var(--dp))`	`min(0.833vw, 1.481vh)`
Text/Header	20	`calc(20 * var(--dp))`	`min(1.042vw, 1.852vh)`
Text/Title	28	`calc(28 * var(--dp))`	`min(1.458vw, 2.593vh)`

Line height is set as a unitless value (for example 1.2) so it follows the font size. Fixed px values are not permitted in any font size, padding, or border width on operator screens.

A.6 Resolution Behavior

Element	1920 × 1080	3840 × 2160
Shell	1920 × 1080	3840 × 2160
Main process area	1400 × 860	2800 × 1720
Trends / dynamic window	400 × 860	800 × 1720
A 200 × 120 template	200 × 120	400 × 240
Text at `calc(14 * var(--dp))`	14 px	28 px
Text at a fixed 14 px (not permitted)	14 px	14 px (half the relative size)

4K monitors running Windows display scaling (150% or 200%) report a smaller CSS viewport to the browser. The dp unit handles this automatically; no per-scaling configuration is required.

A.7 Templates

Each template is designed at the size it will occupy on the screen, with a percent-mode Coordinate root and no aspect ratio. All template text uses the dp style classes. Templates are placed at their designed size so text stays proportional to the graphic, which also supports consistent symbol sizing across screens.

When the same template must appear at a different size, it takes a `scale` parameter (default 1) and its font sizes are bound with an expression such as:

```
"calc(" + (14 * {view.params.scale}) + " * var(--dp))"
```

A.8 Components With Internal Pixel Sizing

Several components set fonts, row heights, or padding internally in pixels and must be overridden explicitly with dp values: the Table and Alarm Status Table (header style, row style, and row height), the Power Chart and Time Series Chart (axis and legend fonts), and the Menu Tree, Dropdown, and Tab Container (font and padding). Icons are sized in dp so they stay proportional to adjacent labels. Images use `fit.mode = contain`, and SVG graphics are preferred over raster images.

A.9 Known Limitations and Testing

The Designer evaluates vw and vh against the whole workspace panel, so text sizing appears incorrect there and must be judged in a browser session. Readability at the smallest display is the limiting factor; the smallest routinely used design size should be 14 dp or larger, and any floor applied with `max(10px, ...)` must be checked for text overflow.

Before rollout, a test screen containing a template at two sizes, a table, and a chart is opened in Chrome with the device toolbar set to 1920 × 1080 and 3840 × 2160 (plus a non-16:9 size such as 1920 × 1200 to confirm letterboxing). Text should occupy the same proportion of its box at every size; any element that does not still has a fixed pixel value or an unset component-internal size.

B. Ignition Perspective vs. Edge Panel: Differences and Integration (8.1 / 8.3)

B.1 Overview

Perspective and Edge Panel are not competing products: Perspective is a visualization module, and Edge Panel is a license for a lightweight Edge gateway that can run Perspective (or Vision) with only two clients. The real comparison is a Standard Ignition gateway running Perspective versus an Edge gateway with the Panel product. The two integrate over the Gateway Network, mainly through a remote tag provider on the central server and Edge Sync Services for history, alarms, and audit data.

- **Ignition Standard + Perspective:** the full platform. Unlimited tags, sessions, and database connections, multiple projects and tag providers. This is the central SCADA server.
- **Ignition Edge Panel:** a local HMI at a machine or skid. It includes everything in Edge IIoT plus local visualization, limited to one project, no databases, and short local history. It keeps running if the link to the central server drops.

Mixed versions work with limits: an 8.3 main server can take tags, history, and alarms from an 8.1 Edge Panel, but an 8.3 Edge cannot store data to an 8.1 main server. Details are in the version sections below.

B.2 Key Differences

The biggest practical limits of Edge Panel are two clients, one project, one tag provider, and no database connections; everything else is close to the standard platform.

Table B.1: Standard Ignition vs. Edge Panel

Area	Standard Ignition + Perspective	Edge Panel (running Perspective)
Clients	Unlimited sessions (per license)	2 Perspective sessions, or 2 Vision clients (1 local, 1 remote); one module at a time
Projects	Unlimited, with inheritance	Exactly one project (default name "Edge"), cannot be deleted
Tags	Unlimited, multiple tag providers, remote providers	Unlimited tags, a single realtime tag provider

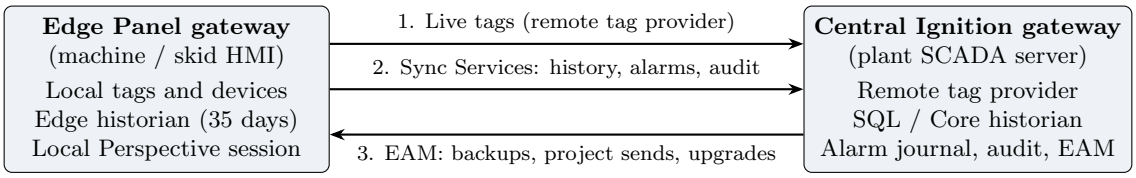
Table B.1 continued

Area	Standard Ignition + Perspective	Edge Panel (running Perspective)
Databases	Unlimited SQL connections, named queries, DB user sources	None; DB queries and bindings do not work
Tag history	SQL Historian; 8.3 adds the Core Historian	Built-in Edge historian: 35 days or 10 million points, not configurable
Alarm journal / audit	Database-backed, long retention	7 days local each
Long-term storage	Native	Via Sync Services to a central gateway
Redundancy	Standard redundant pairs	Edge-to-Edge only; cannot fail over to a standard gateway
Scripting, WebDev, EAM agent, OPC UA server	Yes	Yes (bundled into Panel and IIoT since 8.1.33)
Third-party modules	Yes	Only if the vendor states Edge compatibility
Graphics	Perspective components, Symbol Factory add-on	Perspective or Vision, Symbol Factory included

Views built on a standard Perspective project can be exported and imported into the Edge project and vice versa, as long as they avoid database features. Sources: Ignition Edge (8.3), Edge Panel (8.3).

B.3 Integration Options

The standard pattern is a Gateway Network connection from the Edge Panel to the central server, then a remote tag provider for live data and Sync Services for history, alarms, and audit records.



The Edge Panel keeps the local HMI running if the network drops; the central gateway gives plant-wide views and long-term storage. All three links run over the Gateway Network.

Table B.2: Edge Panel Integration Methods

Method	What it does	Needs on central	Notes
Remote tag provider (Gateway Network)	Central Perspective views read and write Edge tags live	Nothing extra	Main integration path; write/edit access set in Edge security

Table B.2 continued

Method	What it does	Needs on central	Notes
Edge Sync Services	Pushes Edge tag history, alarm journal, and audit log to central storage	Historian, alarm journal, and audit profiles that allow remote storage	Edge buffers up to 35 days if the link is down
EAM (agent on Edge)	Central backups, project sends, upgrades, license management	Enterprise Administration module	Edge acts as an agent
Gateway messaging	Scripts call `system.util.sendMessage` / `sendRequest` between gateways	Message handlers in a central project	Good for events and recipes
Remote alarm notification	Edge alarms trigger central notification pipelines	Alarm Notification module	Uses the Remote Gateway Notification profile
MQTT (Sparkplug B)	Edge publishes tags to a broker; central subscribes	MQTT Engine module and a broker	Avoids Gateway Network version coupling
OPC UA	Central connects to the Edge's OPC UA server	OPC UA client connection	8.3 blocks anonymous writes by default

Shared screens: export views from the central Perspective project and import them into the single Edge project, or the reverse. Point bindings at the right tag provider on each side, and strip any database-backed features before importing to Edge.

B.4 Step-by-Step Setup

The procedure is the same on 8.1 and 8.3; mainly the gateway menu locations and security defaults change. 8.1 paths are approximate from the 8.1 manual layout, so confirm them on the gateway.

1. **Set Edge Panel to Perspective.** 8.1: Config > Gateway Settings > Edge Settings > Visualization Module. 8.3: Platform > System > Ignition Edge > Edge Settings tab. Switching faults the other module; restart the Designer after.

2. **Create the Gateway Network connection.** On the Edge, add an outgoing connection to the central server's address, port 8060 (SSL). Approve the certificate on the outgoing side, then approve the connection and certificate under Incoming Connections on the central server.

   - 8.3 defaults **Require Two-Way Authentication** to true, so both ends must trust each other.

   - Mixed 8.1/8.3: **Allow Java Serialization** must be enabled on every 8.3 gateway.

3. **Allow Edge storage on the central server.** 8.1: Config > Security > Service Security, set Query and Storage for History, Alarm Journal, and Audit on the Edge's zone. 8.3: Platform > Security > Zones > Manage Policy > Query and Storage for the same three sections.

4. **Configure Sync Services on the Edge.** 8.3: Platform > System > Ignition Edge > Sync Services. Pick the central gateway, then the remote history provider, alarm journal, and audit

profile, and enable each. Set batch size, frequency, and an optional schedule.

5. **Create the remote tag provider on the central server.** 8.3: Services > Tags > Create Tag Provider > Remote Tag Provider (Gateway Network). 8.1: Config > Tags > Realtime > Create new Realtime Tag Provider > Remote Tag Provider. Choose the Edge gateway and its provider.

6. **Grant write access on the Edge.** In the Edge's security zone policy (Service Security in 8.1, Zones > Manage Policy in 8.3), set the Edge tag provider's access level to ReadWrite, or ReadWriteEdit to edit tags from a remote Designer.

7. **Build the central views.** Bind Perspective components on the central server to the Edge tag provider, for example `[EdgeProvider]Area/Tag`. Query synced history through the central historian, not the Edge.

8. **Optional: EAM.** Install Enterprise Administration on the central server, make it the controller, and register the Edge as an agent for scheduled backups.

9. **Test failure modes.** Pull the network link and confirm the Edge HMI keeps running, the central server shows bad quality, and synced history back-fills when the link returns.

Source: Configuring Sync Services (8.3).

B.5 8.1 vs. 8.3 Differences

Edge Panel's limits are the same in both versions; the differences are in gateway security, serialization, historians, and menu layout.

Table B.3: Ignition 8.1 vs. 8.3 Differences Relevant to Edge Panel

Area	8.1	8.3
Edge Panel limits	2 clients, 1 project, 35-day history (since 8.1.34; 7 days before)	Same
Sync, EAM, Compute	Bundled into Panel and IIoT since 8.1.33	Bundled
Gateway Network serialization	Java serialization	Protobuf; Java serialization only allowed for 8.1 peers
Two-way authentication	Off by default	On by default; both ends approve certificates
Security settings	Service Security page	Moved into Security Zones > Manage Policy
Gateway web pages	Config / Status menus	Platform / Services / Connections layout; role-based access, 8.1 permissions not carried over
Central historian	SQL Tag Historian	SQL Historian plus new Core Historian (QuestDB); qualified path uses `sys:` and `prov:` instead of `drv:` for Core

Table B.3 continued

Area	8.1	8.3
History scripting	`system.tag.queryTagHistory` etc.	`system.historian.*` preferred; old functions still work
OPC UA	Anonymous clients can write by default	Anonymous clients read-only by default; new certificates on upgrade
Modules	Hot swap allowed	Restart required to install or upgrade a module
Launchers / Workstation	8.1 launchers	8.3 launchers work with 8.1 gateways, not the reverse
Perspective	—	Adds Drawing and Form components and Offline Mode; File Upload capped at 20 MB by default
Store and Forward	Max Records setting	Multi-threaded; renamed setting defaults to unlimited

If the Edge uses Perspective's newer 8.3 components, those views cannot be moved back to an 8.1 gateway. Source: 8.1 to 8.3 Upgrade Guide.

B.6 Mixed Versions: Can They Be Integrated?

Yes, with one direction that works and one that loses data: an 8.3 central server with 8.1 Edge gateways is supported, but an 8.1 central server cannot store data synced from an 8.3 Edge. Inductive Automation's rule is to upgrade the central gateway that hosts history, alarm journal, and audit first, then the Edge and remote gateways.

Table B.4: Mixed-Version Integration

Feature	Central 8.3, Edge 8.1	Central 8.1, Edge 8.3
Gateway Network link	Works; enable Allow Java Serialization on the 8.3 side	Works; enable Allow Java Serialization on the 8.3 side
Live tags (remote tag provider)	Read and write work	Read and write work
Editing remote tag configs from the other version's Designer	Not supported	Not supported
Sync Services (history, alarms, audit)	Works	Fails; data sits in the Edge buffer (35 days) then is lost
Remote history / alarm journal / audit	Works	Fails, same reason
EAM	8.3 controller manages 8.1 agents; some resources (Gateway and Perspective event scripts) cannot be sent to 8.1	Not recommended; upgrade the controller first

Table B.4 continued

Feature	Central 8.3, Edge 8.1	Central 8.1, Edge 8.3
MQTT Sparkplug B	Works; each gateway needs Cirrus Link modules matched to its own version	Works, same condition
7.9 gateways	Cannot connect to any 8.3 gateway, even through a proxy	Same

Practical notes for a mixed period:

- Put the latest 8.1 patch on every 8.1 gateway before connecting it to 8.3; Inductive Automation strongly advises this before upgrading.
- An 8.3 gateway upgraded from 8.1 turns on Allow Java Serialization automatically; a fresh 8.3 install does not, so set it by hand.
- Plan to turn Allow Java Serialization off once every gateway is on 8.3, since 8.3-to-8.3 traffic uses Protobuf.
- Treat mixed versions as a transition state, not a long-term design; tag editing and project pushes across versions are limited.

For the planned 8.1 to 8.3 upgrade of the plant SCADA server, the order works in our favor: upgrade the central Perspective server first, confirm the Gateway Network shows the Edge as connected and sync is flowing, then upgrade the Edge Panel gateways.

Sources: 8.1 to 8.3 Upgrade Guide; 8.3 SDK Upgrade Guide.

B.7 Checklist

Use this list when commissioning an Edge Panel against the central server or moving through the upgrade.

- ☐ Gateway backups (`.gwbk`) of central and every Edge before changes
- ☐ Licenses confirmed 8.3-ready with the sales rep before upgrading
- ☐ Third-party modules confirmed compatible with 8.3, and with Edge where used
- ☐ Central server upgraded to 8.3 before any Edge gateway
- ☐ Allow Java Serialization on for every 8.3 gateway while any 8.1 gateway remains
- ☐ Certificates approved on both ends (two-way authentication)
- ☐ Security zone on central allows Query and Storage for history, alarm journal, and audit
- ☐ Sync Services enabled for all three data types and back-fill verified after a link outage
- ☐ Remote tag provider created; Edge write access set deliberately (ReadWrite only where operators need it)
- ☐ OPC UA write permissions re-mapped after 8.3 upgrade if any client connects anonymously

- ☐ Edge HMI tested with the network unplugged
- ☐ Allow Java Serialization turned off after the last gateway reaches 8.3

Sources

- Ignition Edge, 8.3 manual
- Edge Panel, 8.3 manual
- Configuring Sync Services, 8.3 manual
- 8.1 to 8.3 Upgrade Guide
- 8.1 to 8.3 SDK Upgrade Guide
- Edge Architectures, 8.1 manual

C. Important Equations and Calculations

This appendix collects the main design calculations referenced in Chapter 6. Each one is written as a short, step-by-step method with a worked example, and every equation links to a source that explains it in more detail.

C.1 Short-Circuit Current Rating (SCCR)

Used in: Section 6.3.2. The panel's SCCR must be equal to or greater than the available fault current where the panel is installed.

How to work it out (simple version). UL 508A Supplement SB uses a "weakest link" approach: the panel is only as strong as its lowest-rated power component.

1. **Get the available fault current** at the panel location. This comes from the facility's electrical engineer, a short-circuit study, or the utility. Write it down; it goes on the drawings.
2. **List every component in the power circuit** — main disconnect, fuses and breakers, power distribution blocks, motor starters and overloads, contactors, VFDs, and the primary side of control transformers. Control-circuit parts on the secondary of a control transformer usually do not count.
3. **Write down each component's SCCR** from its nameplate, datasheet, or the manufacturer's tested-combination table. If a part has no rating, UL 508A assigns it a low default value (often 5 kA), which usually ends up being the weak link.
4. **Take the lowest number.** That is the panel SCCR:

$$\mathrm{SCCR}_{\mathrm{panel}} = \min\left(\mathrm{SCCR}_1,\ \mathrm{SCCR}_2,\ \ldots,\ \mathrm{SCCR}_n\right) \ \geq\ I_{\text{available fault}}$$

5. **Compare it to the available fault current.** If the panel SCCR is too low, replace the weak component with a higher-rated one or use a manufacturer's tested combination (for example a specific fuse class protecting a VFD or a motor starter).

UL 508A also lets you raise some component ratings when they are protected by current-limiting fuses or breakers. That is an advanced step — ask before relying on it.

Example. The facility says 22 kA is available at the panel.

Component	SCCR
Main fused disconnect (Class J fuses)	100 kA
Power distribution block (stand-alone rating)	10 kA
Motor starter (manufacturer combination rating)	65 kA
VFD (with the fuses listed in its manual)	65 kA
Control transformer primary fuse holder (Class CC)	200 kA
Lowest value = panel SCCR	**10 kA**

10 kA is less than 22 kA, so the panel fails. Swapping the distribution block for one rated 100 kA when protected by Class J fuses makes the lowest value 65 kA, and $65 \geq 22$, so the panel passes and is marked "SCCR: 65 kA."

Further reading: Eaton — Short-Circuit Current Ratings (SCCR) resource page; Bussmann — SCCR calculations and marking (UL 508A SB); Mersen — Achieving higher SCCR for industrial control panels.

C.2 Enclosure Heat and Cooling

Used in: Section 6.3.6. Check the heat at the hottest expected ambient temperature.

Step 1 — Add up the heat inside the panel

Add the heat loss (in watts) of every device. Use the "power loss" or "heat dissipation" value from each datasheet. Typical sources:

- **Power supplies:** not all input power reaches the output. Heat $= P_{\text{out}} \left(\frac{1}{\eta} - 1 \right)$, where η is efficiency. A 240 W supply at 92% efficiency gives off about 21 W.
- **VFDs:** roughly 2–3% of the drive's rated power, or use the datasheet value.
- **PLC racks, I/O, switches, PCs:** use the datasheet watts (power drawn is roughly heat given off).
- **Relays, contactor coils, transformers:** a few watts each; transformers list their losses.

$$P_{\text{heat}} = \sum P_{\text{device}} \quad [\text{W}]$$

Step 2 — Work out how much heat the enclosure can shed on its own

A closed steel enclosure loses heat through its outside walls. How much it can lose depends on its exposed surface area and how much hotter you let the inside get compared to the room:

$$P_{\text{natural}} = k \times A \times \Delta T$$

- $k \approx 5.5$ W/(m$^2 \cdot$ °C) for painted steel (about 0.28 W/(ft$^2 \cdot$ °F)).
- A = exposed surface area. Don't count a side that is mounted against a wall or another enclosure.
- ΔT = maximum inside temperature you allow minus the maximum ambient temperature.

Step 3 — Compare, and pick a cooling method

If $P_{\text{heat}} \leq P_{\text{natural}}$, natural cooling is enough. If not, add cooling:

$$\text{Fan airflow (CFM)} = \frac{3.17 \times P_{\text{heat}}\ [\text{W}]}{\Delta T\ [^\circ\text{F}]} \qquad \text{AC size (BTU/hr)} = 3.412 \times P_{\text{total}}\ [\text{W}]$$

For an air conditioner, P_{total} is the internal heat *plus* the heat soaking in through the walls when the room is hotter than the panel ($k \times A \times (T_{\text{ambient}} - T_{\text{inside}})$). Add about 20–30% margin to fan airflow for filter and airflow-path losses.

Situation	Use
Heat load is below what the enclosure sheds on its own	**Natural cooling.** Sealed enclosure, no fan.
Too much heat, but room air is clean, dry, and at least ~5 °C (10 °F) cooler than the allowed inside temperature	**Filtered fan** (fan in, filtered vent out). Cheapest active option; filters need regular cleaning.
Room air is dirty, wet, oily, or corrosive, or the panel must stay NEMA 4/4X/12, but the room is still cooler than the allowed inside temperature	**Air-to-air heat exchanger** (keeps the enclosure sealed).
Room is as hot or hotter than the allowed inside temperature, or the heat load is large	**Air conditioner.** Also the usual choice for outdoor panels in direct sun.
Hazardous (classified) location	Cooling equipment must be rated for the area (see Section 6.5.5).

Example. A wall-mounted 36” H × 30” W × 12” D steel enclosure. Room maximum is 30 °C and we want the inside to stay at or below 40 °C, so $\Delta T = 10$ °C (18 °F).

- Exposed area (front, top, bottom, two sides; back is on the wall) = $1080 + 720 + 864 = 2664\ \text{in}^2 \approx 1.72\ \text{m}^2$.
- Natural capacity: $5.5 \times 1.72 \times 10 \approx 95$ W.
- Heat load: 24 V power supply 21 W + PLC rack 40 W + industrial PC 25 W + network switch 10 W + two small VFDs 80 W = 176 W.
- $176 > 95$, so natural cooling is not enough. The air is clean, so use a fan: $3.17 \times 176/18 \approx 31$ CFM; with 30% margin, choose a filtered fan of at least **40 CFM**.
- If the room could reach 45 °C instead, a fan can’t cool below room temperature, so use an air conditioner: wall heat gain = $5.5 \times 1.72 \times 5 \approx 47$ W, total = 223 W, and $3.412 \times 223 \approx 760$ BTU/hr. Choose the next standard size up (e.g., **1,000 BTU/hr**).

Further reading: nVent HOFFMAN — Calculating heat dissipation in enclosures (white paper); Electrical enclosure cooling — fan airflow formula and example.

C.3 Control Transformer Sizing

Applies to: control power transformers (CPTs) that take plant power (for example 480 VAC) and make 120 VAC for panel control circuits — power supplies, the PLC, contactor coils, fans, lights, and a service receptacle.

C.3.1 The Short Version

1. Decide how the transformer connects (primary and secondary voltage).
2. Add up the **sealed (steady) VA** of everything on 120 VAC.
3. Add 25% and round up to the next standard size.
4. Check that the transformer can handle the **inrush** when coils pull in.
5. Size the primary and secondary fuses.
6. Pick a part and wire it per the nameplate.

C.3.2 Step-by-Step

Step 1 — Decide how it connects

A 480 VAC *delta* system has three hot lines and no neutral. A single-phase control transformer is connected **line-to-line** across two of the phases (for example L1 and L2), so its primary sees 480 V. The secondary is 120 V. One side of the secondary (X2) is bonded to ground to make the grounded (neutral) conductor; the other side (X1) is the fused hot.

Most control transformers have dual-voltage windings (for example 240×480 primary, 120×240 secondary). You set the voltage with jumpers shown on the nameplate.

Step 2 — Add up the sealed VA

"Sealed" VA is what each device draws once it is running. Get it from the datasheet; for power supplies use input VA (output power ÷ efficiency ÷ power factor), not output watts. Only count loads that can be on at the same time.

$$\mathrm{VA}_{\text{sealed}} = \sum \mathrm{VA}_{\text{each device}}$$

Step 3 — Add margin and round up

Add 25% for growth and continuous loading, then pick the next standard size up. Common sizes are 50, 75, 100, 150, 250, 350, 500, 750, 1000, 1500, 2000, 3000, and 5000 VA.

$$\mathrm{VA}_{\text{required}} = 1.25 \times \mathrm{VA}_{\text{sealed}}$$

Step 4 — Check inrush

When a contactor or relay coil first pulls in, it draws several times its sealed VA for a few cycles. If many coils pull in at once, the transformer's output voltage dips; if it dips too far, coils chatter or don't close. Combine the sealed load with the inrush of the coils that can energize at the same moment:

$$\mathrm{VA}_{\text{inrush selection}} = \sqrt{(\mathrm{VA}_{\text{sealed}})^2 + (\mathrm{VA}_{\text{inrush}})^2}$$

Then look up the chosen transformer in the manufacturer's *regulation* (inrush) chart, usually in the 90% secondary voltage column at 40% power factor. The transformer's listed inrush rating must be at least your calculated value. If not, go up one size.

Step 5 — Size the fuses

The National Electrical Code (NEC 450.3(B), and 430.72(C) for motor control circuits) sets the maximum fuse size. For a transformer with *both* primary and secondary fuses:

Where	Rated current	Maximum fuse size
Primary	any (with secondary fuse)	250% of primary current
Secondary	9 A or more	125% (next standard size allowed)
Secondary	less than 9 A	167%
Primary only (no secondary fuse)	9 A or more	125%
Primary only (no secondary fuse)	2 A to 9 A	167%
Primary only (no secondary fuse)	less than 2 A	300% (500% allowed by 430.72(C) for control circuits)

$$I_{\text{primary}} = \frac{\text{VA}}{V_{\text{primary}}} \qquad I_{\text{secondary}} = \frac{\text{VA}}{V_{\text{secondary}}}$$

Use **time-delay** fuses (for example Class CC time-delay) on the primary so the transformer's own magnetizing inrush doesn't blow them. Both primary lines are fused because neither leg of a 480 V delta is grounded. On the secondary, fuse only X1; X2 is the grounded conductor.

C.3.3 Worked Example: 480 VAC Delta to 120 VAC

A panel is fed from a 3-phase, 480 VAC delta supply and needs 120 VAC control power. The 120 VAC loads are (example values — use your actual datasheets):

Load	Sealed VA	Inrush VA
24 VDC power supply, 240 W output (92% efficient, PF 0.95)	275	—
Modicon BMXCPS3500 PLC power supply	75	—
Enclosure fan	40	—
LED panel light	15	—
Service receptacle (NEC counts 180 VA)	180	—
Three contactor coils (10 VA sealed / 100 VA inrush each)	30	300
Total	**615**	**300**

1. **Connection.** Single-phase transformer, primary across L1–L2 at 480 V, secondary 120 V with X2 bonded to ground.

2. **Sealed VA** = 615 VA.

3. **Margin:** $1.25 \times 615 = 769$ VA $\Rightarrow$ next standard size is **1000 VA**.

4. **Inrush check** (all three contactors pulling in together):

$$\sqrt{615^2 + 300^2} = \sqrt{378{,}225 + 90{,}000} \approx 684 \text{ VA}$$

   Look up the 1000 VA model in the manufacturer's regulation chart (90% column). Its inrush rating must be at least 684 VA; 1000 VA control transformers are normally rated well above that, so 1000 VA is fine. Confirm on the chart for the exact part.

5. **Fuses.**

- Primary current: $1000/480 = 2.08$ A. Maximum with secondary protection: $2.5 \times 2.08 = 5.2$ A $\Rightarrow$ **5 A time-delay Class CC**, one in each primary line (2 fuses).
- Secondary current: $1000/120 = 8.33$ A. Less than 9 A, so the maximum is $1.67 \times 8.33 = 13.9$ A $\Rightarrow$ **10 A** fuse on X1 (leaves margin above the 8.33 A rating).

6. **Select a part.** For example, the Hammond Power Solutions PH1000MQMJ: 1000 VA, 240×480 V primary, 120×240 V secondary. Jumper the primary for 480 V and the secondary for 120 V exactly as shown on its nameplate diagram. Add finger-safe fuse blocks for the primary and secondary fuses, or choose the factory fuse-kit version (PH1000MQMJ-FK) if available.

Result: 1000 VA, 480 V to 120 V control transformer; 2 × 5 A time-delay primary fuses; 1 × 10 A secondary fuse on X1; X2 bonded to ground.

C.3.4 Further Reading

- Elliott Electric — Control transformer selection guide (inrush VA equation and regulation chart)
- Rockwell Automation — Proper selection of control circuit transformers (white paper)
- Electrical Engineering Portal — Overcurrent protection of transformers (NEC 450.3)

D. Electrical Drawing Package

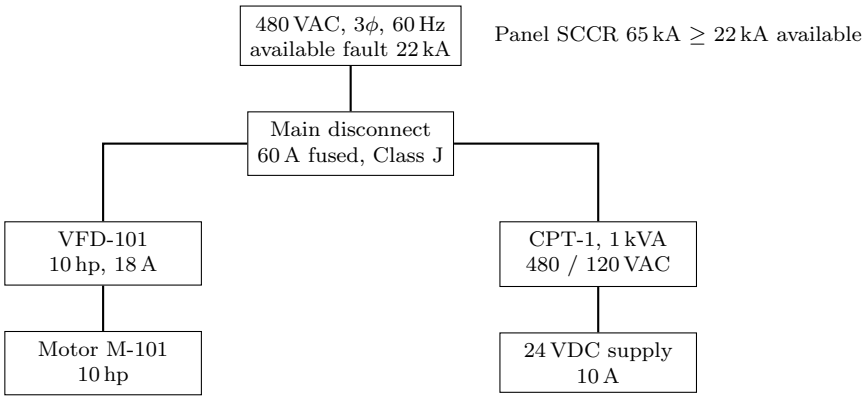
This appendix describes the documents that make up a controls electrical package, with short examples, the differences between them, and when each is used. The package is produced in Stage 4 (Section **??**) and brought to as-built in Stage 8. NFPA 79 lists the technical documentation expected for industrial machinery, and IEC 61082-1 gives the general rules for preparing electrotechnical documents.[1] UL 508A requires a wiring diagram for each listed panel.

D.1 Single-Line Diagram (SLD)

What it is. The power system drawn with one line per circuit, regardless of how many conductors it has. It shows the incoming supply (voltage, phases, available fault current), the main disconnect, protective devices with ratings, transformers with kVA and voltages, and major loads (motors, VFDs, heaters) with their ratings.

What it is not. It does not show control wiring, terminals, or wire numbers.

Quick example.



When it is used. Started in Stage 3–4 to size the feeder and disconnect, check SCCR (Section 6.3.2), and coordinate with the electrical engineer. Used by the electrician and maintenance to understand where power comes from.

D.2 Electrical Schematic

What it is. The detailed circuit drawing of the panel, drawn on numbered lines (rungs) with power on the left and right rails. It shows every device, contact, coil, fuse, terminal, and wire number, and

[1] NFPA 79-2024, Ch. 17 (technical documentation); IEC 61082-1:2014, *Preparation of documents used in electrotechnology — Part 1: Rules*; reference designations per IEC 81346-1:2022 and IEC 81346-2:2019; graphic symbols per IEC 60617 or IEEE 315-1975 (inactive-reserved since 2019).

cross-references between a coil and the contacts it operates. Wire and device numbering follows Section 6.3.7: a wire's number is the line where the signal starts, and a relay's tag is the line where its coil is.

Quick example (one rung):

Line	Content
120	Wire 120 from fuse F120 (X1) → PLC output DO-03 (card A04, terminal 5) → wire 121 → coil of relay CR121 → X2. Cross-reference under CR121: N.O. contact on line 214 (VFD-101 run).

When it is used. Drawn in Stage 4 and issued for the panel build in Stage 5. It is the main troubleshooting document for maintenance and is updated to as-built at closeout.

D.3 Panel Layout

What it is. A scaled drawing of the enclosure door (operator devices, nameplate, disconnect handle) and back panel (DIN rails, wire duct, components, terminal strips), with a bill of materials keyed to the drawing.

Quick example. Back panel 30 × 24 in: top row M580 rack and 24 VDC supplies; second row control relays and fused terminals; bottom row field terminal strips TB1–TB4 with 20% spare space; 2 in wire duct between rows; door shows HMI cut-out, E-stop, reset, and nameplate.

When it is used. Drawn in Stage 4 with the schematics, following the layout-consistency rules in Chapter 6. Used by the panel shop to build and by the designer to check heat (Appendix C.2) and space.

D.4 Wiring and Interconnection Diagrams

What they are. Point-to-point connection drawings or tables. A *wiring diagram* shows the physical connection of each wire inside a panel (terminal to terminal). An *interconnection diagram* shows connections between panels, junction boxes, and field devices, including the terminal plan at each end.

Quick example. PT-210 (field) pins 1/2 → cable C-210 (2-conductor shielded) → JB-2 terminals 7/8 → cable C-JB2 pair 4 → CP-1 TB2 terminals 31/32 → analog input card A06 channel 3. Shield landed at CP-1 only.

When they are used. Produced in Stage 4–5 for the panel shop and the electrician, and used for continuity checks and loop checks in Stages 5–6.

D.5 Cable Schedule

What it is. A table of every field cable.

Quick example.

Cable	Type	From	To	Route	Length
C-210	2C #18 shielded, PLTC	PT-210	JB-2 TB 7–8	Tray T3	12 m
C-101P	4C #10 VFD cable, shielded	CP-1 VFD-101	M-101	Conduit CD-4	25 m
C-ETH3	Cat6 industrial, shielded	CP-1 switch port 3	RIO-2	Tray T1	60 m

When it is used. Preliminary in Stage 4, finalized during construction. Used by the electrician to pull and terminate cable and by the estimator for quantities.

D.6 Supporting Lists

- **I/O list:** every PLC point with tag, description, signal type, range, card, channel, and terminal. Started in Stage 3, approved in Stage 4.
- **Terminal plan:** each terminal strip with terminal number, internal wire, and field wire.
- **Electrical bill of materials:** every part with manufacturer, part number, quantity, and drawing reference.

D.7 How the Documents Differ

Table D.1: Comparison of Electrical Package Documents

Document	Answers the question	Level of detail	Main users
Single-line diagram	Where does power come from and how is it protected?	One line per circuit; ratings only	Electrical engineer, designer, maintenance
Schematic	How does the circuit work?	Every device, contact, coil, and wire number	Designer, panel shop, maintenance
Panel layout	Where does each part physically go?	Scaled positions, sizes, and BOM	Panel shop, designer
Wiring / interconnection diagram	What connects to what, terminal to terminal?	Every wire end and terminal	Panel shop, electrician
Cable schedule	Which cables are pulled, where, and how long?	One row per cable	Electrician, estimator
I/O list	What does each PLC point do?	One row per I/O point	Controls, programmer, commissioning

E. Class I, Division 2 Panel Design

This appendix explains what changes when a control panel, or the equipment it serves, is located in a Class I, Division 2 (C1D2) area instead of an unclassified area. It complements the equipment examples in Section 6.5.5. The rules come mainly from NEC Article 501, with NFPA 496 for purged enclosures.[1] The area-classification drawing and the AHJ have the final say on every point below.

E.1 Start From the Area Classification

Before any design work, get the area-classification drawing and record for the panel location and each field device:

- Class and division (or zone) — for example Class I, Division 2.
- Gas group (A, B, C, or D) of the flammable material. Hydrogen is Group B; most hydrocarbons are Group C or D.
- The temperature code (T-code) needed, based on the autoignition temperature of the gas.
- The classified-area boundaries, which determine where seals are needed.

If the panel can be placed in an unclassified room with only rated field devices in the classified area, this is usually the simplest and cheapest design.

E.2 Equipment Selection

Every piece of equipment installed in the area must be listed and marked for it, for example *Class I, Div 2, Groups A–D, T4*. Read the actual label or certificate, not a catalog summary. Common ways equipment qualifies:

- **Nonincendive equipment** (UL 121201): cannot ignite the gas in normal operation. Many PLCs, I/O modules, power supplies, gateways, IPCs, and monitors are available this way.
- **Hermetically sealed or nonincendive contacts**: for relays, switches, and pilot devices.
- **Explosionproof enclosures**: for devices that cannot otherwise be rated.
- **Purged/pressurized (Type Z) enclosures**: allow ordinary equipment inside a protected enclosure.

[1] NFPA 70 (NEC) 2023: 501.10(B) (Division 2 wiring methods), 501.15 (sealing), 501.30 (grounding and bonding), 501.105 and 501.115 (instruments, relays, switches, and other arcing devices). NFPA 496-2024 (purged and pressurized enclosures). UL 121201, 9th ed. (nonincendive equipment). Clause numbers are from the 2023 NEC; confirm them against the edition adopted by the jurisdiction.

Listed equipment usually comes with installation conditions (a suitable enclosure, a maximum ambient, a required barrier). These conditions are part of the rating and must be shown on the drawings.

E.3 Arcing and Sparking Devices

In a normal panel, any relay, contactor, breaker, pushbutton, or fuse may be used. In a C1D2 panel, devices whose contacts open and close in normal operation must use one of the following:[2]

- contacts that are hermetically sealed, oil-immersed, or nonincendive;
- a nonincendive circuit, where the energy available cannot ignite the gas;
- an explosionproof enclosure; or
- a purged (Type Z) enclosure.

Fuses need the same attention: cartridge fuses and non-indicating, current-limiting fuses are generally permitted in general-purpose enclosures under specific conditions; check the listed conditions before using fused terminals or fuse holders.

E.4 Heat and Temperature Code

Surface temperatures of equipment must stay below the T-code limit for the gas. Panel PCs, power supplies, transformers, VFDs, and resistors run hot. Use the T-code on each device's marking, and check the enclosure heat calculation (Appendix C.2) at the highest expected ambient so devices stay within their rated ambient.

E.5 Wiring Methods

A normal panel may use any wiring method suitable for the building. In a C1D2 area the wiring method is part of the protection.[3]

- Permitted methods include threaded rigid metal conduit and IMC, and listed cable types such as PLTC, ITC, MC, and TC (exposed-run versions where required), with fittings listed for the location.
- Where flexibility is needed, use flexible metal conduit, liquidtight flexible conduit, or flexible cord rated for the location, with bonding jumpers.
- Nonincendive field wiring may use any wiring method suitable for unclassified locations, but must be kept separate from other circuits and installed per the control drawing.

E.6 Seals

Seals stop gas or flames traveling through conduit into another area or into an enclosure.[4]

- A **boundary seal** is required where conduit leaves the Division 2 area for an unclassified area, located near the boundary.

[2] NFPA 70-2023, 501.105(B) and 501.115(B).
[3] NFPA 70-2023, 501.10(B).
[4] NFPA 70-2023, 501.15.

- A seal is required where conduit enters an enclosure that is required to be explosionproof.
- Seals are filled with the listed sealing compound only after the conductors are pulled, and the fitting must be accessible.

E.7 Grounding and Bonding Differences

The bonding rules in Section 6.3.5 still apply, with one important difference: locknut-bushing and double-locknut connections cannot be relied on for bonding between the classified location and the point of grounding. Use bonding jumpers or bonding-type fittings instead, including across flexible conduit.[5]

E.8 Nonincendive Field Wiring and Intrinsic Safety

- **Nonincendive field wiring** connects a nonincendive field device to associated nonincendive apparatus. It must follow the manufacturer's control drawing, including cable capacitance and inductance limits.
- **Intrinsic safety (IS) barriers or isolators** are used when field devices are in a Division 1 area or when IS is preferred. The entity parameters of the barrier, cable, and field device must match, and IS wiring must be separated from all other wiring and identified (typically light blue). Panels with IS circuits follow UL 698A.

E.9 Purged and Pressurized Enclosures (Type Z)

A Type Z purge reduces the inside of an enclosure in a Division 2 area to unclassified, so ordinary equipment such as a PC or VFD can be used. NFPA 496 requires a clean air or inert gas supply, an initial purge before power is applied, and an alarm or indication on loss of pressure.[6] The purge system itself must be rated for the location, and the alarm must go to a place where someone will respond.

E.10 Marking, Documentation, and Maintenance

- Mark the panel and drawings with the area classification it is designed for.
- Add warning labels where required, for example "Do not open when an explosive atmosphere may be present" or "Do not disconnect while circuit is live unless area is known to be non-hazardous."
- Keep the control drawings, certificates, and installation instructions of every rated device with the project files.
- Maintenance: do not open energized equipment or disconnect live connectors unless the area is confirmed non-hazardous (gas testing or a hot-work permit), and replace only with parts carrying the same rating.

[5] NFPA 70-2023, 501.30(A) and (B).
[6] NFPA 496-2024.

E.11 Side-by-Side Comparison

Table E.1: Normal Panel vs. Class I, Division 2 Panel

Topic	Normal (unclassified) panel	Class I, Division 2 panel
Equipment	Any listed industrial equipment	Only equipment listed and marked for C1D2 (or Zone 2), right group and T-code
Relays, switches, breakers	Any type	Hermetically sealed, nonincendive, oil-immersed, or in an explosionproof or purged enclosure
Enclosure	Chosen for environment (NEMA 4X, 12, etc.)	Same, unless an explosionproof or purged enclosure is needed for unrated devices
Temperature	Rated ambient of devices	Also below the T-code limit for the gas
Wiring method	Any suitable method	Listed methods (conduit, PLTC, ITC, MC, TC) with listed fittings
Seals	Not required	Boundary seals; seals at explosion-proof enclosures
Bonding	Locknuts acceptable	Bonding jumpers or bonding fittings; locknuts not relied on
Field devices	Any	Rated devices, nonincendive field wiring, or IS barriers
PCs and HMIs	Any IPC	Rated IPC/monitor, thin client in a rated display, or PC in a purged enclosure
Documentation	Standard package	Plus area classification, control drawings, certificates, warning labels
Maintenance	Normal LOTO	Plus gas testing or permit before opening live equipment

E.12 Design Checklist

- ☐ Area classification, gas group, and T-code recorded for the panel and every field device
- ☐ Option of locating the panel in an unclassified area considered
- ☐ Every device in the area listed and marked for C1D2 (or Zone 2) with the correct group and T-code; installation conditions shown on drawings
- ☐ Arcing devices hermetically sealed, nonincendive, or protected by enclosure or purge
- ☐ Heat calculation done at the highest ambient; T-codes checked
- ☐ Wiring methods, fittings, and seals shown on the drawings
- ☐ Bonding jumpers specified where locknuts would otherwise be relied on

- ☐ Nonincendive and IS circuits follow control drawings and are separated and identified
- ☐ Purge system (if used) rated, with loss-of-pressure alarm to a monitored location
- ☐ Warning labels, certificates, and control drawings included in the turnover package

F. Modbus Addressing on the M580 and Productivity2000

This appendix explains how Modbus works, how to set up Modbus addressing on our two PLC platforms — the Modicon M580 (Control Expert Classic) and the AutomationDirect Productivity2000 (Productivity Suite) — and how to match those addresses in Ignition. Software versions are listed in Section 7.1.1.

When Modbus is used: EFC's standards are OPC UA between the M580 and SCADA and EtherNet/IP for device communications (Section 8.2). Modbus is used for the backup and conversion cases: a device without EtherNet/IP, an EtherNet/IP option that is not available, an existing Modbus RTU system being converted to Ethernet, and Productivity2000 SCADA links (which have no OPC UA server).

Scope: Modbus TCP, Modbus RTU where a device requires it, and Modbus/TCP Security.

F.1 Modbus Basics

Modbus is a request/response protocol: a client asks and a server answers. Servers never send data on their own, so every value SCADA shows was polled. All data lives in four tables.

Table	Size	Access	Legacy prefix	Read FC	Write FC
Coils	1 bit	Read/write	0xxxx	01	05 single, 15 multiple
Discrete inputs	1 bit	Read-only	1xxxx	02	—
Input registers	16-bit word	Read-only	3xxxx	04	—
Holding registers	16-bit word	Read/write	4xxxx	03	06 single, 16 multiple

Function codes in common use

FC	Name	Per-request limit
01 / 02	Read coils / discrete inputs	2000 bits
03 / 04	Read holding / input registers	125 registers
05 / 06	Write single coil / register	1
15	Write multiple coils	1968 bits
16	Write multiple registers	123 registers
22	Mask write register (set bits inside a word)	1
23	Read/write multiple registers in one transaction	125 read / 121 write
43/14	Read device identification	—

Framing

- **Modbus RTU (serial):** binary frames on RS-485/RS-232 with a slave address (1–247) and CRC. One master per bus.
- **Modbus TCP:** the same message with a 7-byte header (transaction ID, protocol ID, length, unit ID) on TCP port 502. No CRC.
- **RTU over TCP:** raw RTU frames (with CRC) over a TCP socket, common on low-cost serial-to-Ethernet converters. Not the same as Modbus TCP; the client driver must match.
- **Unit ID:** ignored by most native TCP devices (use 255 or 1). Through a gateway it selects the serial slave address.

Exception codes

Code	Name	Usual cause
01	Illegal function	Device does not support that FC (for example FC16 or FC22)
02	Illegal data address	Off-by-one, or a request spanning an undefined address
03	Illegal data value	Value out of range, or quantity above the device limit
04	Server device failure	Internal device fault
06	Server busy	Long command in progress; retry
0A / 0B	Gateway path / target failed	Gateway cannot reach the serial device: unit ID, wiring, baud

F.2 Addressing and Data Types

Most Modbus problems are an off-by-one address or a swapped word order. Settle both in writing for every device before building tags.

Address notation

The protocol itself only carries a 0-based offset (0–65535) within a table. Documentation adds other numbering on top:

Notation	First holding register	Where it is used
Offset (0-based)	0	Wire captures, Modicon %MW numbers, Ignition with zero-based on
Register number (1-based)	1	Most device manuals, Ignition default
Legacy 5-digit	40001	Older manuals; the leading 4 names the table
Extended 6-digit	400001	Devices with more than 9999 registers; Productivity Suite

40001, 400001, 1-based HR1, and %MW0 all point to the same register (offset 0). The conversion is:

$$\text{offset} = \text{register number} - 1 = \text{legacy address} - 40001 = \text{6-digit address} - 400001$$

Confirm any vendor map by reading a known value before building the full tag set (Modbus Application Protocol Specification).

Data types and word order

- Modbus only defines bits and 16-bit words. INT/UINT use one register; DINT, UDINT, and REAL use two; LREAL and LINT use four; strings use one register per two characters.
- Bytes within a register are big-endian. The order of the two words in a 32-bit value is not standardized: high word first (ABCD) or low word first (CDAB).
- **The Modicon M580 stores 32-bit values low word first** in consecutive %MW. Ignition must have *Reverse Word Order* on for these devices.
- Test with known values: 123.456 in a REAL and 70000 in a DINT. A word swap shows up as an obviously wrong number.

F.3 Platform Comparison

	Modicon M580 (Control Expert Classic)	**Productivity2000 (Productivity Suite)**
Where Modbus addresses come from	You assign them: located variables at fixed %MW / %M addresses	Assigned per tag in the Tag Database (MOD Start column), automatically or by hand
Address numbering	%MWn = offset n (0-based)	6-digit, 1-based (e.g., 400001) as shown in the Tag Database
32-bit word order	Low word first	Confirm with a test value (see Section F.5)
Modbus TCP server	Built into the CPU, port 502	Built into the CPU, port 502; enable on the Ethernet port
Modbus RTU	Through an X80 serial module (BMXNOM0200) or a gateway	Built-in serial ports
PLC as Modbus client	I/O scanning (DTM Browser) or READ_VAR / WRITE_VAR	MRX (read) and MWX (write) instructions
Ignition settings	Zero-based on, Reverse Word Order on	Zero-based off (1-based); set word order to match the test value

F.4 Step-by-Step: Modicon M580 in Control Expert Classic

The M580 is a Modbus TCP server on port 502 and can also act as a client. Only **located** variables (%M, %MW) can be read over Modbus; unlocated variables and DDT instances cannot. These steps are written for Control Expert Classic v16.2.

Setting up the M580 as a Modbus server

1. **Size the located memory.** In the Project Browser, open the CPU in the PLC bus configuration and go to its *Configuration* tab. Set the number of %M bits and %MW words large enough for the full SCADA map plus room to grow (see the register map layout in Section F.9).

2. **Declare located variables.** In the Data Editor, create each variable SCADA needs and enter its address in the *Address* column, for example `PT101_PV` (REAL) at `%MW200`. Put REAL, DINT, and UDINT values on even addresses and never locate anything on the second word of a 32-bit value.

3. **Copy between your DDTs and the map.** DDT instances (DDT_Process_Value, DDT_Device, and so on) can't be read over Modbus directly. In a dedicated section of the program, copy status values from the DDTs into the located map each scan, and copy commands and setpoints from the map back into the DDTs after checking limits.

4. **Lock down access.** On the CPU's *Security* tab, enable the access control list and list only the Ignition servers and engineering stations allowed to connect. Enforce Security disables FTP, TFTP, HTTP, EtherNet/IP, SNMP, and DHCP/BOOTP; re-enable only what the project needs.

5. **Test.** Write known test values into a REAL and a DINT, then read them with a standalone Modbus client (Modbus Poll, QModMaster) before setting up Ignition.

6. **Ignition device settings.** Port 502, Zero-based Addressing **on**, Reverse Word Order **on**, timeout 2000 ms, 1 concurrent request.

Do not use FC04 (input registers) against the M580. Its behavior depends on the %IW configuration and changed in Control Expert 15.0. Use FC03 (holding registers).

The M580 as a Modbus client

Method	Setup	Function codes	Use for
I/O scanning	DTM Browser: add a generic Modbus device and define request lines with a repetition rate	FC03, 16, 23 only	Cyclic exchange with drives, meters, remote PLCs
Explicit messaging	READ_VAR / WRITE_VAR with ADDM, or DATA_EXCH	Coils, holding and input registers	Input registers, coils, occasional reads/writes

Both methods can run against the same device at the same time. Wrap explicit messaging in a standard DFB with timeout, retry, and diagnostics rather than hand-building each call. The M580 CPU has no serial Modbus port: use an X80 serial module (BMXNOM0200) or an external TCP-to-RTU gateway (Section 6.4.5).

F.5 Step-by-Step: Productivity2000 in Productivity Suite

The P2-622 CPU has a built-in Modbus TCP server and Modbus RTU on its serial ports. In Productivity Suite, Modbus addresses belong to tags.

1. **Enable the server.** In Hardware Configuration, open the CPU's Ethernet port settings and make sure the Modbus TCP server is enabled on port 502. For Modbus RTU, set the serial port's protocol to Modbus and set the address, baud rate, and parity to match the bus.

2. **Assign Modbus addresses to tags.** Open the Tag Database. Select the tags SCADA needs, right-click the *MOD Start* column, and choose *Auto Assign Modbus Address*; or type the address in yourself. Addresses appear in 6-digit, 1-based form (e.g., `400101`).

3. **Keep the map stable.** Assign addresses deliberately, in blocks that follow the register map layout in Section F.9, so adding tags later doesn't shift existing addresses. Export the Tag Database to CSV and keep it with the project as the register map.

4. **Check 32-bit word order.** Put known values in a 32-bit integer and a float tag and read them with a Modbus client. Set Ignition's Reverse Word Order to whichever setting gives the correct values, and record the result in the register map.

5. **The P2 as a Modbus client.** Use the *MRX* (Modbus Read) and *MWX* (Modbus Write) instructions to read from or write to other Modbus devices, such as a meter or a VFD. Set the IP address (or serial port and slave address), function code, starting address, and word-swap option in the instruction. Trigger them on a timer and check the instruction's status bits for errors.

6. **Ignition device settings.** Port 502, Zero-based Addressing **off** (the P2 uses 1-based numbers), Reverse Word Order set from the test above.

[Verify the exact menu names and word-order behavior against the Productivity Suite v4.7 help files on the first project, and update this section.]

F.6 Worked Example

A pressure reading `PT101` (REAL, 32-bit) is read by Ignition from each platform.

Step	Modicon M580	Productivity2000
PLC location	`PT101_PV` located at `%MW200`–`%MW201`	Tag `PT101_PV`, MOD Start `400201` (registers 400201–400202)
Protocol offset	200	200
Ignition device	Zero-based on, Reverse Word Order on	Zero-based off, word order per test
Ignition tag path	`[M580_R23]HRF200`	`[P2_Skid1]HRF201`

Both tags read the same offset (200). The M580 path uses 200 because zero-based addressing is on; the Productivity path uses 201 because Ignition is set to 1-based numbering for that device.

F.7 Ignition Settings Reference

Perspective does not talk to Modbus directly. The Ignition Gateway's OPC UA module runs the Modbus driver as a device connection; OPC tags reference that device and Perspective views bind to the tags. On 8.1, reach serial devices through a TCP gateway.

Setting	Default	Function	M580 value
Port	502	Modbus TCP port	502
Communication Timeout	2000 ms *	Response wait	2000 ms
Concurrent Requests	1 *	Requests in flight	1–2
Max Holding Registers per Request	125	Read size (8.3.1+: separate write, 123)	125 / 123
Zero-based Addressing	Off *	On: HR0 = offset 0	**On**
Reverse Word Order	Off *	Swaps words of 32/64-bit values	**On**
Span Gaps	On *	Reads across unused addresses	On if map contiguous
Reverse String Byte Order	Off	Swaps bytes in string words	Test per device

[Defaults to be verified on the company Gateway.]*

Address syntax. Item paths take the form `[Device]HRF200`. Designators: `C/DI` (bits), `HR/IR` (INT), `HRUS` (UINT), `HRI/HRUI` (DINT/UDINT), `HRF` (REAL), `HRD` (LREAL), `HRS`*n*:*len* (string). A bit in a register is `HR100.3` (requires FC22). A unit ID is prefixed as `[GW1]3.HR100`.

M580 to Ignition cross-reference

With Zero-based Addressing and Reverse Word Order both on, the Ignition address number equals the %MW number.

M580 variable	Type	Offset	Legacy	Ignition (0-based)	Ignition (1-based)
%MW100	INT	100	400101	`HR100`	`HR101`
%MW101	UINT	101	400102	`HRUS101`	`HRUS102`
%MW200–201	REAL	200	400201	`HRF200`	`HRF201`
%MW202–203	DINT	202	400203	`HRI202`	`HRI203`
%MW300.3	BOOL in word	300	400301	`HR300.3`	`HR301.3`
%M50	EBOOL	50	000051	`C50`	`C51`
%MW400–409	STRING[20]	400	400401	`HRS400:20`	`HRS401:20`

[Confirm whether Modicon strings require Reverse String Byte Order on the first project (R23).]

F.8 Modbus TCP in Practice

- Performance depends on the number of requests needed to cover the data. Design maps so the client reads a few large contiguous blocks (up to 125 registers per read, 123 per write).
- Reading across unused addresses (span gaps) saves requests, but some devices return exception 02 for any undefined address in the block.
- Embedded devices often accept few simultaneous clients. Count SCADA servers, redundant servers, historians, laptops, and other PLCs before adding a client.
- Poll tiers: 250–500 ms for operator-critical status, 1 s for analogs, 5–30 s for configuration and diagnostics.
- Serial gateways are as slow as the serial bus. Use 19,200 baud or higher and keep gateway devices on their own driver connection.
- **Security:** Modbus TCP has no authentication or encryption; any host that can reach port 502 can read and write every register. Modbus/TCP Security (TLS on port 802) exists but device support is limited, so protection relies on network segmentation, controller IP allow-lists, and firewall rules (Chapter 8).

F.9 Company Rules

Device connections

- One Ignition device per PLC, named after the PLC tag on the System Layout drawing (e.g., `M580_R23`).
- Every M580 device: Zero-based Addressing on, Reverse Word Order on, port 502, timeout 2000 ms, concurrent requests 1. These are fixed by this standard, not chosen per project.

Register map layout

The SCADA map is a dedicated, contiguous block, separate from PLC working memory. The ranges below are %MW numbers on the M580; on the Productivity2000 use the same offsets (add 400001 for the 6-digit address).

Range	Area	Written by	Contents
0–99	System	PLC	Heartbeat, PLC mode, first-out alarm, map version, comm status of scanned devices
100–999	Status	PLC	Fixed-size blocks per UDT instance
1000–1499	Setpoints	SCADA and PLC	Setpoints and alarm limits, REAL, engineering units
1500–1999	Commands	SCADA	Command words, cleared by the PLC after action
2000+	Spare	Either	Reserved growth

- 32-bit values start on even offsets; nothing overlaps the second word of a 32-bit value.

- Analogs are REAL in engineering units. Scaled integers only where a third-party device requires them, with the factor documented.
- The map version word is incremented on every layout change; Ignition alarms on a mismatch.

[Decide fixed block sizes for DDT_Process_Value and DDT_Device (e.g., 16 and 8 words) so Ignition UDTs need only one base-address parameter.]

Commands and writes

- SCADA never writes directly to an output or interlock. It writes a request; the PLC checks permissives and interlocks (DDT_Interlock) and acts.
- One command word per device (bit 0 start, bit 1 stop, bit 2 reset, ...). The PLC clears the word after processing it.
- Setpoint writes are clamped in the PLC to engineering limits; out-of-range values are rejected and flagged.

Heartbeat and communication health

- The PLC increments a heartbeat word every second; Ignition alarms if it is unchanged for 10 s.
- Ignition writes a watchdog word every 5 s; the PLC alarms and puts remote commands in a defined safe state if it goes stale.
- Every Modbus device scanned by the PLC has a comm-healthy bit in the System area, alarmed in SCADA.

Third-party devices

Vendors and panel builders supply a register map spreadsheet with, for every point: name, table, address, **whether it is 0- or 1-based**, data type, word order, units and scaling, read/write, and poll tier. The header states supported function codes, maximum registers per request, connection limit, and unit ID. This is a Stage 1 input alongside the Standard Hardware Spreadsheet.

Security

- Modbus stays inside the control-network zone. Firewall rules allow port 502 only from named Ignition servers and engineering stations.
- The M580 access control list is enabled on every project and lists those hosts explicitly.
- No Modbus over remote-access or vendor VPN paths.

[Decide whether to trial Modbus/TCP Security or a BMENUA0100 OPC UA module on a future project.]

F.10 Troubleshooting

Symptom	Likely cause	Check / fix
Device stays "Connecting"	Route, IP, firewall, or M580 allow-list	Ping; test port 502 from the Gateway; check the CPU Security tab
All tags Bad	Address outside map, or timeouts	Gateway log exception code
Exception 01	FC not supported	Disable write-multiple or bit writes for the device
Exception 02	Off-by-one or spanned gap	Toggle zero-based to confirm; disable Span Gaps
Values shifted by one	Zero-based mismatch	Fix at the device setting
REAL wrong magnitude	Word order, or HR used for REAL	Use HRF; check Reverse Word Order
Strings pairwise swapped	Byte order in word	Toggle Reverse String Byte Order
Intermittent Bad quality	Too many requests or clients	Consolidate blocks, slow tag groups, lengthen timeout
Exception 0A / 0B	Gateway cannot reach slave	Unit ID, RS-485 wiring, baud/parity
FC04 returns inputs on M580	Control Expert 15.0 behavior	Use FC03

Tools: Wireshark (Modbus/TCP dissector) on a mirrored port; a standalone Modbus client (e.g., Modbus Poll, QModMaster) to test the device independently of Ignition.

F.11 Sources

- Modbus Organization, Modbus Application Protocol Specification v1.1b3 and Modbus/TCP Security v36 (2021)
- Inductive Automation, Ignition 8.1 Modbus driver, Modbus addressing, 8.3 device settings
- Schneider Electric FAQs: M580 Modbus client methods, FC04 behavior, M580 Security tab
- Cybus: Connecting a Modicon M580 (located variables, addressing, and word order)
- AutomationDirect, Productivity Suite — What's New; Productivity Suite built-in help (Tag Database, MRX/MWX instructions)
- PLCtalk: Productivity Suite Modbus address auto-assignment

G. Alarm List and Alarm Documentation

[Future: the alarm documentation described in this appendix is not implemented at EFC yet. It shows what a formal alarm-management practice would look like so it can be discussed and adopted in stages.]

G.1 The Standards

Alarm management is governed by ANSI/ISA-18.2 and its international equivalent IEC 62682. Both define an alarm *lifecycle*: write the rules (philosophy), decide which alarms are justified and how each should be set (rationalization), build them, operate and maintain them, and measure how the system performs. ISA has also published technical reports that explain each step in more depth.[1]

G.2 The Three Main Documents

- **Alarm philosophy** — the site rules: what qualifies as an alarm, how priorities are assigned, colors and sounds, shelving and suppression rules, who approves changes, and performance targets. Written once for the site and reused on every project.
- **Rationalization** — the review of each proposed alarm against the philosophy by a small team (process, operations, controls). Each alarm is either justified and fully defined, or rejected with a reason.
- **Master alarm database (alarm list)** — the result: one record per alarm with all of the fields below. It is the source for configuring Ignition and for testing.

G.3 Alarm Record Fields

Table G.1: Alarm Record Fields

Field	Meaning
Tag / alarm ID	Unique identifier, normally the instrument tag plus the alarm type (PT-210.HI).
Description	Short message shown to the operator.

[1] ANSI/ISA-18.2-2016, *Management of Alarm Systems for the Process Industries*; IEC 62682:2022 (Ed. 2); ISA technical reports TR18.2.1 (alarm philosophy, 2018), TR18.2.2 (identification and rationalization, 2016), TR18.2.3 (basic alarm design), TR18.2.4 (enhanced and advanced alarm methods), TR18.2.5 (monitoring, assessment, and auditing), TR18.2.6 (batch and discrete processes), and TR18.2.7 (packaged systems).

Table G.1 continued

Field	Meaning
Alarm type	High, low, deviation, rate of change, discrete (state), fault, or bad quality.
Setpoint	Value at which the alarm activates, in engineering units.
Deadband	How far the value must return before the alarm clears; prevents chattering.
On / off delay	How long the condition must persist before the alarm activates or clears; filters short spikes.
Cause	What most likely made the condition happen.
Consequence	What happens if the operator does nothing.
Operator response	The specific action the operator takes.
Maximum response time	How long the operator has before the consequence occurs.
Priority	Assigned from consequence and response time (Section G.5).
Operating-state applicability	Process states in which the alarm is meaningful (for example running but not shut down).
Suppression / shelving rules	When the alarm is suppressed by design, and who may shelve it and for how long.
Setpoint basis	Where the setpoint came from (PHA, equipment limit, operating experience).
Test method	How the alarm is proven during FAT and commissioning.

G.4 Worked Examples

Table G.2: Example Alarm List Entries

Tag	Type	SP / DB / delay	Priority	Cause → consequence	Response (time)
PT-210.HI	High process	50 torr / 5 torr / 10 s	Medium	Vacuum pump degraded or header leak → evacuation stalls	Check pump and header (15 min)
TT-305.LO	Low process	−180 °C / 2 °C / 30 s	High	Loss of cooling → column upset and off-spec product	Check refrigeration; reduce feed (5 min)
FIC-120.DEV	Deviation	±10% of setpoint / 2% / 60 s	Low	Valve sticking or loop out of tune → slow drift from target	Check valve position; switch to manual if needed (30 min)
M-101.FLT	Discrete fault	VFD fault bit / — / 0 s	Medium	Drive trip → pump stops, station cannot fill	Read VFD fault code; reset or call maintenance (15 min)
PT-210.BADQ	Bad quality	Signal <3.6 mA or comm loss / — / 5 s	Medium	Transmitter or wiring failure → loss of pressure indication and permissive	Treat value as unknown; check transmitter (10 min)

Table G.2 continued

Tag	Type	SP / DB / delay	Priority	Cause → consequence	Response (time)
PT-210.LO	Low process (state suppressed)	5 torr / 1 torr / 10 s	Low	Only meaningful while venting; suppressed by design in other states	Check vent valve (30 min)
LT-400.HI	*Rejected*	—	—	Duplicates the HH alarm on the same tank and has no separate operator action	Rejected in rationalization; HH alarm retained

G.5 Assigning Priority

Priority comes from two questions: how bad is the consequence, and how much time does the operator have? Equipment importance alone does not set priority.

Time to respond	Minor consequence	Major consequence	Severe consequence
More than 30 minutes	Not an alarm (use a warning)	Low	Low
10 to 30 minutes	Low	Low	Medium
3 to 10 minutes	Low	Medium	High
Less than 3 minutes	Medium	High	High

A typical healthy distribution is roughly 80% low, 15% medium, and 5% high priority alarms. Consequence categories (minor, major, severe) are defined in the site alarm philosophy for safety, environment, product, and equipment.

G.6 Performance Targets

The following benchmarks, commonly cited from ISA-18.2, are guidance for reviewing alarm performance after startup:

- Average alarm rate per operator: about 6 per hour is very likely acceptable; about 12 per hour is the most that is manageable.
- Alarm floods (more than 10 alarms in 10 minutes): less than 1% of the time.
- The 10 most frequent alarms: less than 1 to 5% of all alarms.
- Chattering and fleeting alarms: zero.
- Standing (stale) alarms: fewer than 5 on any day, with a plan to clear them.
- Unauthorized suppression: zero.

G.7 Where It Fits in Ignition

- Alarms are configured on tags, normally in the Ignition UDT for each equipment type (Section 10.4), with setpoint, deadband, delays, priority, and the display message from the alarm

list. Cause, consequence, and response can be stored as associated data and shown in the alarm faceplate.

- State-based suppression is driven by expressions on the alarm's enabled property.
- Shelving is limited by role and time in the alarm status table configuration.
- The alarm journal records activations, acknowledgments, and shelving for the performance review; notification pipelines handle call-outs.

See also Section 5.7 and the Alarms section of Chapter 11.

H. PLC Data Type Reference

This appendix lists the PLC data types (DDTs) used in EFC's Control Expert projects. Section 10.4 explains how they are used. The first three tables are the minimum member lists defined by this standard; the remaining tables document the types in the current program library, taken from the Control Expert data editor. Descriptions marked *TBD* have not been confirmed yet.

[Naming in the current library is not fully consistent: `AI_Sensors` *has no* `DDT_` *prefix, and member capitalization differs between types (for example* `Auto_Perm` *in* `DDT_Valve` *and* `AUTO_PERM` *in* `DDT_Motor_VFD`*). Align these the next time the library is revised.]*

H.1 DDT_Process_Value

Table H.1: DDT_Process_Value Minimum Members

Member	Type	Description
Raw	INT or DINT	Unscaled channel value.
Raw_Min / Raw_Max	DINT	Configured raw endpoints.
PV	REAL	Scaled engineering value.
Eng_Min / Eng_Max	REAL	Engineering range endpoints.
Units	STRING[16]	Engineering-unit text or enumerated unit identifier.
Quality	DINT	Valid, uncertain, bad, stale, simulated, or platform equivalent.
Timestamp	DDT or native time	Time of last valid update where required.
LL / L / H / HH	DDT_AI_Alarm	Analog alarm instances.
HW_Fault	BOOL	Module or channel diagnostic fault.
Underrange / Overrange	BOOL	Signal is below/above valid instrument range.
Simulated	BOOL	Value is under authorized simulation.

H.2 DDT_AI_Alarm

Table H.2: DDT_AI_Alarm Minimum Members

Member	Type	Description
SP	REAL	Rationalized alarm setpoint.
DB	REAL	Deadband in engineering units.
OnDelay / OffDelay	TIME	Qualified activation and clearing delays.
Enabled	BOOL	Alarm is configured for use.
Active	BOOL	Qualified alarm condition exists.
Latched	BOOL	Condition has latched where required.
Acked	BOOL	Acknowledgment state when managed in PLC.
Suppressed	BOOL	State-based designed suppression is active.
Shelved	BOOL	HMI/SCADA shelving indication when exchanged.
Priority	INT	Rationalized priority enumeration.

H.3 DDT_Device

Table H.3: DDT_Device Minimum Members

Member	Type	Description
ModeReq / ModeAct	INT or enum	Requested and active operating mode.
StartReq / StopReq / ResetReq	BOOL	Momentary validated requests.
RunCmd	BOOL	Final logical run command before output mapping.
Running / Stopped	BOOL	Processed equipment status.
AutoPerm / ManPerm	BOOL	Mode-specific permissive summaries.
Interlock	DDT_Interlock	Protective/control interlock summary.
FeedbackFault	BOOL	Command and feedback disagree after timeout.
Fault / Warning	BOOL	Aggregated condition summaries.
Available	BOOL	Equipment is healthy and eligible for control.
RuntimeSec	UDINT	Retentive accumulated runtime with controlled reset.
CommOK / DataValid	BOOL	Communications and exchanged data are valid.
Simulated	BOOL	Authorized simulation is active.

H.4 AI_Sensors

Analog sensor data with its four alarm states.

Table H.4: AI_Sensors Members

Member	Type	Description
`PV`	REAL	Scaled process value in engineering units.
`Raw`	REAL	Unscaled value from the input channel.

Table H.4 continued

Member	Type	Description
`Rate`	REAL	Rate of change of the process value *TBD*(units per time).
`Fraction`	REAL	*TBD*
`LL_Alarm`	BOOL	The low-low alarm is active.
`LL_Alarm_Acked`	BOOL	The low-low alarm has been acknowledged.
`Lo_Alarm`	BOOL	The low alarm is active.
`Lo_Alarm_Acked`	BOOL	The low alarm has been acknowledged.
`Hi_Alarm`	BOOL	The high alarm is active.
`Hi_Alarm_Acked`	BOOL	The high alarm has been acknowledged.
`HH_Alarm`	BOOL	The high-high alarm is active.
`HH_Alarm_Acked`	BOOL	The high-high alarm has been acknowledged.

H.5 DDT_Discrete

Discrete input with an override for testing and maintenance.

Table H.5: DDT_Discrete Members

Member	Type	Description
`Override_Enb`	BOOL	Override is enabled for this input; the PLC uses `Override_Sig` instead of the field signal.
`Override_Sig`	BOOL	Value used for the input while the override is enabled.
`Fault`	BOOL	Input fault (for example a channel or wiring fault) *TBD*.

H.6 DDT_Setpoint

Operator setpoint with a default value.

Table H.6: DDT_Setpoint Members

Member	Type	Description
`SP`	REAL	Active setpoint value.
`Default`	REAL	Default value loaded on first scan or reset *TBD*.

H.7 DDT_Valve

On/off valve with limit switch feedback; used with `DFB_Valve`.

Table H.7: DDT_Valve Members

Member	Type	Description
REQ	BOOL	Open request from the program (automatic mode).
Auto_Perm	BOOL	Permissive summary for automatic operation; all conditions met.
Man_Perm	BOOL	Permissive summary for manual operation; all conditions met.
HMI_Auto	BOOL	SCADA request to switch to Auto.
HMI_Manual	BOOL	SCADA request to switch to Manual.
HMI_Off	BOOL	SCADA request to switch to Off.
HMI_Lockout	BOOL	SCADA request to switch to Lockout.
CMD	BOOL	Final open command to the output.
ZSO	BOOL	Open limit switch feedback (valve is open).
ZSC	BOOL	Closed limit switch feedback (valve is closed).
HMI_CMD	BOOL	SCADA open/close command used in Manual mode.
Interlock	BOOL	Interlock active; the valve is forced to its safe state.
Cycle_Count	DINT	Number of open/close cycles (maintenance counter).
Reset_Count	BOOL	Request to reset the cycle count.
Feedback_Enabled	BOOL	Limit switch feedback is fitted and checked (enables feedback-fault detection).
Auto	BOOL	Valve is in Auto mode.
Manual	BOOL	Valve is in Manual mode.
Off	BOOL	Valve is in Off mode.
LockedOut	BOOL	Valve is locked out.

H.8 DDT_Motor_VFD

Motor driven by a variable frequency drive.

Table H.8: DDT_Motor_VFD Members

Member	Type	Description
REQ	BOOL	Start request from the program (automatic mode).
INTERLOCK	BOOL	Run interlock active; the motor is stopped.
AUTO_PERM	BOOL	Permissive summary for automatic operation.
AUTO	BOOL	Motor is in Auto mode.
MANUAL	BOOL	Motor is in Manual mode.
OFF	BOOL	Motor is in Off mode.
LOCKOUT	BOOL	Motor is in the locked-out (service) mode *TBD*.
HMI_AUTO	BOOL	SCADA command to switch to Auto.
HMI_MANUAL	BOOL	SCADA command to switch to Manual.
HMI_OFF	BOOL	SCADA command to switch to Off.
HMI_LOCKOUT	BOOL	SCADA command to switch to Lockout.
CMD	BOOL	Final motor run command to the drive.
YI	BOOL	Motor running feedback from the drive.

Table H.8 continued

Member	Type	Description
WARNING	BOOL	Motor warning is active.
ALARM	BOOL	Motor alarm (fault) is active.
SPD_AUTO	REAL	Speed command used in Auto mode (from the program).
SPD_MANUAL	REAL	Speed command used in Manual mode (from SCADA).
SPD_CMD	REAL	Output speed command sent to the drive.
RUNTIME	REAL	Accumulated runtime of the motor *TBD*(units).
HMI_RESET_RUNTIME	BOOL	SCADA request to reset the runtime.

H.9 DDT_Scale

Scale or weight transmitter interface.

Table H.9: DDT_Scale Members

Member	Type	Description
GrossWeight	REAL	Gross weight reading.
TareWeight	REAL	Stored tare weight.
NetWeight	REAL	Net weight (gross minus tare).
TareCmd	BOOL	Command to tare the scale.
ZeroCmd	BOOL	Command to zero the scale.
ResetScale	BOOL	Command to reset the scale or its communications *TBD*.
Stable	BOOL	Weight reading is stable (no motion).
Warning	BOOL	Scale warning is active.
Alarm	BOOL	Scale alarm is active.
Fault	BOOL	Scale or communication fault.
Msg	STRING	Status or error message text.

H.10 DDT_Seq

Sequence control and status.

Table H.10: DDT_Seq Members

Member	Type	Description
Disabled	BOOL	Sequence is disabled.
Ready	BOOL	Sequence is ready to start (all start conditions met).
Running	BOOL	Sequence is running.
Msg	string[255]	Current status message for the operator.
Start	BOOL	Start request.
Abort	BOOL	Abort request; the sequence goes to its safe state.

Table H.10 continued

Member	Type	Description
ShowCont	BOOL	*TBD*(likely shows the Continue prompt on SCADA).
Pop_Msg	string[255]	Message shown in an operator pop-up *TBD*.
State	DINT	Current step (state) number.
Continue	BOOL	Operator confirmation to continue past a hold point.
Alarm	DINT	Active sequence alarm code *TBD*.
Complete	BOOL	Sequence has completed.
Percent_Complete	REAL	Progress of the sequence, 0–100%.
HMI_Auto	BOOL	SCADA request to switch to Auto.
HMI_Manual	BOOL	SCADA request to switch to Manual.
HMI_Off	BOOL	SCADA request to switch to Off.
Auto	BOOL	Sequence is in Auto mode.
Manual	BOOL	Sequence is in Manual mode.
Off	BOOL	Sequence is in Off mode.
Ack	BOOL	Operator acknowledgment *TBD*.
HMI_Pause	BOOL	SCADA request to pause the sequence.
Paused	BOOL	Sequence is paused.
HMI_Topfill	BOOL	SCADA request for a top-fill *TBD*.

H.11 DDT_Station

Application-specific type for a cylinder fill station; used with DFB_Station.

Table H.11: DDT_Station Members

Member	Type	Description
TargetWeight	REAL	Target fill weight.
ReadyToStart	BOOL	Station is ready to start *TBD*.
ReadyToFill	BOOL	Station is ready to fill *TBD*.
Active	BOOL	Station is filling.
Complete	BOOL	Station has successfully completed the fill *TBD*.
Deadband	REAL	Fill deadband around the target weight *TBD*.
HMI_Select	BOOL	SCADA request to select this station.
HMI_Reset	BOOL	SCADA request to reset the station.
Selected	BOOL	Station is selected.
Venting	BOOL	Station is venting.
Vacuuming	BOOL	Station is being evacuated.
Recovering	BOOL	Station is recovering gas *TBD*.
Vacuumed	BOOL	Evacuation is complete.
Press_Balanced	BOOL	Pressures are balanced *TBD*.
Disabled	BOOL	Station is disabled.
Failed	BOOL	Station fill has failed.
PT_Valid	BOOL	Station pressure transmitter reading is valid.
PT_Diff	REAL	Pressure difference *TBD*.
InletREQ	BOOL	Inlet valve request.

Table H.11 continued

Member	Type	Description
InletCMD	BOOL	Inlet valve command.
SampleREQ	BOOL	Sample valve request.
SampleCMD	BOOL	Sample valve command.
VacuumREQ	BOOL	Vacuum valve request.
VacuumCMD	BOOL	Vacuum valve command.
PT	REAL	Station pressure.
Press_Max	REAL	Maximum allowed station pressure *TBD*.
TareCMD	BOOL	Command to tare the station scale.
ZeroCMD	BOOL	Command to zero the station scale.
GrossWeight	REAL	Gross weight.
TareWeight	REAL	Tare weight.
NetWeight	REAL	Net weight.
StatusEnum	INT	Station status code (see the StatusEnum values in Appendix I).
Msg	string[255]	Status message for the operator.
ScaleLoaded	BOOL	A cylinder is on the scale *TBD*.
ScaleTared	BOOL	Scale has been tared.
Fill_Rate	REAL	Fill rate *TBD*(units).
HMI_SP_Max_Vac_Press	REAL	Operator setpoint: maximum pressure at which the vacuum lines may be opened (permissive setpoint, Section 5.6).
TotalWeight	REAL	Total weight *TBD*.
SampleWeight	REAL	Sample weight *TBD*.
SamplePress	REAL	Sample pressure *TBD*.
Pressurizing	BOOL	Station is building pressure.
Sampling	BOOL	Station is sampling.
HighPressWarningReq	BOOL	Station maximum-pressure warning request *TBD*.
PressurizeReqd	BOOL	Pressurizing is required *TBD*.
SampleReqd	BOOL	Sampling is required *TBD*.
Filling	BOOL	Station is filling.
Disconnecting	BOOL	Station is disconnecting.
HMI_Disc	BOOL	SCADA request to disconnect.
Purged	BOOL	Station has been purged.
Light	BOOL	Station indicator light *TBD*.

I. Function Block Reference

This appendix documents the commonly used Derived Function Blocks (DFBs) in EFC's Control Expert projects, taken from the Control Expert data editor. Section 10.8 explains how DFBs are designed and managed. For each block, the pin table lists its inputs, outputs, and input/outputs; where a pin is a data type, its members are listed in Appendix H and are not repeated here. Internal timers and working variables are omitted. Descriptions marked *TBD* and names marked *[check]* still need to be confirmed.

I.1 DFB_Analog_Input

Scales an X80 analog input channel into a process value and passes the channel's health and diagnostic bits to the process value structure. Pairs with `DDT_Process_Value`.

Table I.1: DFB_Analog_Input Pins

Pin	Name	Direction	Type	Description
1	`Input`	Input	T_U_ ANA… *[check]* (X80 analog channel device DDT)	The analog input channel from the I/O module; members below.
1	`Raw_Value`	Output	INT	Unscaled channel value.
2	`Eng_Value`	Output	REAL	Scaled value in engineering units.
3	`PV`	Input/output	DDT_Process_ Value	Process value structure, including its LL/L/H/HH alarms (`DDT_AI_Alarm`), `HW_Fault`, `Value_ OOR`, `Underrange`, and `Overrange`.

Public variables: `Overrange` and `Underrange` (BOOL).

Table I.2: X80 Analog Channel Members Used by DFB_ Analog_Input

Member	Type	Description (Schneider)
`FCT_TYPE`	WORD	Function type

Member	Type	Description (Schneider)
CH_HEALTH	BOOL	Channel health
CH_WARNING	BOOL	Channel warning
ANA.VALUE	INT	Analog input value
ANA.FORCED_VALUE	INT	Forced value
ANA.FORCE_CMD	BOOL	Force command
ANA.FORCED_STATE	BOOL	Forced state
ANA.TRUE_VALUE	INT	Physical value
MEASURE_STS	INT	Measurement status
CH_ALIGNED	BOOL	Aligned channel
LOWER_LIMIT	BOOL	Measurement within lower limit
UPPER_LIMIT	BOOL	Measurement within upper limit
INT_OFFSET_ERROR	BOOL	Internal offset error
INT_REF_ERROR	BOOL	Internal reference error
POWER_SUP_ERROR	BOOL	Power supply error
SPI_COM_ERROR	BOOL	SPI communication error

I.2 DFB_Process_Value

Evaluates the four analog alarms (LL, L, H, HH) of a process value, applying each alarm's on-delay, deadband, enable, and inhibit settings. Pairs with DDT_Process_Value and DDT_AI_Alarm.

Table I.3: DFB_Process_Value Pins

Pin	Name	Direction	Type	Description
1	PV	Input/output	DDT_Process_Value	Process value whose LL/L/H/HH alarms are evaluated.
2	Reset	Input	BOOL	Resets latched alarms *TBD*.

No outputs or public variables. Internally, one on-delay timer per alarm (LL_TMR, L_TMR, H_TMR, HH_TMR) uses each alarm's DLY setting.

I.3 DFB_Device

Controls a motor or other on/off device: Off/Manual/Auto/Lockout modes, run command, feedback, runtime tracking, and stall detection. Pairs with DDT_Device.

Table I.4: DFB_Device Pins

Pin	Name	Direction	Type	Description
1	DEV	Input/output	DDT_Device	Device data structure (requests, permissives, command, feedback, alarms, runtime).

Table I.4 continued

Pin	Name	Direction	Type	Description
?	Inputs *[check]*	Input	—	The block's input pins were not visible in the reference screenshots; add them here.

Public variables: `HMI_CMD` (BOOL), `Mode` (INT; Off, Auto, and Manual values from `DDT_Mode`), `Runtime` (REAL), `Reset_Runtime` (BOOL), and `Motor_Stalled` (BOOL). Code sections: `Runtime_Tracking` (structured text) and `Commands`.

I.4 DFB_Valve

Controls an on/off valve with optional open/closed limit switch feedback: modes, open command, interlock and E-stop handling, feedback checking, and cycle counting. Pairs with `DDT_Valve`. Written in ladder diagram (one `Commands` section).

Table I.5: DFB_Valve Pins

Pin	Name	Direction	Type	Description
1	`Valve`	Input/output	DDT_Valve	Valve data structure.
2	`Valve_Output`	Output	BOOL	Output to the valve solenoid.
3	`Interlock`	Input	BOOL	Interlock; when active the valve goes to its safe state.
4	`Estop`	Input	BOOL	E-stop status; when active the valve goes to its safe state.
5	`Sys_Auto`	Input	BOOL	System (unit) is in Auto mode.
6	`Sys_Manual`	Input	BOOL	System (unit) is in Manual mode.
7	`Sys_Off`	Input	BOOL	System (unit) is in Off mode.

No public variables. Internally, on- and off-delay timers handle feedback checking.

I.5 DFB_Station

Runs one cylinder fill station: selection, venting, evacuation, pressurizing, sampling, filling, and disconnecting, with its scale, sequence, pressure transmitters, and three valves. Pairs with `DDT_Station`. Code sections: `DataMapping` and `Monitoring` (structured text).

Table I.6: DFB_Station Pins

Pin	Name	Direction	Type	Description
1	`Station`	Input/output	DDT_Station	Station data structure.
2	`Scale`	Input/output	DDT_Scale	Station scale.
3	`Seq`	Input/output	DDT_Seq	Station fill sequence.
4	`PT`	Input/output	DDT_Process_Value	Station pressure transmitter.
5	`InletValve`	Input/output	DDT_Valve	Inlet valve.
6	`SampleValve`	Input/output	DDT_Valve	Sample valve.
7	`VacuumValve`	Input/output	DDT_Valve	Vacuum valve.
8	`Estop`	Input	BOOL	E-stop status.
9	`Upstream_PT`	Input	DDT_Process_Value	Upstream (supply) pressure transmitter.

Public variables: `ONS_Auto` and `ONS_Manual` (BOOL; one-shots on a change to Auto or Manual *TBD*), and the station status values below, which are written to `DDT_Station.StatusEnum`.

Table I.7: DFB_Station Status Values (StatusEnum)

Status	Value
`Disabled`	0
`ReadyToStart`	10
`ReadyToFill`	20
`Filling`	30
`Complete`	40
`Failed`	50
`Venting`	60
`Vacuuming`	70

J. Software Version Control with Git

[Future: this appendix describes the planned Git integration from Section 7.2. It is not current practice. Until it is adopted, the manual file-naming rules in Section 7.2 apply.]

J.1 Git Basics for Controls Engineers

Git is a version control tool. It keeps the full history of a set of files in a *repository* ("repo"), so you can see what changed, when, by whom, and why, and go back to any earlier version.

- **Commit:** a saved snapshot of the files with a short message describing the change. Commit at meaningful points, not every save.
- **Tag:** a permanent label on a commit. Tags mark revisions: `rev-C` during development, `rev-0` at turnover, `rev-1` and later after turnover.
- **Branch:** a parallel line of work, used to try a change without disturbing the released version. Most projects need only a main branch.
- **Remote:** a shared copy of the repo on a company server (for example GitHub, GitLab, or Azure DevOps, as chosen with IT), so the history is not lost with one laptop.
- **.gitignore:** a list of files Git should not track (temporary, generated, or machine-specific files).

How the EFC rules map onto Git:

- One repository per project (for example `014_R23-Column`), holding the PLC and SCADA folders.
- Lettered development revisions and Rev 0 become tags; the date suffix in file names is no longer needed because Git records the date of every commit.
- A change made under MOC is committed with the MOC number in the message (for example `MOC-2027-014: raise PT-210 HH limit`) and tagged as a new revision.
- The companion revision log (Section 7.2) can be generated from the tag and commit messages.

J.2 Control Expert Classic

The problem. The normal project file (`.STU`) is binary, so Git can store it but cannot show what changed inside it. Control Expert has no built-in Git support.

The approach. Commit the binary project together with a full text export of the same version:

- `.STU` — the working project file (or `.STA`, the archive, which can only be created after the project is built).
- `.XEF` — a full project export in XML. It can be made at any stage and imported to rebuild the project (then saved as `.STU`, rebuilt, and downloaded). Because it is text, Git can show changes to logic, variables, and configuration.
- `.ZEF` — the XEF export plus the DTM (device) configuration; use it when the project has DTMs, such as EtherNet/IP devices.

XML differences are readable but noisy. Schneider's Control Expert DIF utility compares two exports more clearly, and third-party tools (for example Copia) provide Git-based version control with graphical LD/ST/FBD differences for Control Expert.[1]

Routine:

1. Save the project and build it.
2. *File* > *Export Project* to `.XEF` (or `.ZEF`) in the repo folder, using the same name as the project.
3. Commit both files with a message describing the change; tag it if it is a revision.

Exclude: `.BAK` automatic backups and temporary files.

J.3 Productivity Suite

The problem. The project file (`.adpro`) is also binary, and Productivity Suite has no Git support.

The approach. Commit the `.adpro` file together with the tag database CSV export (*Tag Database* > *Export*, which produces `_basic.csv` and `_extended.csv` files) so tag changes are visible in Git. Logic changes are not visible as text; use Productivity Suite's built-in **Project Compare** (available since v2.0) to compare two committed versions, and describe the logic change in the commit message.

Routine: save the project, export the tag database to the repo folder, commit both, and tag revisions. Reusable User Defined Instructions (UDIs) exported to the UDI library can be kept in a shared library repository.

J.4 Ignition 8.1

Ignition projects are stored on the gateway as folders of files, which suits Git well.

- **Where:** `<Ignition install>/data/projects/<project name>`. Views, scripts, named queries, and other resources are stored as folders containing `resource.json` plus content files (for example `view.json` or `code.py`). Create the repository in the project folder (or in `data/projects` for several projects).
- **Ignore:** the `.resources` folder and other generated files.

[1] Schneider Electric, Control Expert help, *User Application and Project File Formats*; Copia Automation, *Schneider Control Expert* support documentation.

- **Pulling changes onto a gateway:** after a `git pull`, tell the gateway to pick up the files with `system.project.requestScan()` (gateway scope), or a scan endpoint module, so the Designer and clients see the update.
- **Not in the project folder:** gateway settings such as tag providers and tags, device connections, database connections, users, and alarm pipelines are stored in the gateway's internal database. Back these up with a gateway backup (`.gwbk`), which is all-or-nothing, and export tags to JSON so tag changes can also be committed.

Inductive Automation recommends Git on the project folder to move work between development, test, and production gateways.[2]

J.5 Changes With Ignition 8.3

Ignition 8.3 (released September 16, 2025, a long-term support release) was designed with Git-based version control in mind:[3]

- **Whole-gateway configuration as files.** All gateway configuration moves from the internal database to files under `data/config` (for example tags under `data/config/resources/core/ignition/tag-definition`), so devices, tags, connections, and other settings can be committed along with projects.
- **Scan endpoints.** The REST API provides `/data/api/v1/scan/config` and `/data/api/v1/scan/projects`, and the gateway has a *Scan File System* button, to load pulled changes.
- **Deployment modes.** One configuration can hold settings for development, test, and production, selected on each gateway with the JVM argument `-Dignition.config.mode=<mode>`. Environment-specific values (addresses, credentials) no longer need separate copies of the project.
- **REST API** with a generated OpenAPI description at `/openapi`, and **secrets management** so passwords are not stored in committed files.
- **.gitignore template** provided by Inductive Automation; exclude `valueStore.idb` and some Vision binary resources.

What changes for us at the upgrade: the repository can cover the whole gateway, not just projects; gateway backups become a recovery tool rather than the only record of configuration; and moving a tested change from a development gateway to production becomes a Git pull plus a scan. Note that 8.3 no longer allows hot-swapping modules, so module installs still need a restart.

J.6 Checklists

Control Expert:

- ☐ Project saved and built; `.XEF`/`.ZEF` exported with the same name
- ☐ `.STU` (or `.STA`) and export committed together with a clear message

[2] Inductive Automation, *Ignition 8 Deployment Best Practices* (8.1 documentation); `system.project.requestScan` (8.1 scripting reference).

[3] Inductive Automation, *Ignition 8.3 Version Control Guide* and *New in 8.3* documentation; *8.1 to 8.3 Upgrade Guide*.

☐ Revision tagged; MOC number in the message where applicable

Productivity Suite:

☐ `.adpro` saved; tag database CSVs exported

☐ Logic change described in the commit message (checked with Project Compare)

☐ Revision tagged

Ignition:

☐ Project folder committed; `.gitignore` in place

☐ 8.1: tags exported to JSON and a gateway backup taken; 8.3: `data/config` committed

☐ After a pull, project (and 8.3 config) scan triggered and the change tested

K. Change Test Record

This form documents and tests a change to a system that has already been built (Section 7.6). Use one form per change. Scale the test steps to the change: a screen text edit may need one line, while an interlock change needs a full functional test. Changes to interlocks and safety-related items require an approved MOC (Section 7.3) and, under process safety management, a pre-startup safety review before the equipment returns to service.[1]

K.1 Change Identification

Change / MOC no.		**Date**	
Project no. and system		**Requested by**	
Software before (file name)			
Software after (file name)			

File names follow Section 7.2 (for example `014_R23-Column-PLC_1`).

K.2 Change Type

☐ I/O (added, removed, or reassigned)	☐ Electrical (wiring, devices, power)	☐ Sequence
☐ Interlock or permissive	☐ Setpoint limit (permissive/protective)	☐ Alarm
☐ SCADA screen or tag	☐ Network	☐ Other: _____

MOC required? ☐ Yes (interlock, protective setpoint, permissive limit, bypass, E-stop or safety circuit) ☐ No

[1] OSHA 29 CFR 1910.119(l) (management of change) and (i) (pre-startup safety review); IEC 61511-1:2016+A1:2017 (modification of safety instrumented functions); ISA/IEC 62443-2-1:2024, SPE 2 (change control); IEC 60204-1:2016+A1:2021, 18.7 and NFPA 79-2024, 18.7 (retesting after modification).

K.3 Description and Impact

What is changing and why	
Affected equipment	
Affected interlocks, permissives, and alarms	
Affected drawings and documents	

K.4 Before the Change

☐ Backup of the running PLC/SCADA taken
☐ Operations notified and agreed timing
☐ Bypasses or forces needed are recorded below
☐ MOC approved (if required)
☐ Equipment in a safe state for the work
☐ Test steps reviewed by a second person

Bypasses/forces applied (tag, reason, applied by): _____

K.5 Test Steps

Table K.1: Change Test Steps

#	Action	Expected result	Actual result	Pass / Fail	Init.
1	*Example:* With header pressure simulated at 40 torr, set `HMI_SP_Max_Vac_Press` to 45 torr and request the vacuum valve to open	Valve opens; permissive shown as met on SCADA			
2	*Example:* Raise simulated pressure to 50 torr and request the valve again	Valve does not open; permissive shown as not met with the reason			
3	*Example:* Enter 120 torr (above the hard-coded 100 torr maximum)	PLC clamps to 100 torr; SCADA shows the limit			
4					
5					

#	Action	Expected result	Actual result	Pass / Fail	Init.
6					
7					
8					

K.6 Regression and Abnormal-Condition Checks

Confirm that nothing connected to the change was affected. Check each item that applies:

☐ Normal operation of related equipment
☐ Alarms and first-out indication
☐ Mode changes (Off/Manual/Auto/Lockout)
☐ Operator security roles
☐ Bad-quality / out-of-range signal behavior
☐ Communication loss and recovery
☐ Power loss and restart behavior
☐ Other interlocks sharing the same inputs

K.7 Documentation Updated

☐ Electrical drawings
☐ I/O list
☐ Control Narrative
☐ Table of Interlocks and Permissives
☐ Alarm list
☐ Revision log / commit message
☐ Backup saved (after change)
☐ Released file name (no date)
☐ N/A

K.8 Return to Service and Sign-Off

☐ All bypasses and forces removed ☐ Equipment returned to normal mode ☐ Pre-startup safety review complete (if MOC)

Role	Name	Signature	Date
Performed by			
Independently verified by			
Accepted by Operations			

L. Sample FAT Document

This appendix is a **sample** Factory Acceptance Test (FAT) document, laid out as a template with a few example rows completed for project `014_R23-Column-PLC`. It is the "FAT Checkout document" of Chapter 3: outlined in Stage 2, procedures and expected results completed by late Stage 5, and actual results recorded in Stage 5A. The structure follows IEC 62381, and the electrical tests follow the testing and verification chapter of NFPA 79.[1] Remove the examples and adapt the sections to each project.

L.1 Cover and Approval

Project no. / name	014 / R23 Column	**System**	Column control panel CP-1
FAT document revision	C	**Test dates**	
PLC software (file)	`014_R23-Column-PLC_C_261002`	**SCADA project**	
Panel builder		**Test location**	

Witness / approval	Name	Signature	Date
EFC Controls			
Panel builder / vendor			
Operations			

L.2 Scope, References, and Test Setup

Scope: the CP-1 control panel, its PLC program, and the SCADA screens for the R23 column, tested at the panel builder's shop before shipment. Field devices are simulated; tests that need the field are listed in Section L.12 for commissioning (SAT).

References (record the revision used): electrical schematics and panel layout; I/O list; Control Narrative; Table of Interlocks and Permissives; alarm list; network map and address list.

[1] IEC 62381:2024 (Ed. 3), *Automation systems in the process industry — Factory acceptance test (FAT), site acceptance test (SAT), and site integration test (SIT)*; NFPA 79-2024, Ch. 18 (18.2 continuity of the effective ground-fault current path, 18.3 insulation resistance, 18.4 voltage tests, 18.6 functional tests); IEC 60204-1:2016+A1:2021, Cl. 18; UL 508A, 3rd ed. (panel construction and markings).

Test setup: how field signals are simulated (loop calibrator, switches, PLC simulation), the SCADA computer used, and the test equipment with calibration dates (multimeter, insulation tester, loop calibrator).

L.3 Entry Criteria

☐ Drawings at the revision under test

☐ Builder's own pre-FAT checks complete and signed

☐ PLC and SCADA software revision recorded

☐ Test procedures reviewed and approved

☐ Test equipment calibrated

☐ Open items from earlier reviews listed

L.4 Documentation and Visual Inspection

Table L.1: FAT — Documentation and Visual Inspection

#	Check	Result / remarks	P/F	Init.
V1	Nameplate: voltage, phase, frequency, FLA, SCCR, enclosure type, drawing number			
V2	UL label present (if a listed panel is required)			
V3	Components match the bill of materials and panel layout			
V4	Wire numbers and terminal markers at both ends, per schematic			
V5	Wire colors per the project scheme; externally powered wiring identified			
V6	Bonding: back panel, door jumper, DIN rails, ground bar (Section 6.3.5)			
V7	High-voltage separation, barriers, and warning labels			
V8	Spare space and spare terminals as specified (at least 20%)			
V9	Workmanship: duct fill, strain relief, no damaged insulation, torque marks			

L.5 Electrical Tests

Table L.2: FAT — Electrical Tests

#	Test	Acceptance	Measured	P/F	Init.
E1	Continuity of the protective bonding circuit from the PE terminal to each exposed metal part	Low resistance per NFPA 79 18.2			
E2	Insulation resistance, power circuits to PE (electronics disconnected)	$\geq 1\,\text{M}\Omega$ at 500 VDC			

#	Test	Acceptance	Measured	P/F	Init.
E3	Power-up: incoming voltage, control transformer secondary, 24 VDC supplies	Within ±5% of nominal	*Example:* 24.1 VDC	*P*	*AZ*
E4	UPS (if fitted): transfer on loss of input	PLC and network stay powered			
E5	Fuses and breakers: ratings match the schematic	As drawn			

L.6 I/O Point-to-Point Checks

Each point is checked from the terminal (simulated field signal) through the PLC to the SCADA screen. Analog inputs are checked at 0, 25, 50, 75, and 100% of range.

Table L.3: FAT — I/O Checks

Tag	Description	Card / ch.	Simulated / expected	PLC / SCADA reading	P/F	Init.
ZSC-110	*V-110 closed limit (DI)*	*A02 / 4*	*Switch closed → 1*	*1 / Closed*	*P*	*AZ*
SV-110	*V-110 solenoid (DO)*	*A04 / 2*	*Command on → 24 V at TB3-12*	*24.0 V*	*P*	*AZ*
PT-210	*Vacuum header (AI, 0–100 torr)*	*A06 / 3*	*4 / 8 / 12 / 16 / 20 mA → 0 / 25 / 50 / 75 / 100 torr*	*0.1 / 25.0 / 50.0 / 75.1 / 99.9*	*P*	*AZ*
FCV-120	*Feed valve (AO)*	*A07 / 1*	*0 / 50 / 100% → 4 / 12 / 20 mA*			

L.7 Communications and Network

☐ IP addresses match the address list

☐ SCADA connects over OPC UA with security enabled

☐ Loss of a device: alarm and fail-safe action

☐ All EtherNet/IP devices connected and healthy

☐ NAT mappings and firewall rules as designed

☐ Loss of SCADA: PLC continues to control safely

L.8 E-Stop and Safety Circuit Tests

Table L.4: FAT — E-Stop and Safety Circuits

#	Test (Section 6.3.3)	Expected result	P/F	Init.
S1	Press each E-stop in turn	Safety relay drops out; outputs and STO removed; E-stop lights; PLC/SCADA show which E-stop		

#	Test	Expected result	P/F	Init.
S2	Open each channel separately (jumper removed)	Safety relay trips and will not reset (cross-fault / channel fault detected)		
S3	Release E-stop; observe reset light	Light blinks (reset required); no restart		
S4	Press and release reset	Safety relay resets; light steady; equipment does not start by itself		
S5	Hold reset button pressed during E-stop release	No reset until button is released (monitored manual reset)		
S6	PLC drops the force-guided relay output	System trips as for an E-stop		
S7	VFD STO status read by PLC during E-stop	SCADA shows drive in STO		

L.9 Functional Tests From the Control Narrative

Table L.5: FAT — Functional Tests

#	Test	Expected result	P/F	Init.
F1	Mode changes Off / Manual / Auto / Lockout for each device	Mode follows request only when permitted; SCADA shows mode		
F2	Start the fill sequence with all permissives met	Steps run in order per the Control Narrative		
F3	Permissive: set the max vacuum-open pressure setpoint to its minimum and maximum limits, and try a value outside them	Values within limits accepted; outside values clamped; limits shown on SCADA		
F4	Each interlock in the Table of Interlocks and Permissives	Correct action; first-out shows the initiating condition; reset only after the cause clears		
F5	Each alarm in the alarm list	Activates at setpoint after delay; clears with deadband; correct priority and message		
F6	Bypass (if used)	Requires authorization; visible on SCADA; logged		

L.10 Abnormal and Recovery Tests

☐ Power loss and restore: no unexpected start
☐ Analog bad quality (open loop, <3.6 mA)
☐ Download/restart: sequences restart as designed
☐ PLC stop/run: outputs to defined state
☐ SCADA disconnect and reconnect
☐ Device communication loss

L.11 SCADA/HMI Tests

☐ Navigation and every screen opens
☐ Faceplates match the device data types
☐ Trends and history recording
☐ Screen scaling at deployed resolutions (Appendix A)
☐ Security roles: operator vs. engineer rights
☐ Alarm display, acknowledge, shelving

L.12 Punch List and Field Tests

Table L.6: FAT — Punch List

#	Item	Severity	Owner	Due	Closed
P1	*PT-210 faceplate shows wrong units (psig instead of torr)*	*Minor*	*Controls*	*Before shipment*	
P2	*Field test: verify V-110 stroke time (needs air at site)*	*Field*	*Controls*	*Stage 6*	

Severity: *Critical* (must be fixed before shipment), *Minor* (may ship; fix by a date), *Field* (can only be tested on site).

L.13 Final Acceptance

☐ All tests passed or recorded on the punch list ☐ No open critical items ☐ Software revision and backup recorded

FAT accepted by	Name	Signature	Date
EFC Controls			
Panel builder / vendor			

M. Parent Company Network and Remote Access Platforms

[Future: EFC is still adopting these platforms from the parent company. The details below describe the products in general; how they apply to EFC sites depends on how the parent company's IT/OT team has deployed them, and should be confirmed with that team.]

This appendix summarizes the two platforms used by the parent company: **Cato Networks**, the network and security platform, and **Claroty xDome Secure Access**, used for remote access to PLCs and other plant equipment. (Claroty is sometimes heard as "Clarity.")

M.1 Cato Networks (Cato SASE Cloud)

M.1.1 What It Is

Cato is a Secure Access Service Edge (SASE) platform: a cloud service that replaces traditional site-to-site VPNs, branch firewalls, and remote-access VPNs with a single managed network and security service.[1]

- **Cato SASE Cloud:** a global network of points of presence (PoPs) that carries traffic between sites and users and applies security inspection: SD-WAN, next-generation firewall, secure web gateway, threat prevention, and zero-trust network access (ZTNA).
- **Cato Socket:** an appliance installed at each site that connects the site to the Cato cloud over encrypted tunnels and provides a local (LAN) firewall.
- **Cato Client:** software for Windows, macOS, Linux, iOS, Android, and Chromebook that connects remote users, with multi-factor authentication and device posture checks. Clientless browser access is also available.

M.1.2 IoT/OT Features

In December 2024 Cato added IoT/OT security to the platform:

- automatic discovery and classification of connected devices (type, manufacturer, and version);
- policies by device type or group, for example limiting what a PLC or HMI may talk to; and
- threat prevention for those devices through DNS security, intrusion prevention, and anti-malware.

Cato's published material does not describe industrial-protocol (Modbus, EtherNet/IP) inspection; protocol-level control inside the plant remains the job of the controls network design in Chapter 8.

[1] Cato Networks, Knowledge Base, *Welcome to Cato Networks* and *What is Cato's ZTNA solution*; Cato Networks press release, *Cato Introduces Industry's First SASE-native IoT/OT Security Solution*, December 10, 2024.

M.1.3 What It Means for Controls

- The plant and business networks connect to the rest of the company through the Cato Socket. The controls networks in Chapter 8 stay behind their own NAT/firewall boundary (or the M580 gateway); Cato does not replace that boundary.
- Any connection between the controls network and the plant network or the wider company (SCADA clients, historian replication, remote access) should be agreed with corporate IT as a defined rule, not left open.
- Remote engineering access to PLCs should use Claroty (below), which is built for that purpose, rather than general Cato remote access straight into a controls network.

M.2 Claroty xDome Secure Access

M.2.1 What It Is

Claroty is an OT cybersecurity company. **xDome Secure Access** (formerly Claroty SRA) is its remote access product for cyber-physical systems: it lets engineers and vendors reach PLCs, HMIs, and engineering workstations in a plant under control and with a full record.[2]

- **Zero-trust access:** each user is given access to specific devices only, with multi-factor authentication and a credential vault so device passwords are not shared with the user.
- **Approval workflows and just-in-time access:** a user requests a session; an approver (for example the site engineer) grants it, including from a mobile device, for a set time.
- **Session recording and live monitoring:** sessions are logged and can be video-recorded; an administrator can watch a live session and disconnect it.
- **No inbound firewall openings:** the site component connects outward, so no inbound ports are opened into the plant network.
- **Access methods:** browser-based RDP, SSH, VNC, and HTTP/S without installing an agent, plus "micro-tunnels" for OT-specific protocols so thick-client engineering software can reach a PLC.
- **Deployment:** as a cloud (SaaS) service or on premises.

M.2.2 How We Would Use It

- An EFC engineer or a vendor requests access to a specific PLC or engineering workstation for a stated reason and time window.
- After approval, they connect either to an engineering workstation inside the plant (by RDP) that has Control Expert or Productivity Suite installed, or through a micro-tunnel from their own laptop to the PLC.
- Online changes made this way still follow Sections 7.3, 7.4, and 7.6; the session recording becomes part of the change record.
- Vendor accounts are time-limited and named, in line with Chapter 9.

[2] Claroty, *xDome Secure Access* product page and solution brief.

M.3 How the Two Fit Together

	Cato Networks	Claroty xDome Secure Access
Layer	Company network and security (WAN, firewall, user access)	OT remote access to specific plant devices
Main users	Corporate IT; all company users	Controls engineers, maintenance, and vendors
At the site	Cato Socket at the plant network edge	Site component inside the plant, connecting outward
Controls role	Carries and protects traffic to and from the plant	Controls, approves, and records sessions to PLCs and HMIs

See also Sections 8.5 (Remote Access) and 8.6 (System Hardening) and Chapter 9 (Accounts and Access Control).